Capstone Project Cover Sheet

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Student's Declaration

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BEAUTY FORMULA DERMOSCAN

OCTOBER 2023

ABSTRACT

The research presented in this paper embarks on addressing a pressing concern in the modern skincare industry: the difficulty faced by consumers and individual stakeholders in comprehensively understanding the intricate chemical compositions of skincare products. Recognizing this gap, our study leverages the integration of Optical Character Recognition (OCR) technology, machine learning algorithms (which includes, TF-IDF text matching technique and Rule-based text classification algorithm), and Power BI visualization techniques. The objective of our project is to provide an innovative solution that will culminate in a user-centric website where users can effortlessly assess skincare ingredients. Simply uploading a photo of their product, users are presented with detailed yet easily digestible insights into its composition. The inception of "Beauty Formula DermoScan" aims to revolutionize how consumers and the industry perceive and interact with skincare ingredients, fostering a more transparent, efficient, and knowledge-driven ecosystem.

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Abbreviations	Meaning
API	Application Programming Interfaces
BALC	Business Analytics Life Cycle
CER	Character Error Rate
CRISP-DM	Cross-Industry Standard Process for Data Mining
CSS	Cascading Style Sheets
DFD	Data Flow Diagram
DTC	Decision Tree Classifier
EDA	Exploratory Data Analysis
GUI	Graphical User Interface
HTML	Hypertext Markup Language
LR	Logistic Regression
ML	Machine Learning
MNB	Multinomial Naive Bayes
N/A	Not Applicable
NLP	Natural Language Processing
OCR	Optical Character Recognition
PowerBI	Power Business Intelligence
RF	Random Forest
SDLC	Software Development Life Cycle
SVM	Support Vector Machine
TF-IDF	Term Frequency-Inverse Document Frequency
UAT	User Acceptance Test
VS code	Visual Studio Code
WER	Word Error Rate

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CHAPTER 1

INTRODUCTION

1.1 Background of Study

The pursuit of flawless and radiant skin has been a timeless aspiration for individuals across cultures and generations. Healthy skin has long been considered a symbol of well-being, which innates individual desire to showcase good health through a clear, even-toned complexion. In contemporary society, this quest for perfect skin knows no age boundaries. People of all generations, from diverse backgrounds, are on a perpetual quest to discover the "best" skincare products that can help them achieve their skin goals. The beauty and skincare market offers a vast array of options, from local pharmacies and physical stores to online platforms. However, this abundance of choices often leaves consumers needing help with a dilemma on how to select the right products that truly deliver on their promises. Many consumers seek guidance from friends, healthcare professionals, or influential online personalities to navigate this sea of products. Unfortunately, these well-intentioned recommendations can sometimes lead to investments in expensive skincare products that fall below expectations.

It is widely acknowledged that daily skincare routines play a pivotal role in determining the long-term health and appearance of one's skin. The presence of synthetic chemicals in numerous skincare products can lead to various adverse effects for consumers, which may result in skin allergies. Consequently, a deep understanding of skincare products and their ingredients becomes essential. This knowledge empowers individuals to craft personalized skincare regimens that effectively target their unique concerns and complement the professional care they receive. To provide a thorough understanding of the profound connection between skin health and skin care ingredients, this project aims to develop an Optical Character Recognition (OCR) system tailored explicitly for skincare ingredient analysis, which allows consumers with the knowledge needed to make well-informed decisions regarding the skincare products they apply.

1.2 Research Motivation

The motivation of the study is to address customer concerns and enhance their knowledge about the ingredients in skincare products. In this day and age, customers face a wide range of options.

Still, many are unclear and concerned about how certain skincare ingredients may affect their skin, particularly with regard to harmful ingredients and allergic reactions. By using OCR to develop an online skincare ingredient checker, this research aims to bridge the knowledge gap and enhance customers' ability to perceive skincare ingredients. With this tool, consumers will be able to make the right and definitive decision to meet their unique skincare needs while also having the ability to avoid potentially harmful ingredients when purchasing skincare products.

1.3 Problem Statement

In the ever-expanding skincare industry, consumers face the daunting task of selecting products for their skin needs and preferences. These challenges originate from the complexity and opacity of skincare product ingredients, which leads to the following significant problems:

A. Ingredient Complexity

There are numerous skincare products on the market, each with a distinctive combination of ingredients. According to Mischa (2019), roughly 12,500 unique chemical compounds are permitted for use in the manufacture of personal care products in the United States alone, where 15 to 50 ingredients can be found in an average skincare product. The contents listed on skincare product labels are often complex and filled with scientific terminology, making it difficult for consumers to determine which ingredients are safe, beneficial, or potentially damaging to their skin.

B. Safety Concerns

Regarding skincare products, safety is of the utmost importance for consumers. Sad to say, for cost-saving reasons, certain beauty products contain restricted and prohibited ingredients known to cause health issues such as cancer, congenital disabilities, and developmental or reproductive disorders (Pratiwi et al., 2022). Fortunately, consumers are becoming increasingly concerned about allergens, irritants, and adverse effects caused by chemical ingredients in their skincare products. However, the need for more user-friendly tools to check the safety of these ingredients continues to add to consumers' worries and leave them uncertain.

C. Product Effectiveness Uncertainty

Skincare encompasses more than just safety; it is also about effectiveness. Consumers are eager to find out which products can effectively treat their unique skin conditions, such as fighting acne, minimizing the effects of ageing, moisturizing dry skin, or offering other skincare advantages. However, skincare products may only sometimes deliver on their marketed promises. As a result, consumers must scrutinize the ingredient lists to ensure efficacy.

D. Information Accessibility

Access to accurate and understandable skincare ingredient information needs to be more comprehensive in an era of readily available information. It is not easy for consumers to learn about skincare ingredients, as they have to turn to the internet and do lots of clicking to find reliable sources. Unfortunately, this pursuit frequently leads to contradictory or untrustworthy information, generating confusion rather than clarity.

1.4 Research Question and Objective

In this proposal, our team recommends the development of a web-based system to provide skincare ingredient analysis services as an education and auxiliary platform to help customers better understand and select suitable skincare products.

1.4.1 Research Objectives

An explicit, succinct, and targeted goal is what is meant by a research objective. It identifies the goal of the study and explains what the researcher wants to achieve with it. As a framework for the research, research objectives direct the researcher's efforts and aid in the development of research questions, methodologies, and data analysis.

The proposed Beauty Formula DermoScan using OCR engine has the following research objective:

1. Integrate the OCR engine into the Beauty Formula DermoScan application, ensuring seamless communication and data exchange between the OCR engine and the application's backend.

2. Implement machine learning algorithms to process OCR-extracted text data.
3. Develop machine learning algorithms for real-time matching of OCR-extracted ingredients with the database entries. Implement an intelligent matching system that considers variations in ingredient names and provides accurate results promptly.
4. Design an intuitive and user-friendly interface for the Beauty Formula DermoScan, allowing users to easily upload images, receive OCR-generated ingredient information, and access detailed descriptions.

1.4.2 Research Questions

Research questions are particular areas that researchers seek to address through their studies or investigations. These inquiries help to define the study's emphasis and shape the methodology, research design, and data collecting and analysis. Research questions are important because they give the study a clear direction by defining its scope and goal.

The proposed Beauty Formula DermoScan using OCR engine will address the following research questions:

1. How can the OCR engine be effectively integrated into the beauty formula DermoScan application to accurately extract text data from various skincare product labels?
2. What machine learning algorithms can be implemented to effectively process the text data extracted by OCR?
3. How can algorithms for real-time matching of OCR-extracted ingredients with database entries be developed, considering variations in ingredient names?
4. What design principles and interactive elements are essential for creating an intuitive and user-friendly interface for the Beauty Formula DermoScan?

1.5 Significance of Research

Our study on skincare ingredient analysis, using Machine Learning (ML) and OCR, has profound implications for the skincare industry.

1.5.1 Implications for the Skincare Industry

The profound implications for the skincare industry we have identified include empowering informed decision-making, enhancing safety and allergen identification, making skin care ingredients more transparent and safer for everyone, catalyzing industry innovation, data-driven insights for strategic decisions, and advancement in research methodologies.

A. Empowering Informed Decision-Making

One key outcome of this research is the empowerment of consumers and stakeholders in the skincare market. With the vast selection of skincare products available, individuals often face difficult choices. Our research provides a robust solution by enabling the swift and accurate analysis of ingredient lists, allowing consumers to make informed decisions aligned with their unique skincare needs and preferences.

B. Enhancing Safety and Allergen Identification

Skin allergies and sensitivities are common concerns among consumers. Our research serves as a vital tool for identifying potential allergens and irritants in skin care products, thereby mitigating the risk of adverse reactions. This aspect of our research is critical in promoting consumer safety and overall well-being.

C. Making skincare ingredients more transparent

Transparency about product ingredients is essential for product safety and consumer trust. In an industry where ingredient lists can be long and complicated, our research is a beacon of transparency. It gives consumers the information they need to make informed choices about the safety of the ingredients in their skincare products. It also encourages manufacturers to be more transparent about their product formulations.

D. Making skincare products safer for everyone

Our research could help to raise the bar for skincare safety across the industry. As more consumers demand safer products, manufacturers may be more likely to remove potentially harmful ingredients from their formulas. This could lead to a more comprehensive and systematic approach to skincare product safety, which would benefit everyone.

E. Catalyzing Industry Innovation

The incorporation of machine learning and OCR technology into skincare product analysis has the potential to catalyze innovation within the industry. Companies can harness these technologies to develop tailored products that meet the distinct needs and preferences of their target demographics, thus fostering a new era of effective and personalized skincare solutions.

F. Data-Driven Insights for Strategic Decisions

Additionally, our research generates valuable data regarding ingredient trends and consumer preferences within the skincare industry. These insights are invaluable to both researchers and industry players, aiding in data-driven decisions regarding product development and marketing strategies. This, in turn, encourages an environment of continuous improvement and adaptability within the industry.

G. Advancement in Research Methodologies

Finally, our project contributes to the broader landscape of machine learning and OCR applications. The techniques and methodologies developed serve as a foundation for similar projects in various domains, promoting interdisciplinary research and technological advancements beyond the scope of skincare. This underscores the versatility and potential of machine learning in real-world contexts.

1.5.2 Summary of Significance of Research

In conclusion, our research on skincare ingredient analysis, leveraging machine learning and OCR technology, has far-reaching implications. It empowers consumers, enhances safety, fosters transparency, supports efficacy-based choices, drives innovation, generates valuable data, and advances research methodologies. It represents a significant stride towards enhancing skincare practices and exemplifies the transformative potential of technology in diverse domains.

CHAPTER 2

LITERATURE REVIEW

2.1 Overview

The extensive information and pertinent knowledge in the current business climate that is provided in the literature review is relevant to or adopted into the project that is being proposed. In this chapter, several research studies from other scholars in adjacent domains or related industries were gone through. Literature enables us to pinpoint the precise project domain and scenario, which aids in choosing the tools and techniques to be used. Key theories and concepts for various types of components, including domain and problem statement, tools and techniques used, data analysis, machine learning (ML) model, evaluation metrics, and conceptual framework, will be further explained in this chapter.

2.2 Domain and Problem Scenario

Skincare ingredient analysis is the study of how safe and effective different ingredients in skincare products are. It covers a wide range of chemicals and natural substances used in cosmetics and personal care products, such as moisturizers, cleansers, serums, and sunscreens. The goal is to understand how these ingredients affect different skin types and assess any potential risks or side effects.

Table 2.1: Summary of Domain

Author	**Title**	**Summary**	**Critical Ideas**
Mayer (2022)	Healthline’s Evidence-Based Skin Care Ingredients Dictionary	● List of ingredients and their details. ● Medically reviewed. ● Which ingredients are harmful/prohibited.	Variables (Only dimension variables): ● Safety rating ● Potential hazards ● Potential side effects ● Skin type ● Skin concerns ● Benefits ● Descriptions
SkinSAFE	Ingredients: Essential & Glossary	● Provides information about the safety of skin care product ingredients. ● Collects information on skin care product ingredients from a variety of sources, including scientific studies, government databases, and product labels. ● Assigns each ingredient a safety rating of "Safe," "Caution," or "Unsafe”.	

Hivarkar et al. (2022)	Product Analysis using Computer Vision	Checking Ingredient List User Interface: ● Provides an interface for uploading scanned product ingredient images. ● Includes "Choose file" and "Submit" buttons for image selection and upload. Optical Character Recognition (OCR): ● Extracts and analyses text from the uploaded ingredient image. Text Analysis: ● Processes extracted text to remove white spaces, unwanted punctuation, and noisy data. ● Prepares the text for API calls. Edamam Food Database API: ● Uses Natural Language Processing (NLP) to extract food entities from unstructured text.	● While this article primarily focuses on food ingredient lists (cosmetic products have been mentioned as well), its insights can similarly apply to skincare products, which also feature ingredient labels on their bottles.

		● Allows users to search for food by nutrient quantity, brand, keyword, name, or UPC/Bar-code. ● Provides nutrition facts, allergen labels, and other food information. Displaying Results: ● Displays the uploaded image at the top. ● Shows fetched ingredient details in a tabular form. ● Details include labels, energy, protein, cholesterol, and more.	

Pratiwi et al. (2022)	Analysis of Prohibited and Restricted Ingredients in Cosmetics	● Discussion of prohibited and restricted ingredients in cosmetics. ● Highlighting potential adverse effects of certain ingredients. ● Emphasis on the importance of ingredient monitoring in cosmetics. ● Overview of recent analytical methods for ingredient detection. ● Compliance with FDA and EU regulations regarding cosmetic ingredient.	● Discuss the importance of skin care product ingredient analysis for various reasons, including developing new products, improving existing products, and ensuring the safety and efficacy of products. ● Discuss the latest trends and developments in skin care product ingredient analysis, such as the use of artificial intelligence and machine learning. ● Machine-learning they adopted: Deep-Learning.
Bouslimani et al. (2019)	The impact of skin care products on skin chemistry and microbiome dynamics	● Research article investigates skin care product impact on skin chemistry and microbiome. ● The study involves four beauty products and 11 volunteers over 9 weeks. ● Specific beauty product compounds detectable on skin for weeks. ● Different products had individualized effects, altering steroid, pheromone levels, and microbiome.	

		• Implications for potential personalized beauty products and skin disorder therapies.	

2.2.1 Domain

Based on a website article by Healthline that was written by Mayer (2022), and SkinSAFE, a database of beauty and skincare products, we have managed to identify our dimension variables, which are safety rating, potential hazards and side effects, skin type and concerns, benefits, and ingredient descriptions. Besides, in Hivarkar et al. (2022)'s study, a similar business process has been identified, where they have also adopted the use of OCR technology for their food ingredient analysis. In addition, our user interface will be very much similar to theirs, where the user has to upload a photo in order to get the analysis as the end result. In Pratiwi et al. (2022), and Bouslimani et al. (2022)'s studies, their objectives align with ours, which are providing ingredient safety assessments in order to contribute to the skin care safety assurance to the society. Their articles will also be used to support our findings regarding skin care ingredients.

2.2.2 Problem Scenario

Consumer safety remains a top concern, with a strong emphasis on ensuring skincare products contain non-toxic ingredients to prevent adverse reactions (Pratiwi et al., 2022). Personalization is key, as individuals have unique skin types and concerns, necessitating tailored product selection. Transparency regarding ingredient composition is increasingly vital, enabling informed choices and reducing the risk of hazards or allergens. The extensive array of skincare offerings can be bewildering, especially for newcomers, adding complexity to the selection process. Consumers also seek comprehensive scientific explanations for understanding how skincare ingredients influence skin health and appearance, aiding them in selecting products that align with their desired outcomes (Bouslimani et al., 2019). The challenge lies in deciphering complex ingredient labels, often laden with technical terms and jargon, making it challenging for consumers to discern the ingredients present and identify potential allergenic or harmful components (Mayer, 2022).

In response to the challenges of choosing skincare products, a new project is developing a website that uses OCR to scan ingredient labels and ML to assess product safety. Beauty Formula DermoScan will provide users with safety ratings, cater to their specific skin types and concerns, offer insights into potential hazards and side effects, benefits, and ingredient descriptions.

Additionally, it aims to provide understandable, scientific descriptions of ingredients to empower consumers to make informed choices about their skincare products.

2.2.3 Summary of Domain and Problem Scenario

We derived key dimension variables from reputable sources, such as Healthline and SkinSAFE. We will also be adopting OCR technology similar to Hivarkar et al. (2022)'s study for food ingredient analysis. Our project, Beauty Formula DermoScan, addresses the challenge of consumer safety, personalization, and transparency in skincare product selection. We use OCR and ML to provide safety ratings, cater to individual skin needs, and offer clear insights into ingredients, empowering users to make informed skin care choices.

2.3 Tools and Techniques

During the development of Beauty Formula DermoScan, four main tools and techniques will be used to achieve our project goals, namely, (1) Data preprocessing using Python, (2) Tesseract - open-source OCR engine, (3) Web development using Visual Studio Code, (4) PowerBI for visualization.

2.3.1 Python Language - Data Preprocessing

To ensure data accuracy, our project has utilized Python as the primary tool for data preprocessing. As Python is a dynamic and widely used programming language, it offers a wealth of libraries and modules specifically designed to streamline the critical phase of data preprocessing. Data preprocessing is pivotal in the research, as it involves cleansing, transforming, and preparing raw data to ensure its suitability for in-depth analysis and the application of various machine learning algorithms.

Table 2.2: Summary of Python Language - Data Preprocessing

Author	**Title**	**Summary**	**Critical Ideas**
Maye (2021)	Developing Techniques for Data Cleaning and Data Visualization On COVID-19	Python is a valuable tool for data cleaning due to its versatility and a wide range of libraries and packages designed for data cleaning and manipulation. ● Python's code capabilities make it a popular choice for data cleansing in the context of machine learning and deep learning, helping ensure data meets the required structure for algorithm fitting.	● The modularity of Python makes it adaptable for various data preprocessing tasks, catering to diverse project requirements.
Bansal & Chopra (2020)	Data Cleaning Techniques for Large Data Sets	● Python libraries are employed to remove missing values from the dataset, ensuring it is clean for subsequent analysis. ● The paper utilizes Python libraries for data cleaning as part of the	● Underscores the importance of Python's compatibility with machine learning and data analysis libraries, allowing for a seamless transition from preprocessing to advanced analytical tasks.

		preprocessing step.	
Oni et al. (2019)	A Comparative Study of Data Cleaning Tools	● The Pandas module, which is a part of Python's data analysis toolkit, is a powerful tool for data cleaning. ● Pandas is primarily designed for data analysis, making it suitable for tasks like data cleaning.	● Python's Pandas library is consistently recognized for its significance in data cleaning and transformation, as it provides efficient data manipulation capabilities. ● Python's ability to handle large datasets using modules like Numpy is emphasized, offering solutions to the challenges of big data preprocessing.

Based on the research conducted by Maye (2021), Python emerges as an indispensable tool for data cleaning, owing to its adaptability and a comprehensive suite of libraries and packages tailored for data cleaning and manipulation. According to Bansal & Chopra (2020), Python libraries are strategically employed to eliminate missing values from datasets, ensuring that the data is pristine and ready for subsequent analysis. Oni et al (2019) discussed the Pandas module within Python's data analysis toolkit is highlighted as a robust instrument for data cleaning.

Based on the findings, incorporating the Python programming language into our system serves as a robust mechanism for ensuring precise data accuracy, ultimately elevating the quality of our analysis outcomes. Python's prominent modules, such as Pandas and NumPy, are instrumental in simplifying data cleaning, handling large datasets, and making the data amenable for complex analytics and drawing meaningful results of our project.

2.3.2 Tesseract - open-sourced OCR engine

In the development of our Beauty Formula DermoScan, the system's functionality revolves around the extraction of text content from images containing text. To fulfill our specific project requirements and objectives, we have chosen to implement an OCR engine. Tesseract, as an open-source OCR engine, proves to be a robust choice, offering automated text extraction and precise document analysis. Its adaptability and customizable features align seamlessly with the needs of our system development.

Table 2.3: Summary of Tesseract - open-source OCR engine

Author	Title	Summary	Critical Ideas
(Ramteke & Al Maamari, 2023)	Tesseract OCR recognition based on Arabic machine-printed document	Process: ● Convert the image into binary format. ● Image resizes. ● Noise removal. ● Able to detect the text in images such as line, word and character detection.	● Simplifying the image by converting it into black and white improves the accuracy of subsequent text detection and recognition algorithms.
(Khete & Bakshi, 2022)	Autonomous Assistance System for Visually Impaired using Tesseract OCR & gTTS	● Can be used directly to extract printed text from images. ● Offers an API for programmers to integrate OCR capabilities into applications. ● Capable of recognizing handwritten text.	● API for ease of integration process and allows seamless and easy integration of output.

(Adjetey & Sarpong Adu-Manu, 2021)	Content-based image retrieval using Tesseract OCR engine and Levenshtein Algorithm	● Uses machine learning and pattern recognition techniques to recognize each character's shape and form. ● Group the recognized characters into words. ● Provides a confidence level for each recognized word to ensure the accuracy rate.	● Confidence levels for recognized words and enhanced precise retrieval quality.
(Zacharias et al., 2020)	Image processing-based scene-text detection and recognition with Tesseract	● Recognized text from the image and provided it as output. ● Deep learning-based text recognition model. ● Allow document digitization, data extraction, or text analysis.	● By using Tesseract as an OCR engine, the pre-built engine enables automation for word recognition and extraction using deep learning.

Ramteke & Al Maamari (2023) emphasize the image simplification process using binary format conversion, resizing, and noise removal for improved text detection and recognition accuracy. Khete & Bakshi (2022) highlight Tesseract’s direct usage for text extraction and integration through an API, which provides seamless integration in system development.Adjetey & Sarpong Adu-Manu (2021) focus on content-based image retrieval, providing confidence levels for recognized words and enhancing retrieval quality. Zacharias et al. (2020) demonstrate Tesseract’s capabilities in image processing, allowing automation for word recognition and extraction using deep learning.

The integration of Tesseract into our research represents a pivotal decision in enhancing the effectiveness and precision of our OCR system. Tesseract's deep learning-based text recognition capabilities automate the process of word recognition, simplifying complex tasks and streamlining document digitization and data analysis. Its exceptional accuracy in character recognition and the ability to group characters into words ensure a high-quality output, crucial for our project's success.

2.3.3 Visual Studio Code - Web development

In web development, the choice of development tools plays a pivotal role in shaping the efficiency and effectiveness of the entire development process. Among the plethora of tools available, Visual Studio Code (VS Code) has emerged as a prominent and versatile code editor, widely embraced by developers across the globe. VS Code, developed by Microsoft, offers a comprehensive and user-friendly environment for web developers to craft, test, and deploy web applications which caters to the requirements of our system website user interface development.

Table 2.4: Summary of Visual Studio Code - Web development

Author	Title	Summary	Critical Ideas
(Tan et al., 2023)	Visual Studio Code in Introductory Computer Science Course: An Experience Report	• Simplified User Interface: It features a compact and user-friendly interface, along with workspace configure settings that simplify project-specific configurations. • Cross-Platform Compatibility: VS Code is accessible across various operating systems, including macOS, Windows, and Linux, with a consistent user interface. • Extension Support: VS Code's core feature is its robust extension support, enabling users to add languages, debuggers, and tools to enhance their development experience.	• The simplified user interface of VS Code offers a streamlined and user-friendly experience, with workspace configuration settings that simplify project-specific setups and preferences.

<table>
<tr>
<td>(Wijaya & Azwir, 2020)</td>
<td>Visual Studio Code VDM Support</td>
<td>

Visual Studio Code, often referred to as VS Code, is a freely available source code editor.
Language support: comes with built-in support for languages like JavaScript and TypeScript.
Document Handling: VS Code efficiently handles source code files within a project using a document system.

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Visual Studio Code's exceptional performance, adaptability, rich ecosystem of extensions, regular updates, and strong community support make it the ideal choice for web development.
Its versatility extends beyond web development, which ensures users have a reliable and feature-rich code editor that evolve with our project requirements.
The rich collection of extensions simplifies the integration process of programming language, which includes, front-end development with HTML, CSS, and JavaScript, or backend development with Node.js, Python, or PHP into our workflow.

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<td>(Rask et al., 2020)</td>
<td>Information System Development Using Microsoft Visual Studio to Speed Up Approved Sample Distribution Process</td>
<td>

Programming Language: It is used for programming and developing computer programs in various programming languages.
Code Editor and Debugger: Visual Studio includes a code editor with integrated debugging capabilities, allowing developers to write and debug their code efficiently.

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		• Multi-Language Support: It supports a vast array of programming languages, totaling 36 different languages. • Versatile Capabilities: Users can use Visual Studio for various tasks, such as program creation, code editing, debugging, and design.	

Tan et al. (2023) investigates the practicality of using Visual Studio Code (VS Code) as an integrated development environment (IDE) for introductory-level Python courses. They emphasize its accessibility, ease of use, functionality, and popularity as key factors contributing to its suitability in educational settings. Wijaya & Azwir (2020) discussed Visual Studio Code (VS Code) and its features, emphasizing its role as a freely available source code editor. They specifically highlight its built-in support for languages like JavaScript and TypeScript, along with its efficient document handling system for organized code management. Rask et al. (2020) explore the application of Microsoft Visual Studio in information system development while underlining its versatility, code editing, and debugging capabilities, as well as its support for a wide range of programming languages, making it a comprehensive tool for software development.

In conclusion, Visual Studio Code's adaptability, language support, user-friendly interface, and extension ecosystem align well with the needs of our web development. By incorporating VS Code into our workflow, our team is able to enhance our development process, improve collaboration, and streamline project management. This, in turn, contributes to the overall success and efficiency of our web development endeavor.

2.3.4 PowerBI - Visualization

The visualization of the analysis results holds a crucial position, aiding users in comprehending the final output by presenting a clear, easily understandable visual representation. In our system, we will utilize Power BI for visualization, which can seamlessly integrate with our website, offering a structured review of users' analysis outcomes. Power BI, developed by Microsoft, serves as a business intelligence tool that empowers users to craft interactive dashboards and visually represent datasets according to their customized preferences.

Table 2.5: Summary of PowerBI - Visualization

Author	**Title**	**Summary**	**Critical Ideas**
(Khilari et al., 2022)	Business Intelligence Tool-Power BI for Performance Management	• Power BI's primary function is to create and share interactive reports and dashboards, both within and outside organizations. • Excels in searching, transforming, and visualizing data from various sources, including cloud data and APIs which allows seamless integration.	• Power BI offers powerful data integration capabilities, allowing users to connect and analyze data from various sources, enhancing business intelligence.
(Shoaib & Nandi, 2022)	Power Bi Dashboard for Data Analysis	• Power BI Desktop offers data integration from various sources to create business intelligence reports. • It enables the integration of data sources, albeit with some limitations in modeling flexibility. • Power BI is utilized for tasks like creating dashboards, designing and	• Power BI supports tasks such as dashboard creation, application design, data analysis, and report sharing, contributing to informed decision-making. • The tool is ideal for maintaining data accuracy and security while

		sharing applications, and performing data analysis for uncovering business insights.	integrating multiple data sources, enabling comprehensive big data analysis.
(Lan, 2019)	Building A Sales Dashboard for A Sales Department by using Power BI	• Power BI, developed by Microsoft, is a powerful Business Intelligence (BI) tool known for its ability to collect, organize, and visualize data from various sources. • It empowers end users to create reports and dashboards without the need for IT or data administrators, making it user-friendly for beginners. • Power BI is a cloud-based tool that allows business users to monitor and interact with their data, offering connectivity to various data sources, including local files, SharePoint, and enterprise data warehouses.	• The platform's ability to promote data-driven actions based on strategic insights is a significant advantage for organizations. • Power BI's user-friendly approach and the unique formatting of data enhance its appeal as an enterprise business intelligence solution.

Khilari et al. (2022) discusses Power BI's fundamental capability, which falls under the creation and sharing of interactive reports and dashboards, making it accessible for organizations and individual users. Shoaib & Nandi (2022) focus is on Power BI Desktop's ability to integrate data from multiple sources for creating business intelligence reports. While Lan (2019) emphasizes the prowess and the user-friendly nature of Power BI.

The incorporation of PowerBI within our system holds the transformative potential to empower users in generating interactive reports and dynamic dashboards for robust business intelligence. This powerful tool excels in seamlessly integrating and visualizing data from a wide array of sources, offering a versatile platform for data-driven decision-making. Its capabilities encompass efficient data management, fostering an environment where users can effortlessly create and customize reports and dashboards tailored to their specific needs.

2.3.5 Summary of Tools and Techniques

Each of the tools and techniques introduced will serve as a crucial aid in order to successfully develop and implement our system. Python's robust data preprocessing capabilities enable the reorganization and cleaning of dataset used efficiently, which ensures our data is well-prepared for analysis and further processing. Tesseract plays a pivotal role in text recognition, aiding us in extracting meaningful information from images and documents, enhancing the overall functionality of our system. Visual Studio Code serves as our development environment, facilitating the creation of a user-friendly and versatile web interface. Its cross-platform compatibility and extension support empower us to build an adaptable and feature-rich platform. PowerBI acts as a potent tool for data visualization, enabling us to present our analysis results in an easily understandable and interactive manner. It enhances our capacity to derive valuable insights from the data and supports informed decision-making. Together, these tools and techniques constitute a powerful arsenal for the successful realization of our project's goals.

2.4 Data Analysis

Data analysis is the process of cleaning, transforming, and analyzing raw data in order to obtain usable and relevant information that can assist businesses in making informed decisions. The types of data analysis that will be employed in this project include descriptive, exploratory, and predictive data analysis. The table below summarizes the example of data analysis done by previous literature that is related to this project.

Table 2.6: Summary of Data Analysis

Type	**Author**	**Title**	**Summary**	**Critical thoughts**
Exploratory + Descriptive	Dhaduk (2022)	EDA: Exploratory Data Analysis with Python	• Frequency table can be utilized to study the distribution of categorical variables, where metrics such as count and count percentage against each category can be used to assess the distribution. • Count-plot or bar chart can be used to visualize categorical variables.	• **Frequency distribution table** can be utilized to conduct ingredient frequency analysis. • **Bar chart** can be created to visualize and display the comparison of number of ingredients for each rating category.
Predictive	Purwandari et al. (2021)	Multi-class Weather Forecasting from Twitter Using Machine Learning Aprroaches.	• The authors used SVM, MNB, and LR for text classification. • SVM outperforms the other algorithms for the task of text classification with an accuracy value of 93%.	• The results suggest that **SVM** is more suitable for text classification tasks as compared to other algorithms mentioned in the articles. Limitations of SVM: • SVM works best on complex dataset. • SVM requires substantial labelled data for training.
	Rabbimov and	Multi-Class Text Classification of Uzbek News	• The authors performed text classification with six different machine learning algorithms: Support	

	Kobilov (2020)	Articles using Machine Learning	Vector Machine (SVM), Decision Tree Classifier (DTC), Random Forest (RF), Logistic Regression (LR) and Multinomial Naive Bayes (MNB). • SVM has the highest accuracy at 86.88%.	• It can be challenging to understand the reasoning behind SVM's decisions.
	Popovski et al. (2019)	A Rule-based Named-entity Recognition Method for Food Information Extraction	• The authors proposed a rule-based named-entity recognition method for food information extraction and classification. • The results showed that the rule-based system behaved consistently using different independent evaluation datasets and achieved promising results.	• Rule-based approach can complete text information classification task with high success rate. Advantages of rule-based system: • Quickly classify text based on predefined rules without the need for training on large datasets. • Can be computationally efficient and have low resource requirements.
	Carchiolo et al. (2019)	Medical prescription classification: a	• The authors developed a system to analyze and authorize medical prescriptions.	

		NLP-based approach	• The system used rule-based approach for text information classification, which correctly classified 95% of the 800,000 text rows.	• Can be effective even with limited labelled data.
	Chen et al. (2019)	Clinical trial cohort selection based on multi-level rule-based natural language processing system	• This paper reviewed a Natural Language Processing (NLP) system integrated with rule-based approach to classify patients based on text information from medical records. • The system obtained an overall micro-F1 score of 0.9028, showing high accuracy even with a small training dataset.	

2.4.1 Descriptive and Exploratory Data Analysis

Descriptive analysis summarizes and explains the key characteristics of a dataset. It provides an overview of the data and aids in identifying patterns or trends. On the other hand, exploratory data analysis (EDA) focuses on exploring and comprehending data using visualizations, summary statistics, and data profiling techniques.

Dhaduk (2022) highlighted the use of frequency distribution tables and bar charts for summarizing and visualizing datasets involving categorical variables. For example, frequency tables can be utilized to conduct frequency analysis with the help of metrics such as count and count percentage against each category. Furthermore, visualization tools such as bar charts can be created to visualize and compare variables for each category.

2.4.2 Predictive Data Analysis

Predictive analysis uses historical data and machine learning algorithms to identify patterns and build predictive models.

The previous literature of Rabbimov and Kobilov (2020) and Purwandari et al. (2021) compared different machine learning algorithms and concluded that SVM is more suitable for text classification. However, there are some limitations to using SVM for text classification. Both literatures agreed that SVM produces better results for text classification when performed on complex datasets. Moreover, SVM requires adequate labelled data for model training to obtain a high accuracy value. Besides, it can also be challenging to understand the reasoning behind decisions made by the SVM model.

On the other hand, Popovski et al. (2019) proposed a rule-based system suitable for information classification of textual data. Carchiolo et al. (2019) validated this approach by developing a rule-based system for text information classification, which achieved a success rate of 95%. Moreover, Chen et al. (2019) suggested that a rule-based NLP system can classify text information accurately without needing a large training dataset. In short, all three articles concluded that the rule-based approach works well on smaller datasets and can classify text information more efficiently and faster based on predefined rules without needing large datasets for model training.

2.4.3 Summary of Data Analysis

In conclusion, the above literature provided sufficient evidence that a rule-based system would be more suitable for our project, which involves classifying skincare ingredients based on their ratings and benefits or risks. Therefore, our project envisions a rule-based system as the foundation of our skincare ingredient classification solution.

2.5 Machine Learning Model

The machine learning model part will be separated into two parts for more detailed explanation, namely, text matching and text classification. To further ensure the achievement of desired output, our project will perform text matching via using TF-IDF, while text classification is performed using Rule-Based approach.

2.5.1 Text Matching

Text matching is a specific application of text similarity, and TF-IDF (Term Frequency-Inverse Document Frequency) is one of several techniques and approaches that can be used to measure text similarity. TF-IDF text matching is a technique for assessing how similar two pieces of text are by examining the terms two pieces of text include and their significance within a wider collection of documents.

Table 2.7: Summary of ML for Text Matching

Author	Title	Summary	Critical Ideas
Wang et al. (2020)	Measurement of Text Similarity: A Survey	● Overview of text similarity measurement approaches. ● Classified into text distance calculation and text representation techniques.	● TF-IDF is one of the text similarity measurements under text representation techniques in this paper.
Sitikhu et al. (2019)	A Comparison of Semantic Similarity Methods for Maximum Human Interpretability	● Techniques for calculating semantic similarity: cosine similarity using TF-IDF vectors, cosine similarity using word embedding, and soft cosine similarity using word embedding.	● Among the three techniques, the cosine similarity using TF-IDF vectors has the highest accuracy rate, delivering results that are simple to understand and used in other information retrieval applications.

Shahmirzadi et al. (2019)	Text Similarity in Vector Space Models: A Comparative Study	● Different vector space models' performance for measuring semantic text similarity, including TF-IDF, topic models, and neural models.	● The basic TF-IDF model is a sensible option in terms of performance and cost. ● Advanced techniques like Latent Semantic Indexing (LSI) and Paragraph Vectors (D2V) requires more computational work and is only worthwhile when the text is short and the similarity check is simple.

Wang and Dong (2020) deliver a thorough overview of text similarity measurement techniques and classifies TF-IDF as a corpus-based approach, which corpus is an enormous collection of text that is used to analyze and extract linguistic knowledge from. Sitikhu et al. (2019) conclude that cosine similarity using TF-IDF vectors outperforms other techniques in the paper. Shahmirzadi et al. (2019) highlight that in terms of performance and price, the basic TF-IDF model is a decent choice as greater computational work is required for advanced algorithms.

2.5.2 Text Classification

Text classification is an NLP task where text documents are classified into preset classes or labels according to their content. Rule-based text classification is one of the oldest classification techniques. Its categories text using predetermined rules developed by experts. These rules specify the criteria by which text is categorized.

Table 2.8: Summary of ML for Text Classification

Author	**Title**	**Summary**	**Critical Ideas**
Amishakirti (2023)	Rule Based Approach in NLP	● Explores both the advantages and disadvantages while developing rule-based systems for text-based tasks.	● The rule-based technique entails using a specific set of rules or patterns to extract information, capture certain structures, can be used to complete tasks like text classification.
Ali (2022)	Understanding Text Classification in Python	● Detailed guide on text classification in Python. ● Covering many text classification aspects.	● Rule-based classification uses manual rules to categorize text, while machine learning-based classification uses supervised machine learning to create models for text categorization.

Luo. (2021)	Efficient English text classification using selected Machine Learning Techniques	• Examines machine learning algorithms, SVM and NB, for text classification tasks.	• SVM is an approach that is used to predict and define how to classify new categories for a sample of datasets
Cervantes et al. (2020)	A comprehensive survey on support vector machine classification: Applications, challenges and trends	• An overview of the Support Vector Machine (SVM).	• SVMs are a powerful technique used in data classification and regression analysis

Aubaid and Mishra (2020)	A Rule-Based Approach to Embedding Techniques for Text Document Classification	• Studies the Rule-based technique for classifying text documents.	• By combining rule-based text classification approach with embedding techniques, the authors developed a D2vecRule algorithm, which outperforms other algorithms such as JRip, One R, and ZeroR.

This article written by Amishakirti (2023) discusses rule-based approach in text analysis, including the pros and cons, and the text-based tasks that a rule-based approach can be applied. Ali (2022) discovers the concepts and applications of text classification, while exploring examples of how to create a Python text preparation pipeline and text classification model. Aubaid and Mishra (2020) develop a text classification technique: D2vecRule algorithm, which is a combination of rule-based text classification approach and embedding techniques, and good classification results are provided by the D2vecRule algorithm. As Cervantes et al. (2020) states that SVMs have limitations of (1) parameter selection, (2) trouble solving multi-class issues, (3) limited performance on imbalance dataset; hence, SVM is not chosen to perform text classification in our project.

2.5.3 Summary of Machine Learning Model

Due to its scalability, optimal efficiency, and affordability, the TF-IDF model is seen as an appropriate option for text matching, particularly when more sophisticated algorithms require greater computer resources. Moreover, after comparing different text classification techniques, the rule-based classification technique is chosen due to it performs well in limited data as creating rules does not require a large dataset, and it works well in well-defined and stable text categories.

2.6 Verification and Validation

Verification and validation are significant parts in research, especially in fields that involve calculation and analysis. Verification ensures the model and system is correctly built, while validation ensures the model and system meet the user's expectation and needs. Based on our research, Character Error Rate (CER) and Word Error Rate (WER) are commonly used in evaluating the performance of an OCR engine, and metrics such as accuracy, error rate, precision, recall, and F1 score are used to measure the performance of the classification models.

Table 2.9: Summary of Verification and Validation

Author	Title	Summary	Critical Ideas
Rakshit et al. (2023)	A Novel Pipeline for Improving Optical Character Recognition through Post-processing Using Natural Language Processing	• This paper introduces a process that starts by reading handwritten or printed text using OCR and then enhances its accuracy using NLP.	• CER and WER are measures that can be used to evaluate the performance of an OCR engine.

Minaee et al. (2021)	Deep Learning Based Text Classification: A Comprehensive Review	• Provides a comprehensive review of more than 150 deep learning-based models for text classification. • Discussing their technical contributions, similarities, and strengths	• Metrics such as accuracy, error rate, precision, recall, and F1 score are used to measure the performance of the classification models.
Clausner et al. (2020)	Flexible character accuracy measure for reading-order-independent evaluation	• This paper introduces an edit-distance-based accuracy measure for OCR performance evaluation.	• Ground truth comparison allows comprehensive evaluation of the accuracy OCR systems.

Fraser et al. (2019)	Evaluation of diagnostic and triage accuracy and usability of a symptom checker in an emergency department: observational study	• Aims to assess how accurate and user-friendly a symptom checker tool is when used by patients who visit an emergency department (ED) for diagnosis and triage purposes.	• The accuracy of an ingredients checker can be evaluated by comparing the results with other similar checkers.
Pontes et al. (2019)	Impact of OCR Quality on Named Entity Linking	• Evaluates the performance of named entity linking over digitized documents with different levels of OCR quality.	• CER and WER are measures that can be used to evaluate the performance of an OCR engine.

To evaluate the performance of the OCR engine in their research, both Rakshit et al. (2023) and Pontes et al. (2019) use one of the most popular measures, which is Character Error Rate (CER) and Word Error Rate (WER). CER is the percentage of characters that contain errors when compared to the original text, while WER represents the proportion of words that the OCR system erroneously identifies. Clausner et al. (2020) uses ground truth comparison which aims to assess the quality and accuracy of the OCR output by assessing the precision and alignment of the identified letters and words with the ground truth. Minaee et al. (2021) applied metrics such as accuracy, error rate, precision, recall, and F1 score to measure the performance of the classification models they reviewed in the paper. Fraser et al. (2022) evaluate the symptom checker by comparing the results with other similar checkers, which also can be applied in ingredients checkers to evaluate the performance as it’s also an online checker.

2.6.1 Summary of Verification and Validation

In brief, for a thorough understanding of the accuracy of character and word recognition, CER and WER are applied to evaluate the performance of the OCR system. They are appropriate for fine-grained evaluations as they consider deletions, insertions, substitutions, and other oversights. In text classification, metrics like accuracy, error rate, precision, recall, and F1 score are used to gauge how well a model performs in classifying texts into defined categories as these metrics offer a fair evaluation of the model's capabilities in this area.

2.7 Framework

The system under the proposed project contains a machine learning model that is integrated with a number of system features, such as data visualization and a user-friendly GUI. This means that the SDLC and the Machine Learning Life Cycle should be the focus of our research.

A. Iterative Model

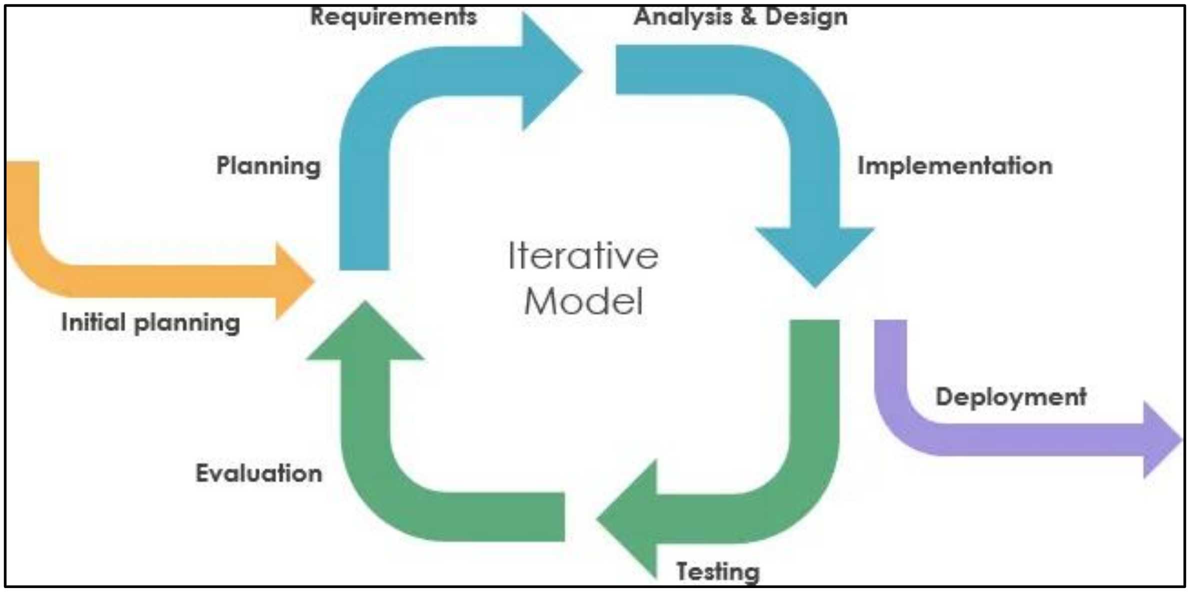

Figure 2.1: SDLC Iterative Model by Jin (2023)

The Iterative Model is one of the software development life cycle (SDLC) models, which emphasizes the repetition of development and testing activities. The software development process is split up into smaller chunks, known as iterations, under the Iterative Model. The SDLC's steps of requirements, design, implementation, testing, deployment, and evaluation are all completed during each iteration.

The paper by Aqeel et al. (2023) states that their project uses an iterative model. The iterative model is used to iteratively complete all the steps of development, and this model helps to complete the project step by step. The project consists of a web platform and a mobile app, both developed using the iterative model. To manage organizations, an admin panel was first created. Designing the user interface and connecting it with a MongoDB database and Node.js backend were the main priorities for the mobile app. The web application used the same technologies and followed a similar procedure.

Jin (2023) claims the iterative model, in contrast to the waterfall model, offers flexibility and higher success rates. It allows simultaneous development of all functions, refining them gradually from rough to fine. This method is very suitable for unclear needs and divides development into short fixed-length iterations. The requirements analysis, system design, implementation, testing, and evaluation are all covered in each iteration. Each iteration includes the study of the requirements, the design of the system, its implementation, testing, and evaluation. User input allows for the iterative refinement of requirements, enabling progressive completion of the system's functionalities or business logic.

As Jin (2023) mentioned earlier, the iterative model is the opposite of the waterfall model. Therefore, the next paper will study the waterfall model and the iterative model to understand the difference between the two. The paper by Kyeremeh (2019) discusses that the waterfall model requires that each stage be finished before moving on to the next. This model suits small projects with clear requirements. A review that follows each stage determines whether or not to continue the project based on its progress. Testing begins only after development is complete. For the iterative model, it doesn't require full initial requirements. A partial software implementation is used initially, and further requirements are found by reviewing it. New versions of the software are produced in each cycle by repeating this procedure.

These two models were further discussed in Gharajeh (2019)'s paper. The Waterfall model is a sequential software development methodology that follows the SDLC framework step by step. It assumes perfect execution of each process without considering past issues. If any problems persist from a previous step, this model may not work effectively. Teams familiar with the software being produced are a good fit because they are aware of the development environment from previous projects. Furthermore, the Iterative model operates through cycles in software development, where steps are repeated. Small-scale software development is used at first to ensure that each phase is completed in turn. The software becomes more capable than in prior versions as additional features and modules are introduced with each iteration.

Table 2.10: Summary of Literature about SDLC Iterative Model

<table>
<tr><th>Author</th><th>Title</th><th>Summary</th><th>Critical Ideas</th></tr>
<tr><td>Aqeel et al. (2023)</td><td>Web-Based Method to Connect Organizations and IT People Using Iterative Model</td><td rowspan="2">Iterative model
• Operates through repeated cycles in the software development process.
• Developed on a small scale, following all steps sequentially.
• Subsequently, additional features and modules are designed, tested, and added in subsequent iterations.
• With each iteration, the software gains more features and capabilities than the previous versions, allowing for continuous improvement and enhancement.</td><td rowspan="2">Advantages of Iterative model
• Flexibility
• Rapid prototyping
• Incremental development
• Early delivery of features
• Continuous feedback
• Quality improvement
• Continuous improvement

Advantages of Waterfall model
• Simplicity and ease of use
• Structured approach
• Clear milestones
• Well-suited for stable requirements
• Quality focus</td></tr>
<tr><td>Jin. (2023)</td><td>Software Development Framework — Iterative Model</td></tr>
</table>

<table>
<tr><td>Kyeremeh

(2019)</td><td>Overview of System

Development Life

Cycle Models</td><td rowspan="2">All functions can be developed together. Its granularity can be from rough to fine and gradually refined.
Waterfall model
Each step of the development process follows in a linear sequence.
Next step begins only after the preceding one is completed. It assumes that each phase is executed perfectly without needing to address past issues.
Consequently, if there are unresolved problems from earlier stages, this model might encounter difficulties.</td><td rowspan="2"></td></tr>
<tr><td>Gharajeh

(2019)</td><td>Waterative model: An integration of the waterfall and iterative software development paradigms.</td></tr>
</table>

Both SDLC models have their own advantages, but these advantages do not mean that they are suitable for our system development. In our opinion, SDLC iterative model is a better option compared to waterfall model. The iterative model is most suited for projects in dynamic situations where needs frequently change because of its flexibility and adaptability, which allow for changes even after work has started. Furthermore, the iterative methodology enables rapid prototype and early feedback, allowing stakeholders to visualize the product early and fine-tune requirements, thus increasing satisfaction. Certain modules can be deployed more quickly, delivering concrete results right away, and successfully fulfilling time-to-market requirements with incremental feature delivery. Continuous feedback loops and active client participation throughout the process guarantee alignment with stakeholder expectations, fostering high customer satisfaction and minimizing the possibility of later, expensive modifications. Additionally, it excels at managing complicated and sizable projects whose initial requirements are not totally apparent. The iterative model is ultimately a preferable option for projects when change is anticipated, client involvement is crucial, and high-quality results are vital due to its adaptability, response to input, and continuing improvement.

B. Business Analytics Life Cycle (BALC)

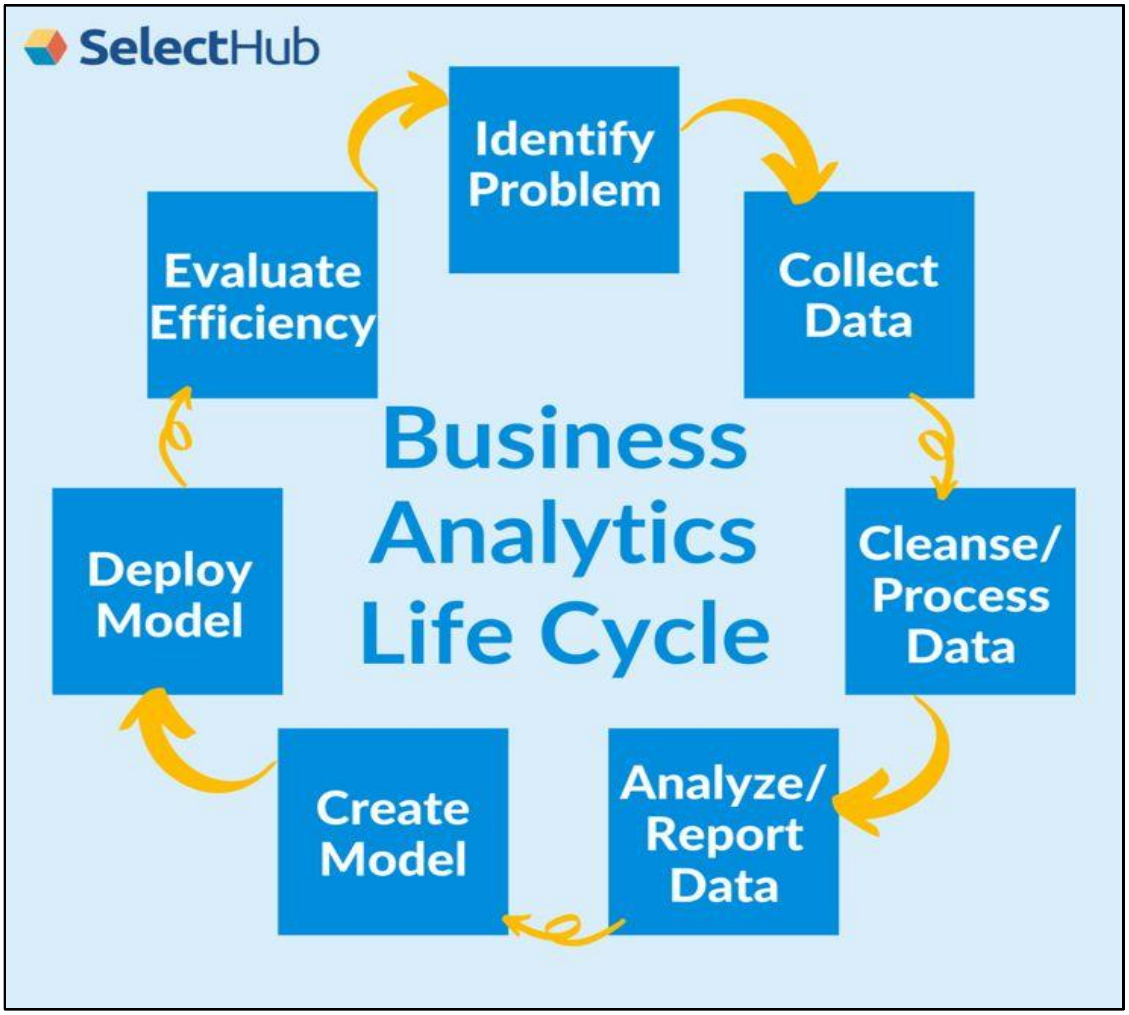

Figure 2.2: Overview of BALC by Allen, R. (2023)

Figure 2.2 provided by Allen (2023) illustrates an overview of BALC as a circular process, which includes a series of steps, including identifying problems, collecting data, cleaning/preprocessing data, analyzing/reporting data, creating model, deploying model, and evaluating efficiency. In our proposed project, we use BALC as our machine learning model framework to develop our machine learning model based on the steps in BALC.

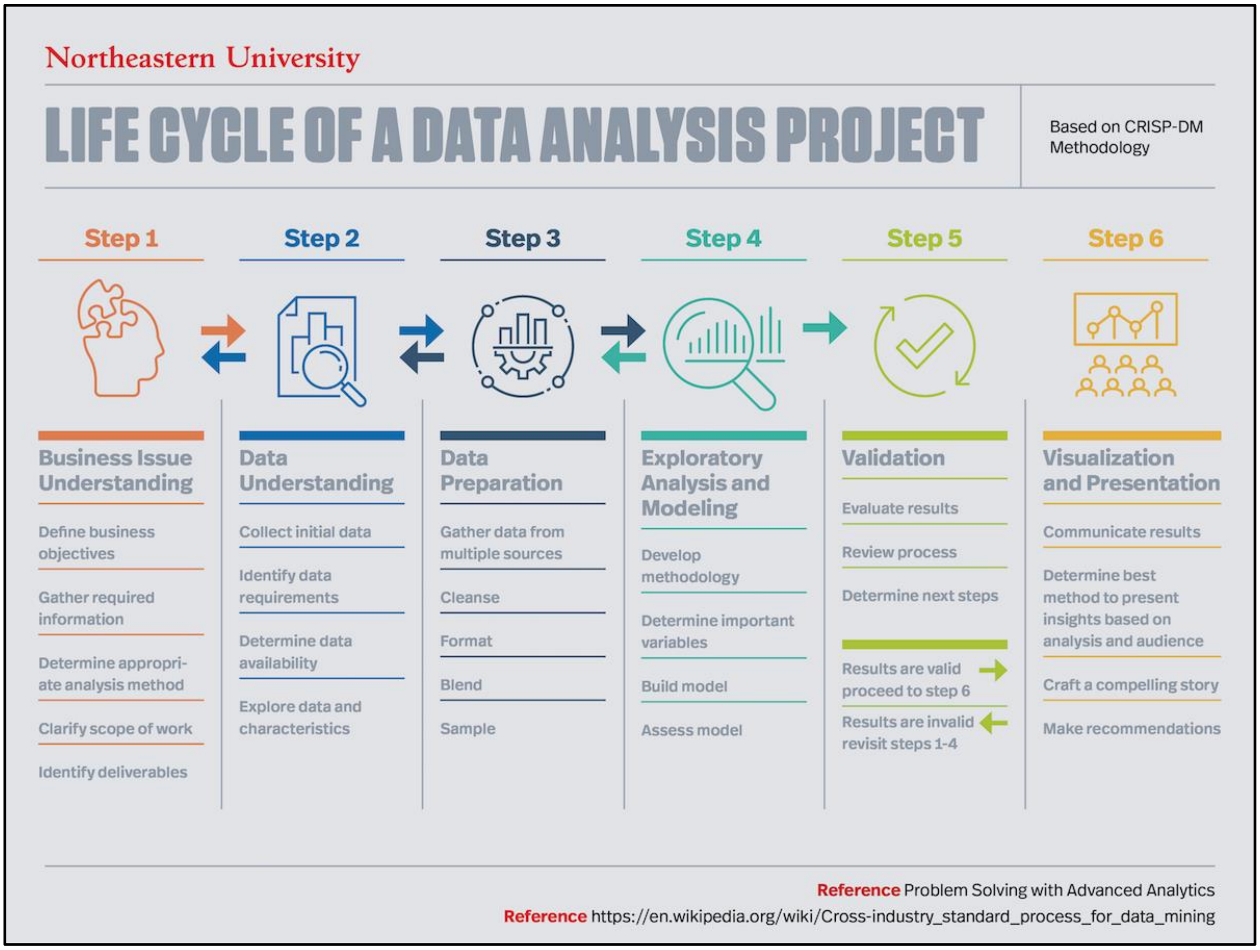

Figure 2.3: Six Steps in the Data Analysis Process by Joubert, S. (2023)

The paper by Joubert (2023) discusses in detail the steps in the data analysis process, which is similar to the BALC process. Figure 2.3 describes the six major phases that make up the CRISP-DM approach for undertaking data analytics projects. According to Paula Muñoz, a Northeastern alumna, these steps include understanding the business issue, understanding the dataset, preparing the data, exploratory analysis, validation and visualization, and presentation.

Table 2.11: Summary of Literature about BALC

<table>
<tr><th>Author</th><th>Title</th><th>Summary</th><th>Critical Ideas</th></tr>
<tr><td>Allen (2023)</td><td>Big Data and Business Analytics: Similarities, Differences, and Interactions</td><td rowspan="2">BALC
1. Identify problem.
2. Collect data.
3. Cleanse / process data.
4. Analyze / report data.
5. Create model.
6. Deploy model.
7. Evaluate efficiency.

CRISP-DM
1. Business issues understanding.
2. Data understanding.
3. Data preparation.
4. Perform exploratory analysis and modeling.
5. Model predictive analytics.
6. Deploy model.</td><td rowspan="2">Understanding domain
● Skincare ingredient information.
● Skincare ingredient rating.
● Potential technological solutions for skincare ingredient checker.

Collect Data
● Dataset from GitHub / Kaggle.
● Web scrapping.

Data Preparation
● Data cleaning.
● Develop text matching model.

Data Exploration Analysis (EDA)
● Bar chart.</td></tr>
<tr><td>Joubert (2023)</td><td>Understanding the lifecycle of a data analysis project</td></tr>
</table>

		7. Evaluate efficiency	• Frequency table. Create Model • Implement machine learning model. Deploy Model • Build a prototype. Evaluate Efficiency • Validation and verification.

After studying both articles from Allen (2023) and Joubert (2023), we found that the BALC lifestyle and the CRISP-DM have similar processes and both processes are suitable for becoming the life cycle of the machine learning model we want to develop. Therefore, our proposed BALC for our machine learning model development will be designed based on the similarity of the two.

Table 2.12: Designing Proposed BALC from Allen (2023) and Joubert (2023)

BALC	**CRISP-DM**	**Our Proposed BALC**
Identify problem	Business Issue Understanding	Understanding domain
Collect data	Data understanding	Data extraction
Cleanse / process data	Data Preparation	Data preprocessing
Analyze / report data	Exploratory Analysis and Modeling	Exploration Data Analysis (EDA)
Create model	N/A	Model development
Deploy model	N/A	Deployment
Evaluate Efficiency	Validation	Testing

2.7.1 Planning and Requirement

In this phase, the business requirements are collected, and the related documents are gathered and analyzed. The collected requirements are then planned to be used to develop the system accordingly. It begins with meticulous planning, where project objectives, scope, resources, and constraints are clearly defined. Requirements analysis involves the collection, refinement, and prioritization of project requirements.

2.7.2 Analysis and Design

The complete set of requirements for the project team to start working in a specific direction is provided during this phase. They employ various diagrams, such as a data flow diagram, class diagram, activity diagram, and state transition diagram, to gain a thorough understanding of the program design and aid in the development process. The design phase, which outlines the high-level architecture and precise design requirements, starts once it is clear what has to be accomplished.

2.7.3 Implementation and Development

Combining the SDLC iterative model with BALC (Business Analytics Life Cycle) makes building the model a more collaborative and iterative. At this stage of the project, the system's actual construction starts. The analysis and design produced during the design stage will serve as the stage's guidelines. Every necessity, strategy, and design plan has been coded. The developer will use established coding and measurement standards to execute the selected design. Coding and version control are implemented to ensure modular development for practical cooperation between components. In our proposed project, the SDLC iterative model will be combined with BALC (Business Analytics Life Cycle) at this stage to make building the model more collaborative and iterative.

2.7.4 Testing and Evaluation

The testing and Evaluation phases are pivotal and continuous. The software is thoroughly tested throughout the testing phase, including unit, integration, and system tests, to find bugs and ensure functionality complies with specifications. Since the model is iterative, this phase gains from constant validation and adjustment as new features are added. After testing, the Evaluation phase evaluates the software's performance in comparison to the original goals while taking into account stakeholder comments acquired over the course of the iteration. This feedback-driven review guides the refinement process and enables modifications to requirements, designs, and implementations. The iterative feedback loop in both phases guarantees a dynamic and responsive development process, where the software goes through rigorous testing and improves depending on real-world usage and stakeholder input, resulting in an improved and user-centric end product.

2.7.5 Summary of Framework

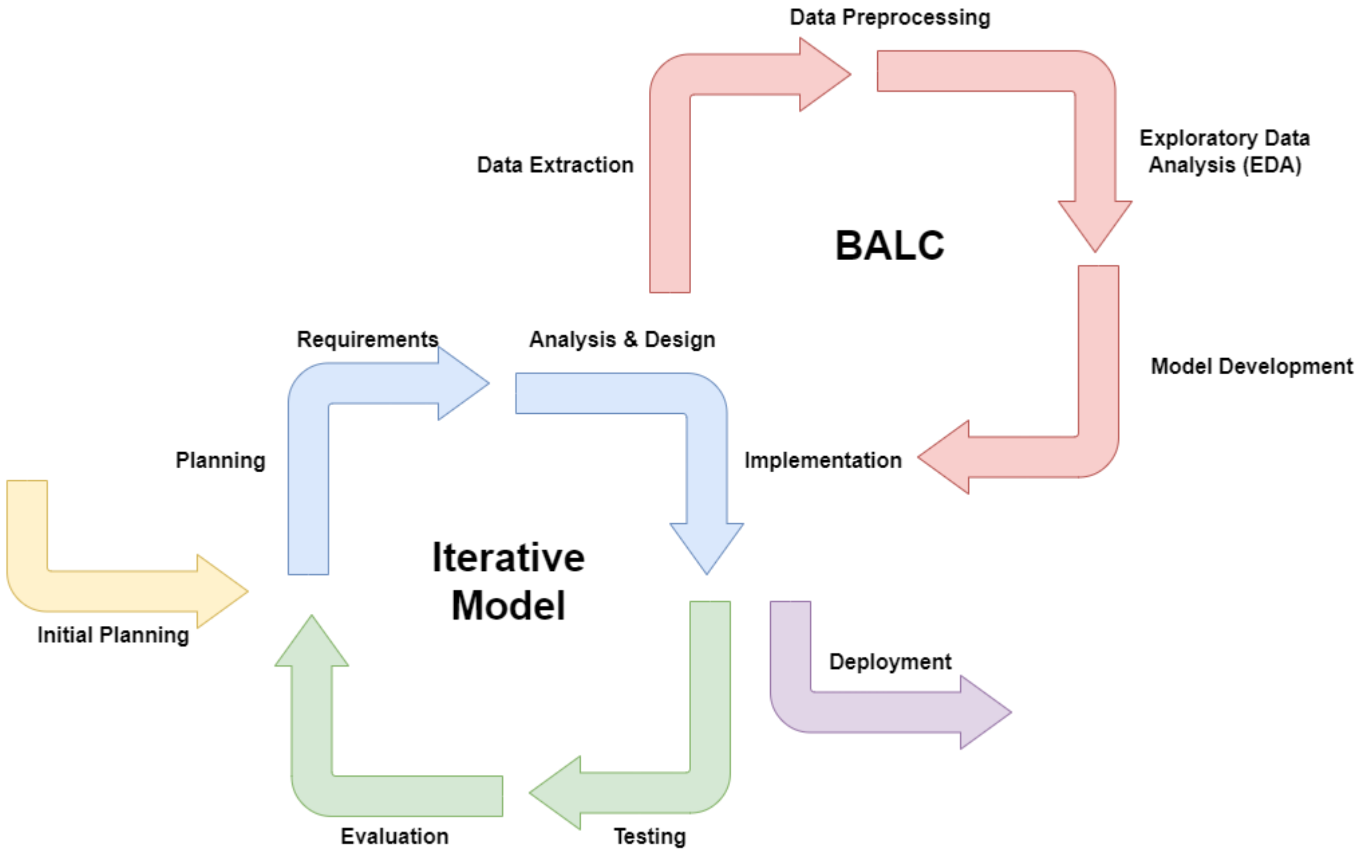

Figure 2.4: Combined Framework of SDLC Iterative Model and BALC by Jin (2023) and Allen (2023)

The combination of the Software Development Life Cycle (SDLC) Iterative Model and Business Analytics Life Cycle (BALC) creates a dynamic and adaptive approach to software development and machine learning development. The SDLC Iterative Model unfolds through iterative cycles, allowing for continuous refinement of requirements, designs, and implementations based on real-time feedback. Meanwhile, the BALC framework guides the machine learning process, emphasizing data collection, preprocessing, modeling, and performance to extract valuable business insights. This combined framework allows for concurrent and symbiotic development. While the SDLC Iterative Model facilitates the iterative and responsive development of software solutions, BALC concurrently enables the data collection, data preprocessing, and development of machine learning models. The iteration cycles in the two models are seamlessly aligned to ensure that the software development and machine learning development process is carried out iteratively and collaboratively. Integrating these models results in a holistic methodology where software development benefits from iterative refinement and closely aligns with stakeholder needs, while machine learning leverages iterative feedback loops to ensure accurate insights and correct results. This combined approach fosters a responsive and efficient development process, ensuring that software solutions and machine learning models are fine-tuned and tailored to meet evolving requirements and objectives.

2.8 Expected Contribution

This project's expected contribution is to empower consumers with transparent and personalized skin care choices through OCR technology and advanced data analysis, fostering consumer awareness.

Table 2.13: Summary of Expected Contribution

Author	Title	Summary	Critical Thoughts
De Alwis (2022)	WellnessCare: OCR-Based Web Application for Cosmetic Product Safety Assurance	● Real-time label ingredient analysis. ● Provides safety assurance (detecting potential allergens). ● Enhance transparency in skin care products	● Presents a promising tool that offers real-time ingredient analysis for cosmetic products, with a focus on detecting potential allergens
Karthikeyan et al. (2020)	An OCR Post-correction Approach using Deep Learning for Processing Medical Reports	● OCR technology facilitates online retrieval of information from printed materials. ● It enhances access to valuable data in documents, forms, and applications. ● This technology unlocks critical information previously trapped in physical documents.	● Improve the accessibility of paper documents through digitalized storage.
Awel & Abidi (2019)	Review on Optical Character Recognition	● Optical Character Recognition (OCR) converts text from printed or handwritten documents into a digital format. ● OCR deals with challenges such as varying fonts and image quality. ● OCR is significant for automating text extraction and processing from images.	● Automated text extraction (to detect skincare components)

		• OCR is essential for the development of the project "Beauty Formula DermoScan".	

Firstly, in De Alwis (2022)'s thesis, they have adopted OCR technology to conduct their research, and they have successfully developed their project with the aim that is similar to Beauty Formula DermoScan, which is to provide real-time ingredient label analysis and safety assurance for the users. Based on the approach of Karthikeyan et al. (2020), OCR can digitize printed materials, making them easier to access, retrieve, and edit. It can also automate the process of reading and storing medical documents, saving time and reducing errors. OCR can also handle unstructured data, such as scanned paper documents, and improve the accuracy of processed medical text. Besides, in Awel and Abidi (2019)'s review, OCR can make text machine-editable, automate data entry, improve accessibility, and support multiple languages. It is commercially available and research advancements continue to improve the technology. OCR is a versatile tool capable of extracting information from diverse documents, enabling real-time analysis for Beauty Formula DermoScan, and enhancing transparency in the skincare industry.

2.8.1 Impact to Industry

Beauty Formula DermoScan will be expected to revolutionize the industry. It will make ingredient information easily accessible, which compels skincare companies to prioritize transparency and safety (De Alwis, 2020). This encourages competition among manufacturers to produce safer and more effective products. Additionally, this platform raises industry standards by fostering consumer trust and accountability among manufacturers, which ultimately leads to safer and higher-quality skincare products.

Beauty Formula DermoScan will be expected to revolutionize the industry. It will make ingredient information easily accessible, which compels skincare companies to prioritize transparency and safety (De Alwis, 2020). This encourages competition among manufacturers to produce safer and more effective products. Additionally, this platform raises industry standards by fostering consumer trust and accountability among manufacturers, which ultimately leads to safer and higher-quality skincare products.

2.8.2 Impact to the framework

Beauty Formula DermoScan's impact on the framework is about making things smoother and more user-friendly. We automate ingredient data extraction with OCR technology, making it quicker and more efficient (Awel & Abidi, 2019). Users can actively participate and provide input, and as more skincare products are added, the system can quickly adapt. Plus, it makes the ingredient information consistent and easy to understand (Karthikeyan et al., 2020). It also ensures that the project follows industry rules and gives valuable insights to skincare professionals.

2.8.3 Summary of Expected Contribution

Our project utilizes OCR technology to empower consumers with transparent skincare choices. Beauty Formula DermoScan not only raises industry standards, encouraging competition for safer products, but also streamlines processes in our framework, providing user-friendly ingredient information and valuable insights for skincare professionals.

2.9 Overall Summary of Literature Review

Research suggests that Beauty Formula DermoScan's efficiency and efficacy can be increased, and customers can more easily grasp the skincare product ingredients by integrating ML models. Four essential technologies are needed for the project: Tesseract for OCR integration, VS Studio for website development, Python for data preprocessing and ML modeling, and PowerBI for visualization. For text classification, the rule-based technique is chosen, while for text matching, the TF-IDF method is selected. Flexibility is required because the development of ML models may involve trial and error. It is recommended to use a hybrid model that combines the business analytics life cycle (BALC) and the iterative approach in our framework. Chapter 3 will go into greater detail about the tools mentioned above and methods.

CHAPTER 3

METHODOLOGY

3.1 Overview

In this chapter, a combined framework integrated with the SDLC iterative model, and the business analysis life cycle (BALC) will be applied in our project. The iterative framework model focused on the web development part, while the BALC framework focused on the machine learning development part. Figure 3.1 shows the overall integration framework.

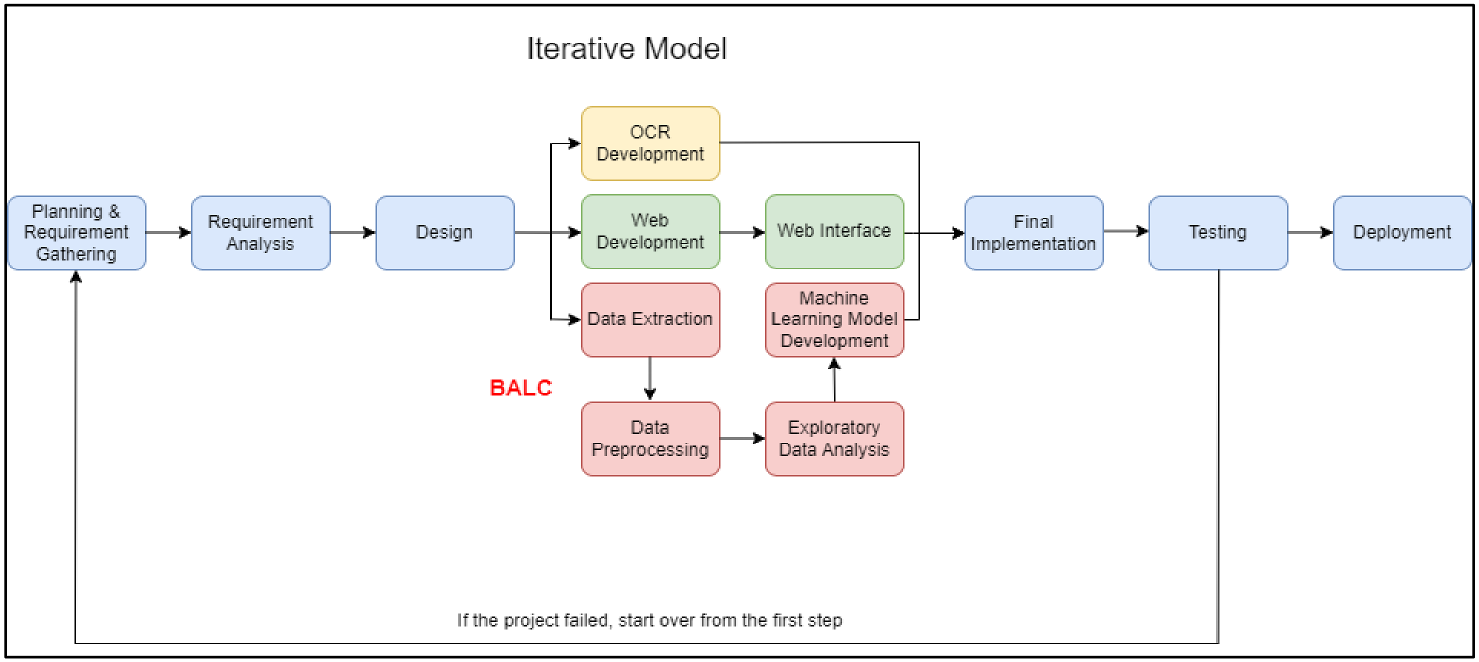

Figure 3.1: Proposed Framework- Integration of Iterative Model and Business Analytics Life Cycle (BALC) by Jin (2023) and Allen (2023)

1. In the initial stage of **planning and requirements gathering**, we will plan the scope of the project, discuss the initial requirements, study the As-Is business process, and analyze stakeholders and audiences.
2. Secondly, in the **requirement analysis phase**, we will propose our proposed business process flow from the existing process flow and discuss the expected output of our project.
3. The user interface of the website is designed during the **design stage**, including the landing page, upload prompts, upload button, analysis result diagram, and detailed description of the result page. Then, we will talk about our system overview from front-end to back-end. We will also design a data flow diagram because it provides a visual representation of how data moves in the system, including processes, data storage, and data flow, which helps to understand, communicate, and make decisions throughout the software development life cycle.
4. Next, in the **development phase**, the system is developed by coding and building user interfaces and modules, and then it is merged and tested. BALC will be involved in this phase, and will follow the following sequence: Data Extraction, Data Preparation, Exploratory Data Analysis (EDA) and Model Development. The web development in SDLC iterative model, the machine learning development in BALC and OCR development are carried out simultaneously before the final implementation.
5. During the **final implementation** process, the website, machine learning model and OCR engine will be integrated as a complete system.
6. Once the code is implemented, the **testing phase** will be carried out to guarantee the robustness, accuracy, and dependability of the system. Some tests will be conducted, including integration testing of OCR systems and machine learning models, as well as user acceptance testing based on the design process. As a result, the project will proceed to the deployment phase if the testing process is successful. If the project doesn't succeed, we’ll need to go back to step one and pinpoint the issue until it is fixed before deploying.
7. After all stages are completed, the system will be **deployed**, and a prototype will be built.

3.2 Planning & Requirement Gathering

Planning and requirement gathering is the first and foremost step of developing Beauty Formula DermoScan. In the planning and requirement-gathering phase of this research paper, a comprehensive analysis of the project scope was conducted, which highlighted the development of the skincare ingredient analysis platform. This platform, designed for web interface, leverages OCR technology for real-time label scanning of skincare products. The project's scope was enriched by dimension variables such as safety rating, potential hazards, potential side effects, skin type, and skin concerns, ensuring that the platform offers holistic skincare ingredient analysis. Additionally, the stakeholder and audience analysis shed light on the diverse groups involved, including skin care companies, beauty consultants/dermatologists, general consumers, and skin care enthusiasts, each playing a crucial role in shaping the project's functionality and design. Understanding the unique needs and expectations of these stakeholders and audiences is essential for delivering a user-focused and comprehensive skincare ingredient analysis platform.

3.2.1 Project scope

Beauty Formula DermoScan aims to create a user-centric skincare ingredient analysis platform that leverages OCR technology, rule-based models, and TF-IDF analysis to empower consumers with accurate ingredient information and safety assessments. This platform will enable users to scan skincare product labels in real-time, assess ingredient safety, and understand skincare ingredients better. The project's focus on usability and continuous data quality ensures a reliable and user-friendly experience, fostering informed skincare choices and enhancing consumer well-being in the dynamic skincare industry.

3.2.2 As-Is Business Process

"As-Is Business Process" refers to the current, existing state of a business process without any modifications or improvements. It captures all the phases, activities, inputs, outputs, interactions, and other aspects of how a certain process works at a particular time. An important first step in business process improvement approaches is to analyze the existing business process.

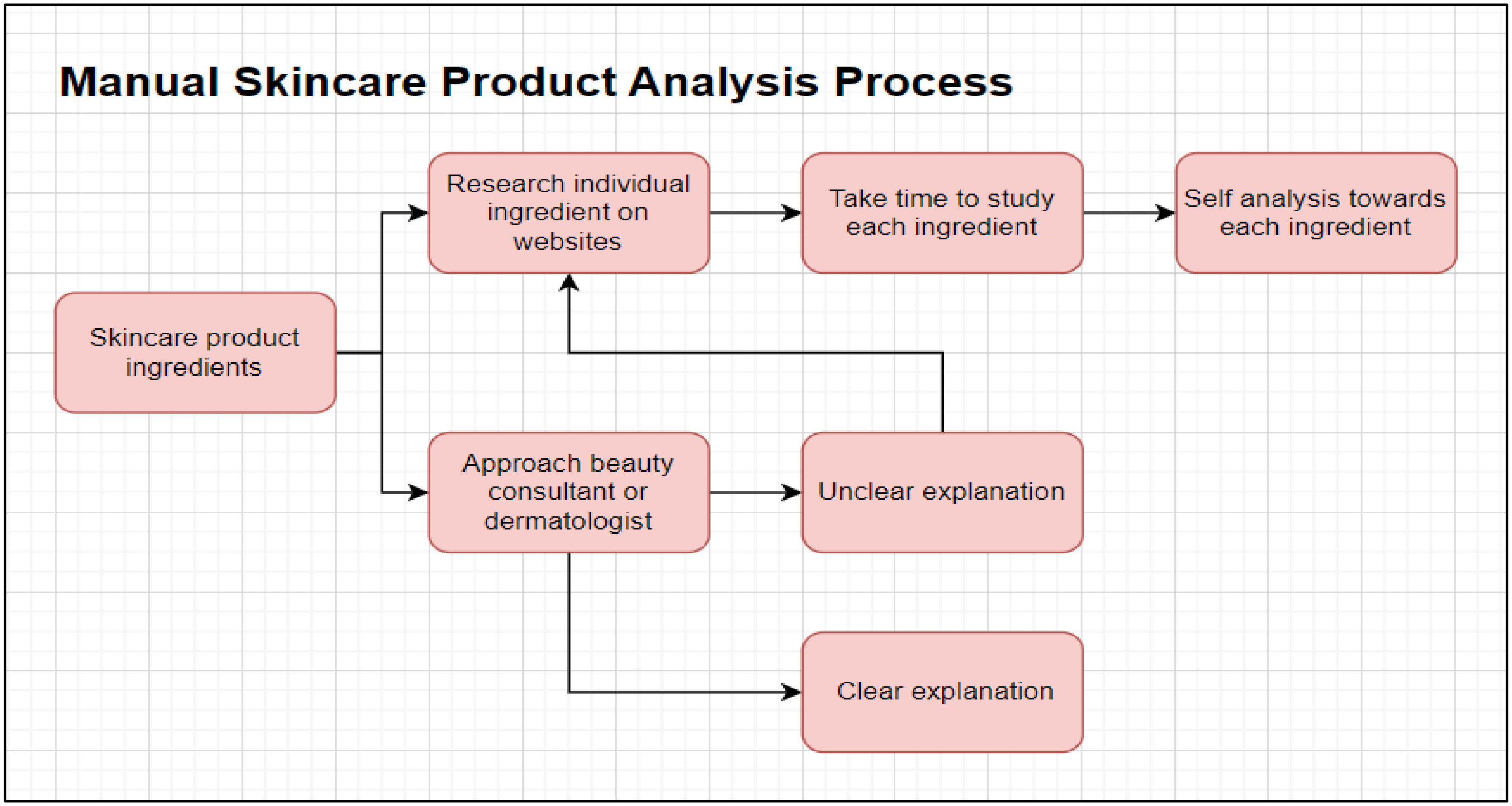

Figure 3.2: As-Is Business Process Flow

The manual skin care product analysis process is the current business process and consists of more cumbersome steps. There are two scenarios involved in the process. When consumers want to learn more about all the ingredients in skin care products, they either look for each ingredient individually on Google or approach a beauty consultant or dermatologist about the ingredient knowledge.

In the first scenario, consumers must take their time to read the ingredient information because there is a lot of relevant information on Google, and consumers must study and filter them by themselves to find the most accurate information. Even more problematically, it is just one of the ingredients. Usually, a typical product will contain 15-50 ingredients *(Mischa, 2019)*, which means that consumers have to spend a lot of time searching for and studying relevant information about ingredients. Then, based on their understanding of the ingredients, they can summarize their analysis results of the product and the product's ingredients.

In the second scenario, if the beauty consultant or dermatologist has enough knowledge to explain all the ingredients of the skin care product well, this problem is solved. However, suppose the beauty consultant is unable to adequately describe all the ingredients. In that case, it will move to the first scenario, where consumers have to search and understand each ingredient online by themselves. Analyzing skin care products with this method is more manual and conventional. As a result, the entire process is time-consuming and energy intensive.

3.2.3 Stakeholders Analysis

Stakeholder and audience analysis is essential for our skincare product analysis project. Potential stakeholders who will benefit from our system include (1) skincare companies, (2) beauty consultants/dermatologists, (3) general consumers, and (4) skincare enthusiasts. Understanding each group's unique needs and expectations is crucial to developing a platform that provides transparency, safety, and valuable insights into the skincare industry.

A. Skin Care Companies

Skincare companies are pivotal stakeholders as they produce and sell skincare products. Their reputation and compliance with industry standards are directly impacted by the platform's assessments. Skincare companies expect the platform to encourage transparency and compliance with safety standards. They may be incentivized to ensure accurate ingredient labelling and transparency to gain consumer trust and maintain a competitive edge.

B. Beauty Consultants and Dermatologists

Beauty consultants and dermatologists are professionals who provide skincare advice to clients and patients. They seek reliable sources of information to guide their recommendations to clients and patients, ensuring they receive products that align with their specific skin types and concerns.

C. Skin Care Enthusiasts

Skincare enthusiasts are passionate about skincare and constantly seeking more information about ingredients and product safety. They value the platform's ingredient analysis because it helps them improve their skincare routines and make better product choices.

D. General Consumers

People who need to learn much about skincare can also benefit from the platform. It provides easy-to-understand explanations of skincare ingredients and safety assessments so that people can make informed choices when shopping for skincare products.

3.2.4 Dimension Variables

Beauty Formula DermoScan will use OCR technology to scan skincare product labels in real time and provide users with information on safety ratings, potential hazards, potential side effects, benefits, and ingredient descriptions.

A. Safety Rating

Implement a safety rating system that assesses the safety level of skincare products based on ingredient analysis. Users can quickly identify products with higher safety ratings, enhancing their confidence in product selection.

B. Potential Hazards and Side Effects

Integrate features to identify and communicate potential hazards and side effects associated with skincare product ingredients. Users receive clear warnings and explanations about possible adverse effects, allowing them to make informed decisions.

C. Benefits

Aimed at providing users with comprehensive information about the positive effects that specific skincare product ingredients can offer. This information will educate users about how an ingredient can enhance their skin's health and appearance.

D. Descriptions

Designed to offer detailed explanations of each ingredient's properties and functions. This ensures that users have a clear and easily understandable source of information.

3.3 Requirement Analysis

The requirement analysis will be divided into 2 sections, namely To-Be Business Process and Expected Output.

3.3.1 To Be Business Process

The proposed Beauty Formula DermoScan Process shown in Figure 3.3 is the "To Be Business Process" of skin care product analysis. "To-Be Business Process" refers to the future or desired state of a business process after improvements and optimizations have been implemented. It reflects the ideal workflow that a business wants to accomplish and incorporates adjustments to cope with inefficiencies and boost production.

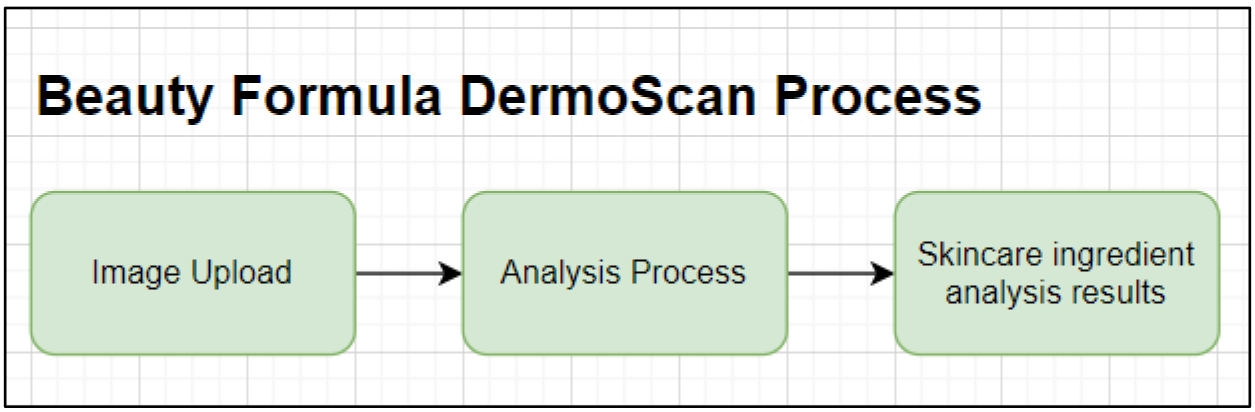

Figure 3.3: To-Be Business Process Flow

In the process of skin care product analysis, the proposed business process flow of the Beauty Formula DermoScan is divided into three stages: (1) image upload, (2) analysis process, and (3) skincare product ingredient analysis results. The OCR technology and machine learning models mentioned in Section 2.5 will be implemented in this process to improve the system's efficiency, consistency, and accuracy. The proposed idea is to establish a more advanced skin care product analysis system where consumers can upload images of product ingredients. The system will extract textual data from the uploaded image and analyze ingredient results based on the pre-defined rules in our database. The detailed analysis results of the skin care product ingredients will be summarized for consumers' reference. Therefore, in terms of time and effort, the proposed process is more practical, convenient, and efficient than the "As-Is Business Process" in Section 3.2.2. A further explanation of how business processes work will be discussed in Section 3.4.2 Data Flow Design.

3.3.2 Expected Output

First of all, the expected performance of the proposed project is to have an integrated Tesseract OCR into the proposed Beauty Formula DermoScan system, which provides more efficient, accurate, and reliable data extraction capabilities, contributing to improved operational efficiency. By integrating Tesseract OCR, the proposed system is expected to process large volumes of documents quickly and extract data in real-time and from various document types, including printed text, handwritten forms, and scanned documents. It is also expected to process unstructured data, such as text within images or scanned documents, and transform it into structured, machine-readable data.

Next, the proposed system must be able to effectively process the text data extracted by OCR and match the ingredients extracted by OCR with its database in real-time. This seamless integration is vital to ensure the data's accuracy and swiftly cross-reference the ingredients with existing records, enabling the system to provide precise and up-to-date skin care ingredient analysis results in a timely manner, thereby enhancing user experience and reliability.

Another expected performance is that the website interface must be user-friendly and easy to operate. Ensuring an intuitive and straightforward user experience is essential for the system's success. Users should be able to navigate the interface effortlessly, access necessary features without confusion, and complete tasks with minimal effort. The interface should be responsive, provide smooth interaction without delay, and ensure that users can easily upload photos or images and quickly access detailed results. In addition, the system should contain useful tooltips and guidance prompts to enhance user understanding and reduce the possibility of errors.

3.4 Design

In the development of our project, two essential design components will be unveiled: the user interface mockup and the data flow diagram (DFD) design for our Beauty Formula Dermo Scan system. These design elements are pivotal in shaping the user experience and the flow of information within our innovative platform. The user interface mockup encapsulates the visual representation of our website, offering a sneak peek into the user-friendly and aesthetically pleasing digital environment we are creating. In tandem with this, our data flow design maps out the intricate pathways through which data and information will be processed and presented within the system.

3.4.1 System Overview

In Figure 3.4, the system overview is divided into two parts: front-end and back-end. The front-end, serving as the user interface, engages users with intuitive designs and interactive elements, ensuring a positive user experience. Meanwhile, the back-end acts as the system's engine, managing data, processing requests, and executing complex algorithms behind the scenes. These components collaborate seamlessly through APIs, forming a dynamic flow of data and commands. When users interact with the front end, their requests are processed by the backend, which retrieves, manipulates, and stores data while ensuring the system's integrity and security.

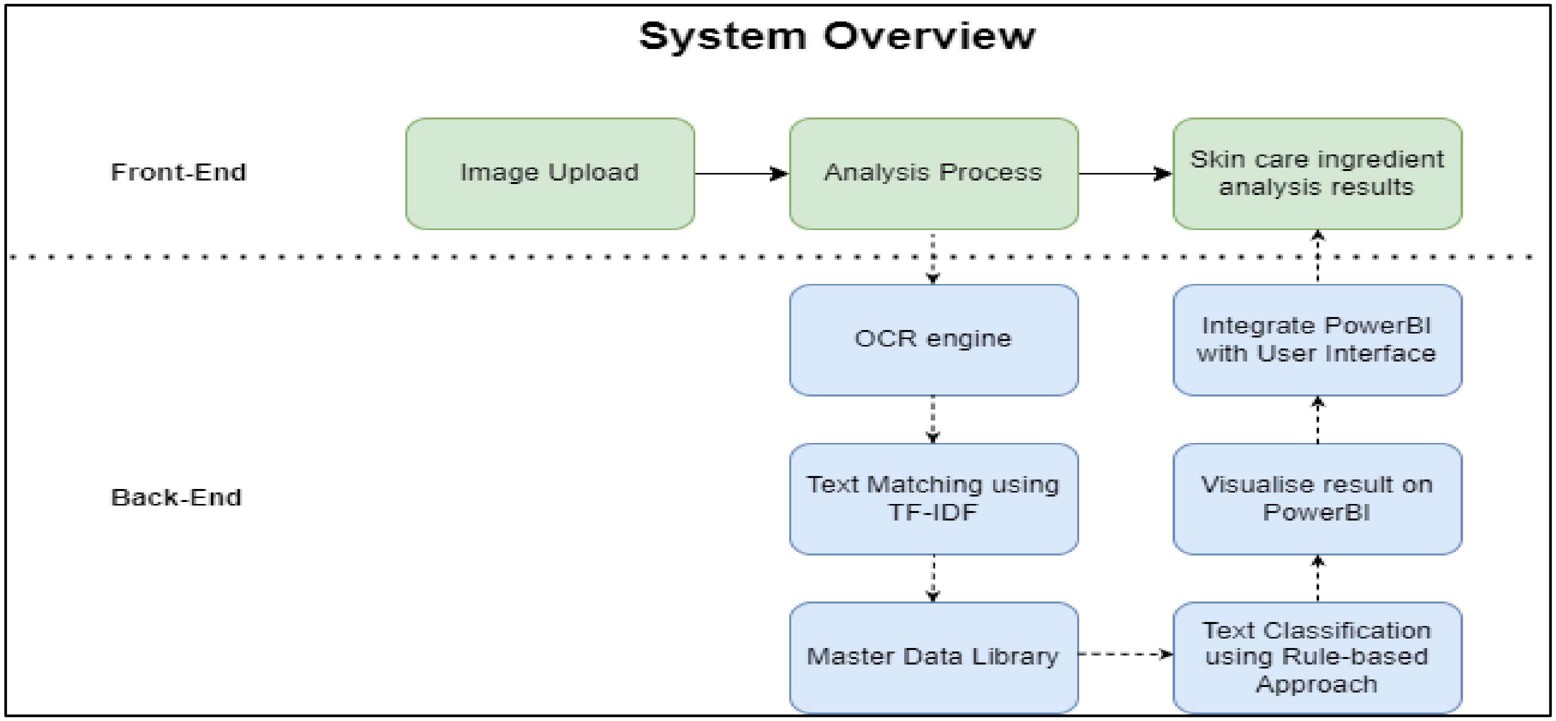

Figure 3.4: System Overview Process

In the system overview of our project, when the user uploads an image of the skincare product ingredient to the system, our system will seamlessly carry out the analysis process. This is where the back-end process takes place. On the back end, the system will run the analysis process in the following order: (1) OCR Engine; (2) Text Matching using TF-IDF; (3) Master Data Library; (4) Text Classification using Rule-based Approach; (5) Visualize result on PowerBI; (6) Integrate PowerBI with User Interface. Once the analysis process is completed, the user will receive the skincare ingredient analysis results in the user interface. Each system overview process will be further explained in Section 3.4.2 Data Flow Design.

3.4.2 Data Flow Design

The data flow design comprehensively illustrates our system's analysis process development, explaining the back-end operations involved in generating our final results. In the envisioned to-be system workflow, three straightforward stages are outlined: photo upload to the system, the analysis process, and the presentation of skincare ingredient analysis outcomes, as depicted in Figure 3.5. Our detailed explanation will focus on the second stage - the analysis process, which prominently incorporates machine learning models, data visualization, and a graphical user interface integration.

To-Be Business Process Flow Overview:

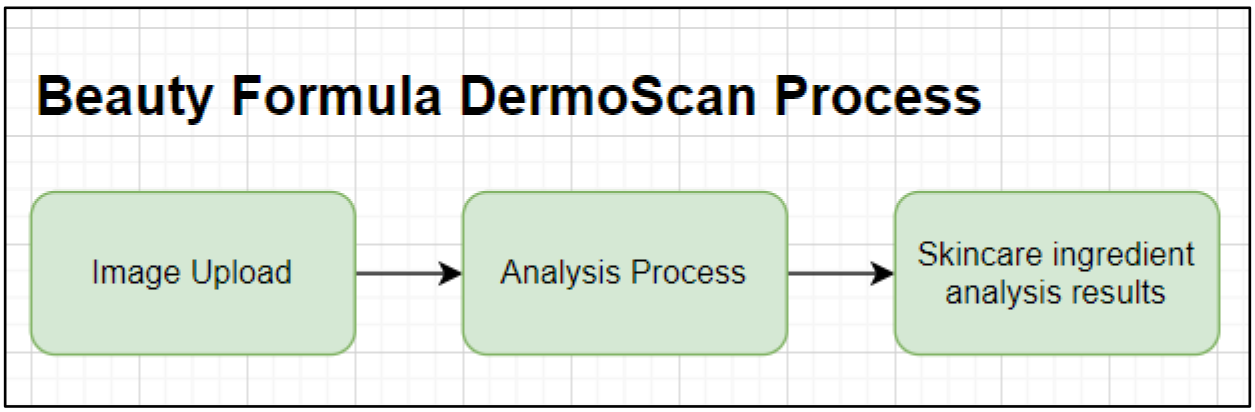

Figure 3.5: To-Be Business Process

Data Flow Diagram for Analysis Process:

The depicted Data Flow Diagram (DFD) provides a comprehensive representation of the Beauty Formula Dermo Scan's data architecture. It serves as a visual roadmap, offering an insightful glimpse into how data flows systematically within our system while playing a pivotal role in achieving the core objectives of our system, as it effectively guides the flow of information, ensuring efficiency and accuracy in data processing.

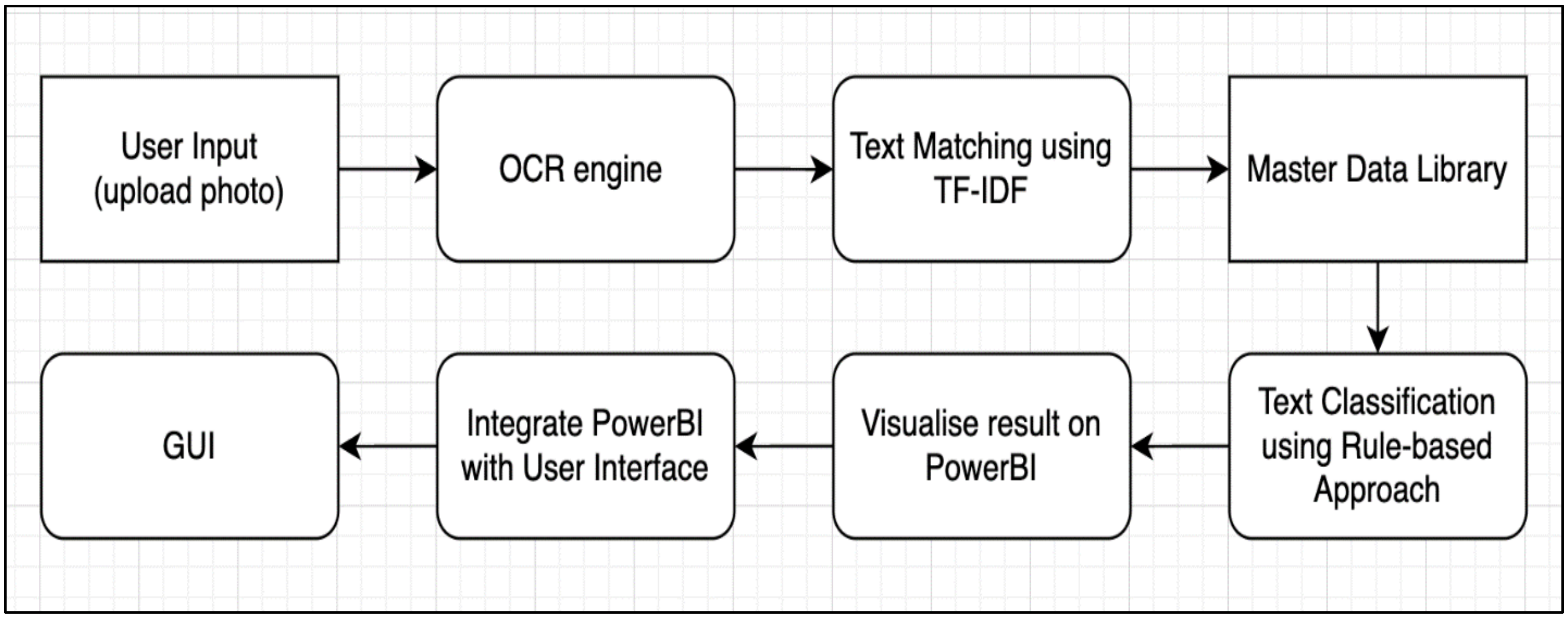

Figure 3.6: DFD

Table 3.1: DFD Flow Explanation

DFD Flow	Explanation
1. User input (upload photo):	Users upload photos of skincare product ingredient labels via the website's graphical user interface (GUI).
2. OCR engine:	The uploaded photos are processed by the OCR (Optical Character Recognition) engine to perform text extraction.
3. Text Matching using TF-IDF:	Extracted textual data will undergo text matching using the TF-IDF (Term Frequency-Inverse Document Frequency) algorithm to identify and process individual ingredient.
4. Master data file:	The data will go through the master data file which contains comprehensive information about skincare product ingredients.
5. Text classification using Rule-Based approach:	A rule-based approach is employed to match the recognized ingredients with the corresponding data in the master data file.
6. Visualise result on PowerBi:	The generated rating results of individual ingredient will be fed into PowerBI to create piechart which provide informative results at a glance.
7. Integrate PowerBI with User Interface:	Visualizations of result will be integrated into system's graphical user interface, where users can access these results seamlessly from the same interface where they uploaded the photos.
8. GUI	Users can view and interact with the results, including ingredient details, product ratings, and other relevant information.

3.4.3 User Interface Design

This part will display the detailed flow of system navigation and GUI for comprehension.

A. Detail Flow Diagram

The detailed flow diagram offers a general overview of the front-end system flow on our website. It illustrates a straightforward process as follows:

Table 3.2: System Flow Summary

	System Flow
1.	When a user lands on the webpage, they have the option to click on the "Tips to Upload" button for guidance, if necessary.
2.	Users can choose an image by clicking on the "Image Upload" button.
3.	After selecting the image, they can proceed by clicking the "Confirm and Upload" button.
4.	The system will then display the analysis results.

Prior to the introduction of our user interface design, we will first introduce an overview diagram of system detail flow diagram to showcase our website navigation.

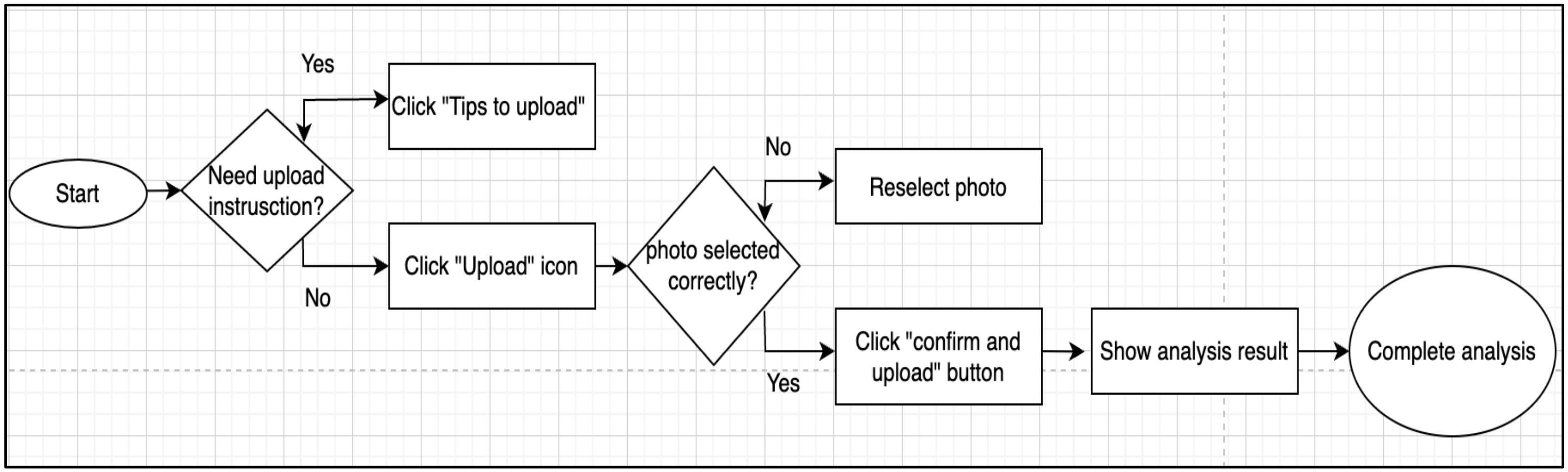

Figure 3.7: Detail Flow Diagram of System Flow

B. User Interface Design Mockup

The following will serve as a mockup user interface generated using Canva which depicts the flow of our system in generating ingredient information.

Figure 3.8: Landing page

On the system's landing page, the website features our system's name, "Beauty Formula Dermo Scan," along with the system slogan, "Analyze any skincare products in seconds!" In the centre of the page, users will find an upload button enabling them to upload photos of their skincare product ingredients. Also, in the lower-left corner, there will be a user-friendly URL link button that guides users to valuable tips and tricks for successfully uploading their photos.

Figure 3.9: Tips to upload

When users click on the "Tips to Upload" button, a pop-up window will appear, providing them with both textual descriptions and a visual example picture. This resource is designed to assist users in understanding and effectively utilizing the upload feature.

Figure 3.10: Upload photo via computer

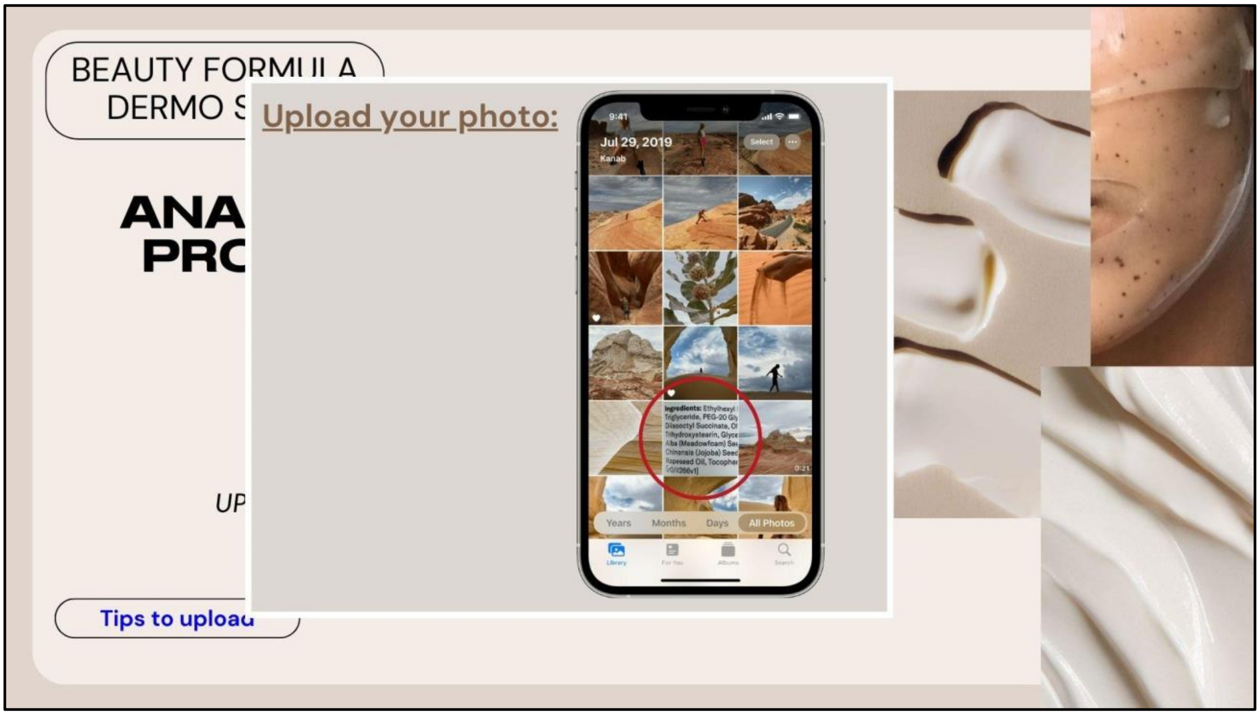

Figure 3.11: Upload photo via mobile device

When users click on the "Upload" button, they can either upload a photo from their laptop or select one from their mobile phone's photo gallery. This flexibility ensures a seamless and user-friendly experience for individuals using various devices to interact with our system.

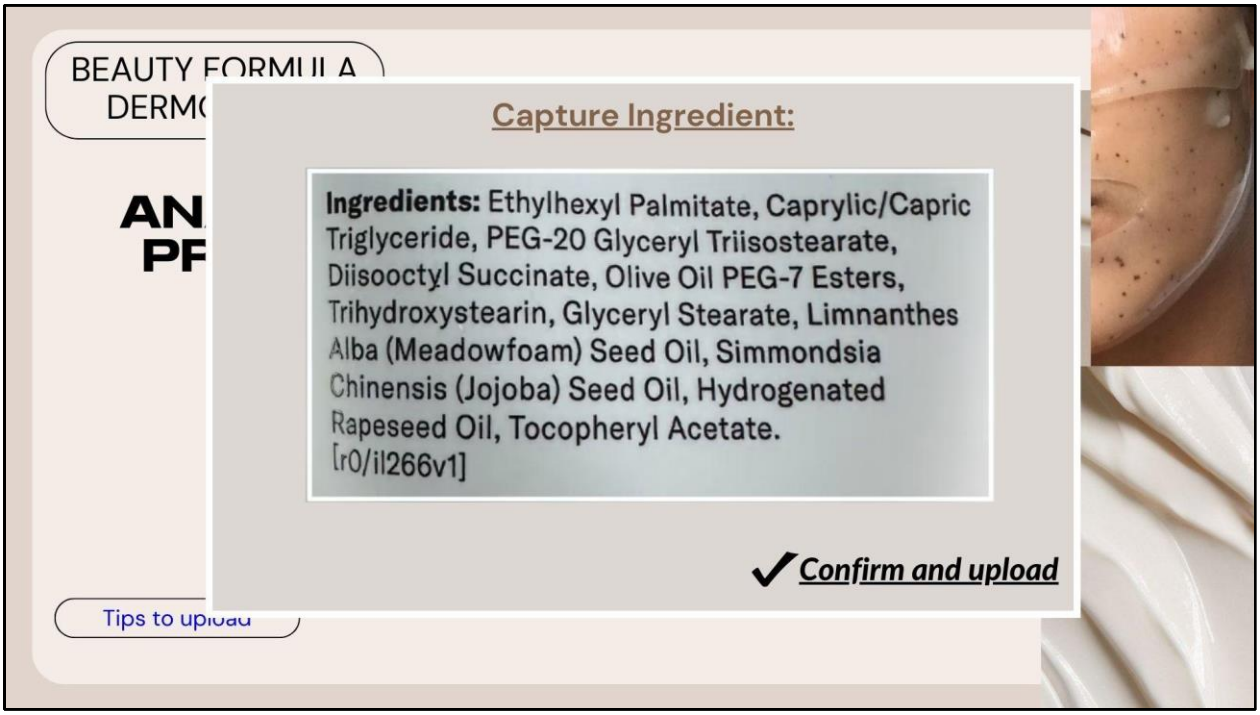

Figure 3.12: Confirm and upload picture

Users will have the option to cross-check the selected picture within the pop-up window. If they confirm that the image is chosen correctly, they can proceed by clicking the "Confirm and Upload" button in the window's lower right-hand corner. This step ensures that users have complete control and confidence in their uploaded images.

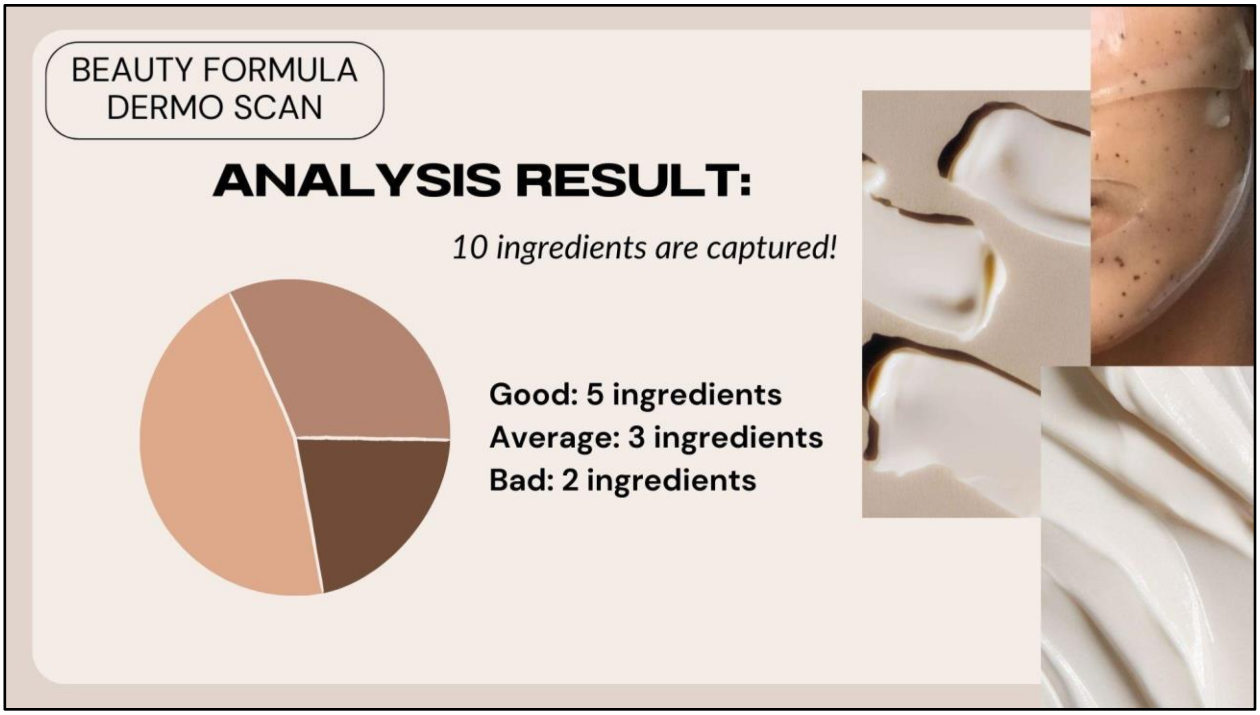

Figure 3.13: Pie chart analysis result

Following the analysis of the ingredient list, the results will be initially presented through a pie chart. This visual representation offers users an immediate overview of the product's rating, which will be categorized into three ratings based on the analysis results. This graphical presentation enhances the user experience by providing a quick and easily digestible snapshot of the product's overall composition and rating.

Figure 3.14: Detailed description of result

Ultimately, the detailed results of each ingredient will be listed and displayed on our user interface. Users can click on each ingredient's name to access more in-depth information.

3.5 Machine Learning Development

This section describes the machine learning development part of our system. Based on our project's system overview discussed in section 3.1, the machine learning development will take place on (1) Data Extraction, (2) Data Preprocessing, (3) Exploratory Data Analysis, and (4) Machine Learning Model Development.

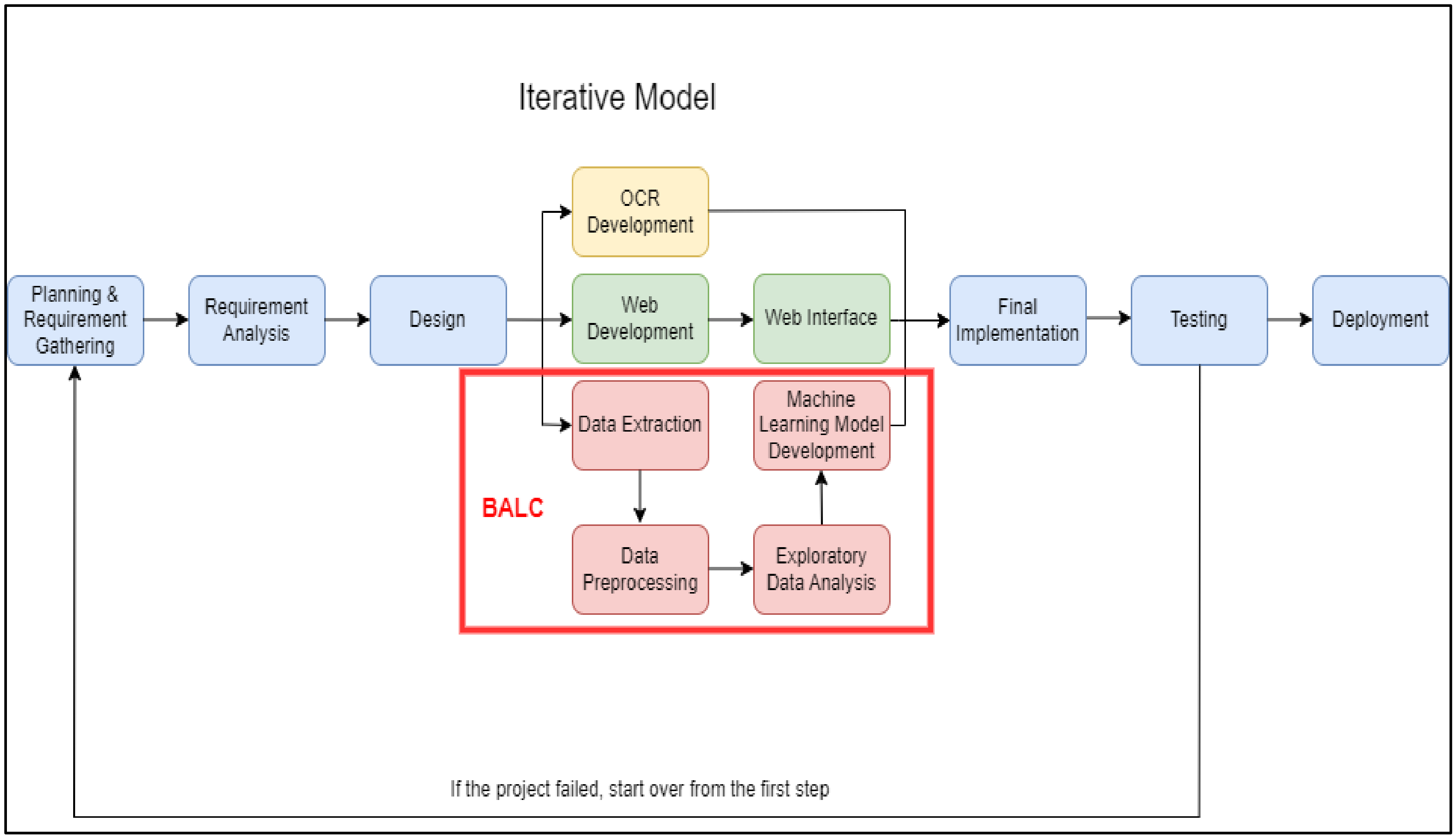

Figure 3.15: Machine Learning Development

Regarding the tool we employ, we have chosen to utilize Google Colaboratory for various stages of our machine learning development, including data preprocessing, exploratory data analysis (EDA), and model development. Google Colaboratory, commonly known as Colab, serves as an indispensable asset in our project's model development. It is a prominent cloud-based resource that is pivotal in fostering machine learning education and advancing research endeavors. This platform seamlessly leverages the capabilities of Jupyter Notebooks, providing a collaborative and easily accessible environment that greatly expedites machine-learning tasks for researchers (Carneiro et al., 2018).

3.5.1 Data Extraction

The first step in data extraction is to identify and gain access to the sources where the required data is available. For our project, we decided to use a dataset we found on GitHub. After identifying the data source, the next step is to retrieve the data from the source. The dataset containing structured data in CSV format was downloaded from GitHub and stored in a local file.

3.5.2 Data Preprocessing

A. Data Cleaning

The first step for data preprocessing in our project is data cleaning. Since data often contains missing values and inconsistencies, data cleaning is necessary to identify and address these issues. Figure 3.16 shows the general outline of codes to be used during the data-cleaning process, while Table 3.3 provides explanations of the significant pseudocode used for data-cleaning.

```
# Import necessary libraries
import pandas as pd
import numpy as np

# Load the dataset
data = pd.read_csv('ingredients.csv')

# Check data types and missing values
data.info('ingredients.csv')

# Handling missing values
data.dropna(inplace=True)

# Removing duplicates
df.drop_duplicates(inplace=True)

# Save cleaned data
data.to_csv('ingredient_cleaned.csv', index=False)
```

Figure 3.16: Pseudocode for Data Cleaning

Table 3.3: Description of Significant Pseudocode used for Data Cleaning

Significant Pseudocode	**Description**
# Import necessary libraries	● Import the required Python libraries. ● ‘Pandas’ for data manipulation and analysis. ● ‘NumPy’ for numerical operations.
# Check data types and missing values	● Provide information about the dataset, including data types and the count of non-null values for each column.
# Handling missing values	● Addresses the issue of missing values in the dataset by removing rows that contain missing values.
# Removing duplicates	● Identifies rows that have the same values in all columns and keeps only one instance of each unique row.

B. TF-IDF for Text Matching

The second step for data preprocessing involves the application of the TF-IDF technique for text matching. The general outline of codes used to perform text matching is shown in Figure 3.17, while Table 3.4 provides explanations of the significant pseudocode used for text matching.

```
# Import necessary libraries
import pandas as pd
from sklearn.feature_extraction.text import TfidfVectorizer
from sklearn.metrics.pairwise import cosine_similarity

# Load your database of ingredient names
database = pd.read_csv('ingredient_cleaned.csv')

# Load the list of ingredient names to match
ingredient_list = ["ingredient1", "ingredient2", "ingredient3"]

# TF-IDF Vectorization
tfidf_vectorizer = TfidfVectorizer()
tfidf_matrix_database = tfidf_vectorizer.fit_transform(database['ingredient name'])
tfidf_matrix_list = tfidf_vectorizer.transform(ingredient_list)

# Text Matching
similarity_scores = cosine_similarity(tfidf_matrix_list, tfidf_matrix_database)

# Retrieve the top matching ingredient for each input ingredient
matching_results = {}

for i, ingredient_name in enumerate(ingredient_list):
    # Find the index of the ingredient in the database with the highest similarity score
    best_match_index = similarity_scores[i].argmax()
    best_match = database.at[best_match_index, 'ingredient name']

    # Store the best matching ingredient in the results dictionary
    matching_results[ingredient_name] = best_match

# Print or use matching_results as needed
print(matching_results)
```

Figure 3.17: Pseudocode for Text Matching

Table 3.4: Description of Significant Pseudocode used for Text Matching

Significant Pseudocode	Description
# Import necessary libraries	• Import required libraries, including Pandas for data manipulation and Scikit-learn for TF-IDF vectorization and cosine similarity calculations.

# TF-IDF Vectorization	• Initialize a TF-IDF vectorizer to transform text data into TF-IDF representations. Then, calculate TF-IDF matrices for both the database and the input list of ingredient names.
# Text Matching	• Calculate cosine similarity scores between the TF-IDF vectors of the input ingredient list and the database.
# Retrieve the top matching ingredient for each input ingredient	• For each input ingredient, find the index of the ingredient in the database with the highest similarity score using argmax().

3.5.3 Exploratory Data Analysis (EDA)

After data cleaning, the cleaned dataset is ready for Exploratory Data Analysis (EDA). Our EDA focuses on creating a bar chart to compare the number of ingredients for each rating category and a frequency distribution table to conduct ingredient frequency analysis. Figure 3.18 shows the general outline of codes used to perform EDA on our cleaned dataset, while Table 3.5 provides explanations of the significant pseudocode used for text matching.

```
# Import necessary libraries
import seaborn as sns
import matplotlib.pyplot as plt

# Count the number of ingredients for each rating category
rating_counts = df['rating'].value_counts()

# Create a bar chart to compare the number of ingredients for each rating category
plt.figure(figsize=(8, 6))
sns.barplot(x=rating_counts.index, y=rating_counts.values, palette='viridis')
plt.xlabel('Rating Category')
plt.ylabel('Number of Ingredients')
plt.title('Number of Ingredients by Rating Category')
plt.xticks(rotation=45)  # Rotate x-axis labels for readability
plt.show()

# Create a frequency distribution table showing how many times each ingredient appears in the dataset
ingredient_freq = df['ingredient'].value_counts().reset_index()
ingredient_freq.columns = ['Ingredient', 'Frequency']
ingredient_freq.head(10)  # Display the top 10 most frequent ingredients
```

Figure 3.18: Pseudocode for EDA

Table 3.5: Description of Significant Pseudocode used for Text Matching

Significant Pseudocode	Description
# Import necessary libraries	• Import necessary libraries for data visualization, including Seaborn (sns) and Matplotlib (plt), which are commonly used for creating charts and graphs.
# Count the number of ingredients for each rating category	• Calculate the frequency of each unique rating category in the 'rating' column of the dataset.
# Create a bar chart to compare the number of ingredients for each rating category	• Create a bar chart using Seaborn's bar plot function. The x-axis displays rating categories, and the y-axis represents the number of ingredients.
# Create a frequency distribution table showing how many times each ingredient appears in the dataset	• Calculate the frequency of each unique ingredient in the 'ingredient' column of the dataset and display the top 10 most frequent ingredients.

3.5.4 Model Development

Text classification will be applied to categorize the entries in our dataset. Leveraging predefined criteria, machine learning algorithms will classify each result as a benefit or a risk, depending on its associated rating. Text classification involves the systematic organization of text data into distinct categories or classes determined by the content and context of the text. For the machine learning used for classification, a rule-based approach will be applied after examination, which will be further discussed below.

A. Classification Using Rule-Based Approach:

For our model development, our project will employ a rule-based approach for text classification, ensuring that each rating is accompanied by a rationale. In essence, the threshold of the ratings will align with either benefits or risks, bolstering the insights garnered from the classification process. Figure 3.19 shows the general outline of codes used to perform rule-based text classification, while Table 3.6 provides explanations of the significant pseudocode used to perform rule-based text classification.

```python
import pandas as pd

# Set Pandas display options to show more rows
pd.set_option('display.max_rows', None)  # This will display all rows

# Load the dataset
file_path = 'benefitsandrisks.csv'  # Replace with the actual path to your CSV file
data = pd.read_csv(file_path)

# Define a function to classify ingredients and create the "reason" column
def classify_ingredients(row):
    if row['rating'] in ['Good', 'Best']:
        return row['benefits_eg']
    elif row['rating'] in ['Average', 'Poor']:
        return row['risks_eg']
    else:
        return ''

# Apply the classification function to create the "reason" column
data['reason'] = data.apply(classify_ingredients, axis=1)

# Save the updated dataset to a new CSV file
data.to_csv('classified_ingredients.csv', index=False)

# Display the updated dataset
data
```

Figure 3.19: Google Colab script for text classification

Table 3.6: Description of Significant Pseudocode for Text Classification

Significant Pseudocode	Description
# Set Pandas display options to show more rows	• Adjust the display options of the Pandas library to show more rows of data
# Load the dataset	• Indicate where a file path should be specified.
# Apply the classification function to create the "reason" column	• Where the actual classification of ingredients takes place based on the "rating" column.
# Save the updated dataset to a new CSV file	• Creates a new dataset with the additional column.

# Display the updated dataset	the dataset with the modification for users to review

B. Dataset Overview

The modified dataset result using python has been presented in the figure below.

name	discription	category	rating	rating_num	benefits_eg	risks_eg	reason
1, 2-Hexanediol	A synthetic preservative and moisture-binding ...	['Preservatives']	Good	2	Hydrates the skin	Skin irritation.	Hydrates the skin
10-Hydroxydecanoic Acid	A synthetic ingredient that functions as a ski...	['Emollients']	Good	2	Reduces fine lines	Allergic reactions.	Reduces fine lines
4-T-butylcyclohexanol	A synthetic fatty alcohol that functions as an...	['Emollients', 'Skin-Soothing']	Good	2	Improves skin elasticity	Acne breakouts.	Improves skin elasticity
Acacia farnesiana extract	A fragrant extract from a type of acacia tree....	['Plant Extracts', 'Fragrance: Synthetic and F...	Poor	0	Minimizes pores	Redness or flushing.	Redness or flushing.
acacia senegal gum	Herb that can have skin soothing properties, b...	['Texture Enhancer', 'Plant Extracts', 'Skin-S...	Good	2	Enhances skin texture	Dryness or peeling.	Enhances skin texture
acai	Acai (pronounced "ah-sigh-ee") is a small berr...	['Antioxidants', 'Plant Extracts']	Best	3	Brightens complexion	Excessive oiliness.	Brightens complexion
acerola fruit extract	Acerola contains vitamin C. However, the dry a...	['Plant Extracts']	Good	2	Promotes collagen production	Sensitivity to sunlight.	Promotes collagen production
acetic acid	Acid found in vinegar, some fruits, and human ...	['Sensitizing']	Poor	0	Fades dark spots	Rash or hives.	Rash or hives.
acetone	Strong solvent that is used in nail polish rem...	['Sensitizing']	Poor	0	Calms skin irritation	Itchy skin.	Itchy skin.
acetylated castor oil	Used as an emollient and thickening agent in c...	['Texture Enhancer', 'Emollients']	Good	2	Prevents breakouts	Burning or stinging sensation.	Prevents breakouts

Figure 3.20: Result based on Google Colab script

C. Remove Additional Columns

The provided diagram represents a sample output generated using the script. As illustrated in the diagram, the script matches ingredient ratings with either their corresponding benefits or risks and neatly organizes this information under the "reason" column. To enhance clarity and reduce potential confusion, the script will also remove the "benefits" and "risks" columns from the dataset. This streamlined presentation aims to provide users with a more straightforward and comprehensible view of the ingredient classifications and their associated reasons.

```
# Remove 'benefits_eg' and 'risks_eg' columns
data.drop(columns=['benefits_eg', 'risks_eg'], inplace=True)

# Save the updated dataset to a new CSV file
data.to_csv('classified_ingredients_without_eg.csv', index=False)

# Display the updated dataset
data
```

	name	discription	category	rating	rating_num	reason
0	1, 2-Hexanediol	A synthetic preservative and moisture-binding ...	['Preservatives']	Good	2	Hydrates the skin
1	10-Hydroxydecanoic Acid	A synthetic ingredient that functions as a ski...	['Emollients']	Good	2	Reduces fine lines
2	4-T-butylcyclohexanol	A synthetic fatty alcohol that functions as an...	['Emollients', 'Skin-Soothing']	Good	2	Improves skin elasticity
3	Acacia farnesiana extract	A fragrant extract from a type of acacia tree....	['Plant Extracts', 'Fragrance: Synthetic and F...	Poor	0	Redness or flushing.
4	acacia senegal gum	Herb that can have skin soothing properties, b...	['Texture Enhancer', 'Plant Extracts', 'Skin-S...	Good	2	Enhances skin texture
5	acai	Acai (pronounced “ah-sigh-ee”) is a small berr...	['Antioxidants', 'Plant Extracts']	Best	3	Brightens complexion

Figure 3.21: Remove benefits and risks column using script and its result

Upon executing the script on Google Colab, which entails the removal of both the "benefits" and "risks" columns, the outcome of the classification process using a rule-based approach is shown in Figure 3.21. Table 3.7 provides a brief explanation for the main pseudocode used to remove the columns and their descriptions.

Table 3.7: Description of Significant Pseudocode for Removing Additional Columns

Significant Pseudocode	**Description**
# Remove 'benefits_eg' and 'risks_eg' columns	• Delete additional columns to avoid confusion
# Save the updated dataset to a new CSV file	• Download the modified CSV file
# Display the updated dataset	• Display modified dataset on the console to review on the final output

3.6 Web Development

The web development process plays a pivotal role in our system development as the environment will provide the end user with a graphical user interface to visualize our entire system. To provide an overview of our website, the GUI will allow users to upload images and review their detailed analysis results.

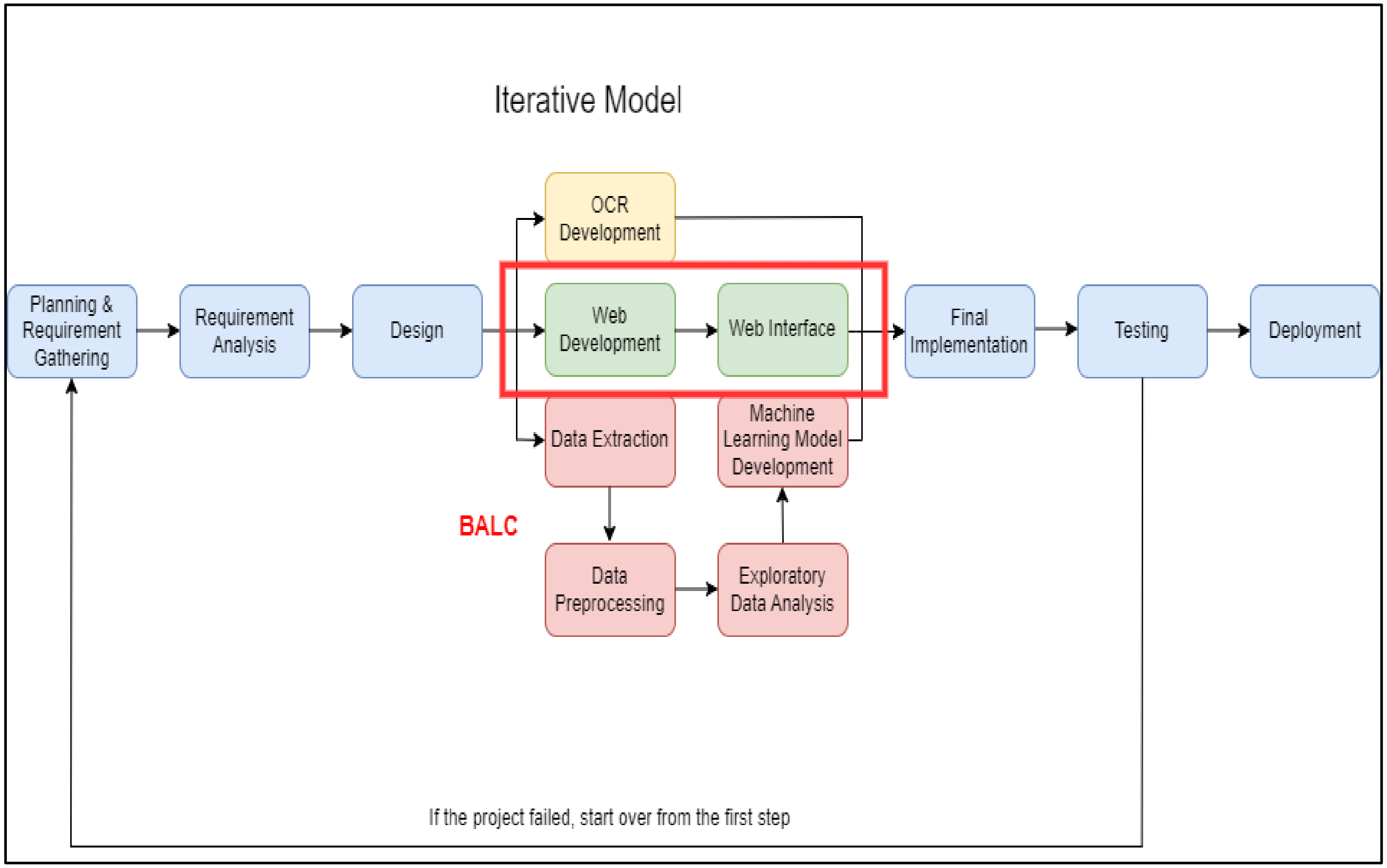

Figure 3.22: Web Development

A. Web Development Process Overview

In the web development section, we will comprehensively explain the three web development processes. This discussion will encompass the creation of three vital components:

1. The landing page, where end users can submit skincare ingredient images.
2. A general overview of the integration process.
3. The crafting of the result page.

These steps collectively constitute a critical aspect of our web development, ensuring a thorough understanding of how our graphical user interface (GUI) will be developed.

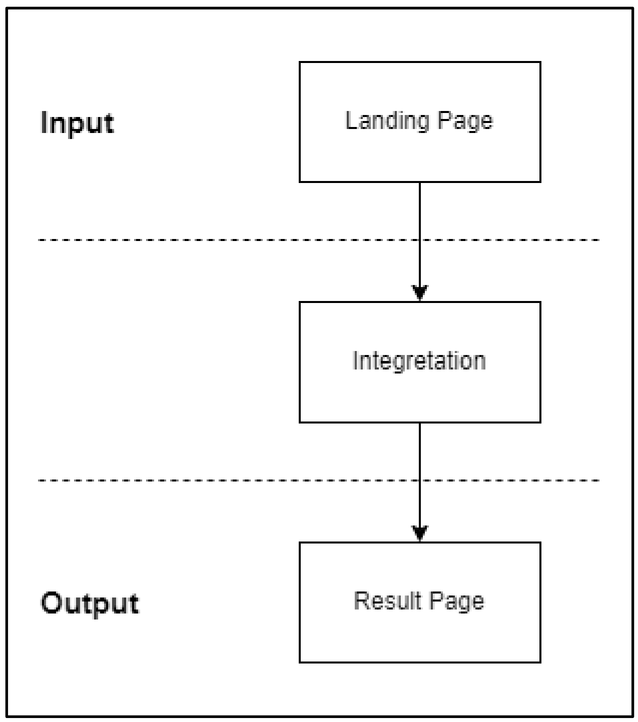

Figure 3.23: Web Development Process Overview

As mentioned in Section 2.3.3, we will use Visual Studio Code as the main tool for web development; HTML (HyperText Markup Language) and CSS (Cascading Style Sheets) will be created in Visual Studio Code to create dynamic and visually appealing web applications. Visual Studio Code offers a robust development environment where developers can create, edit, and integrate HTML and CSS into web applications. HTML is the standard language to develop and design websites and web applications. It uses a system of markup tags to describe the structure of web pages. CSS is a stylesheet language used for expressing the look and formatting of a document written in HTML. It controls a web page's layout, colours, fonts, and other visual aspects.

3.6.1 Landing Page

The landing page of a website is the main or introductory page visitors see when they navigate to the site's root domain. On our website, the main purpose of the landing page is to allow users to upload images of skincare product ingredients directly and conveniently. According to Figure 3.8, the landing page design of our webpage will include our logo, title, upload button, and upload prompt. We will code each feature, layout, color, and image of the landing page accordingly. The following is the reference code for the landing page:

A. HTML Reference Code

I. Head

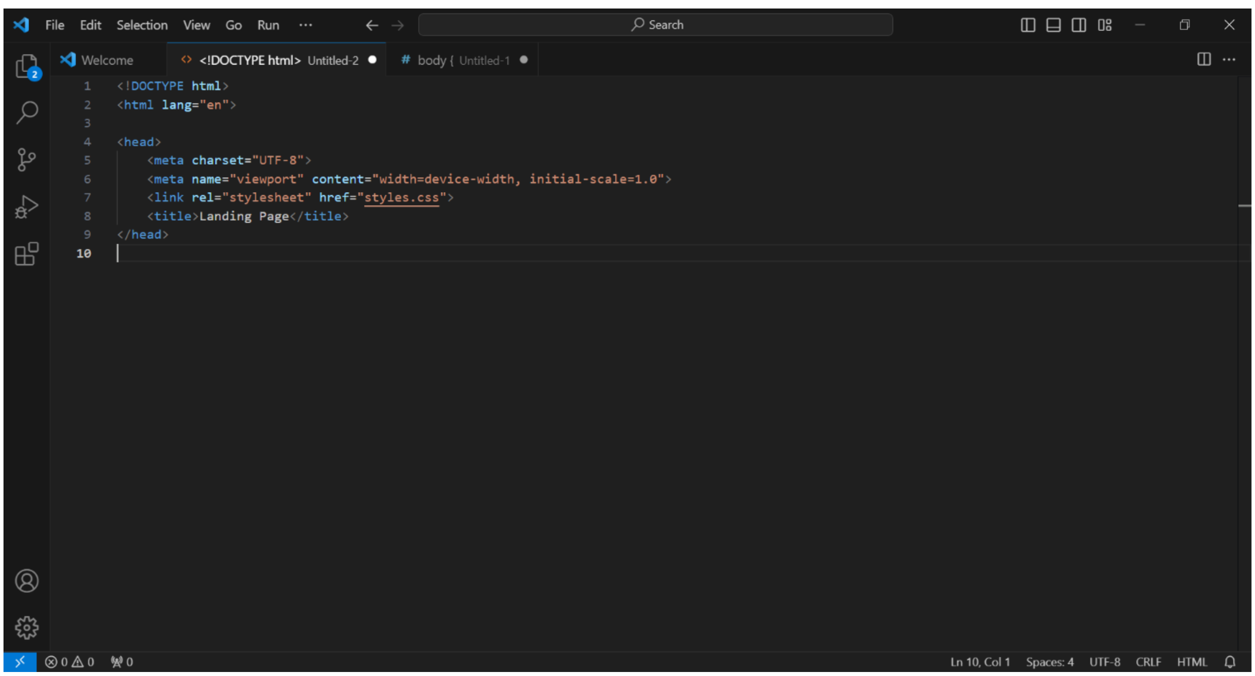

Figure 3.24: HTML For Head of Landing Page

II. Body

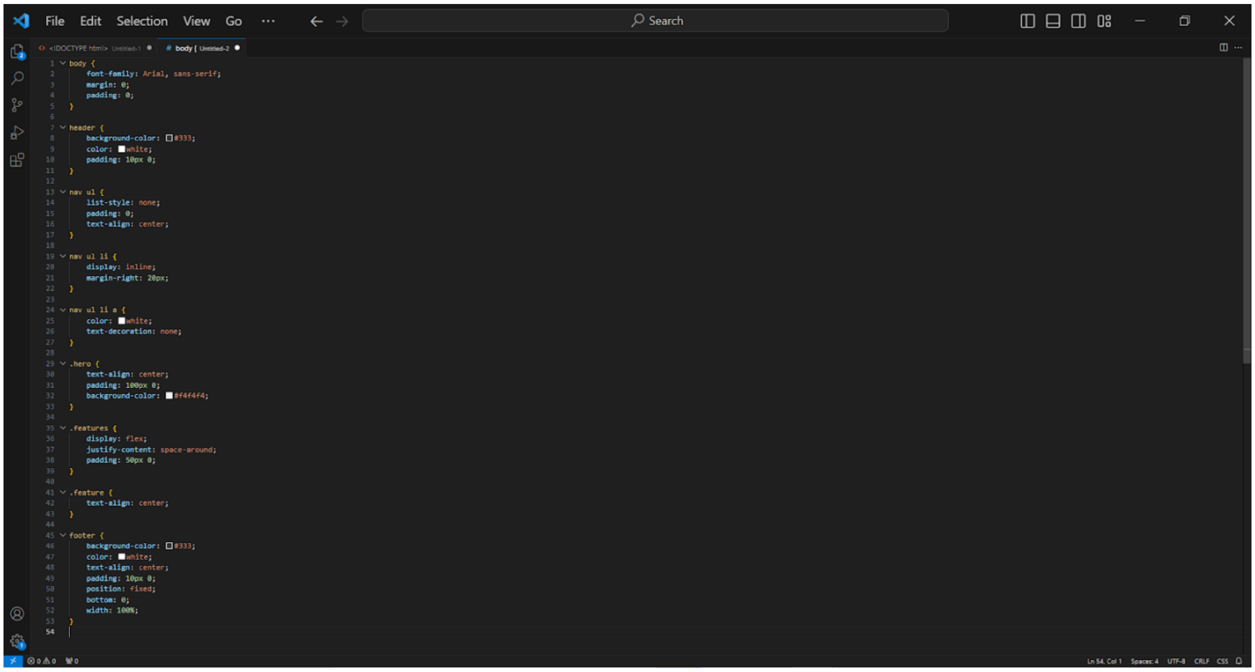

Figure 3.25: HTML For Body of Landing Page

B. CSS Reference Code

I. Head

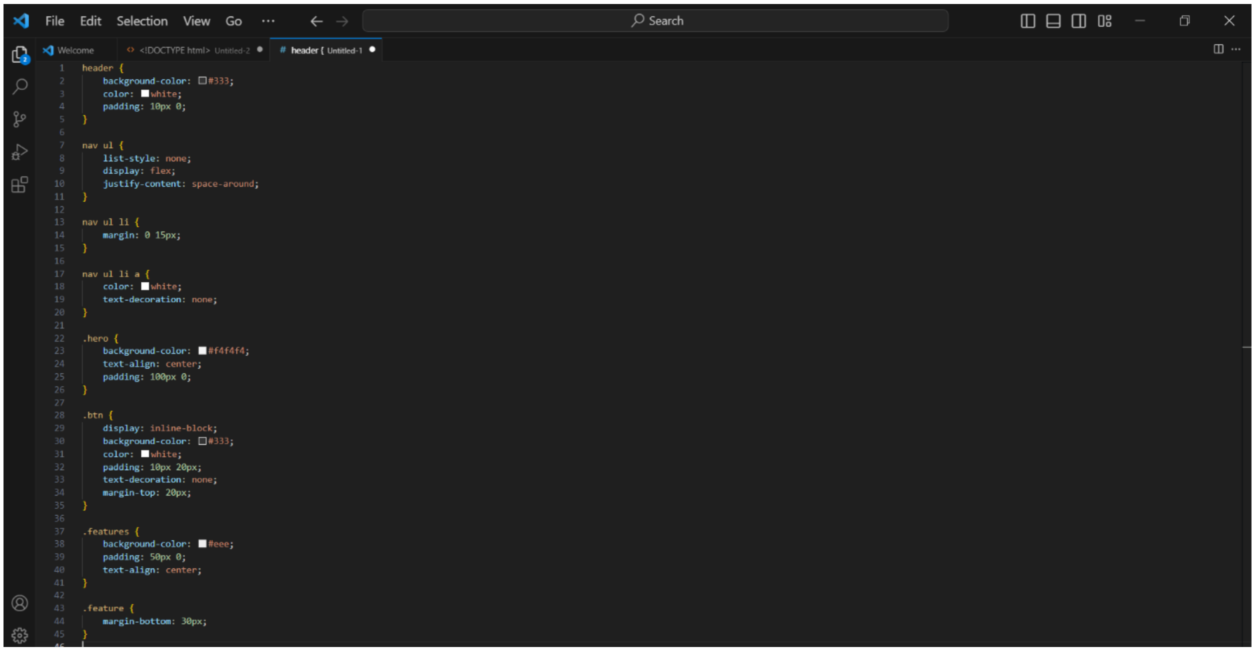

Figure 3.26: CSS For Head of Landing Page

II. Body

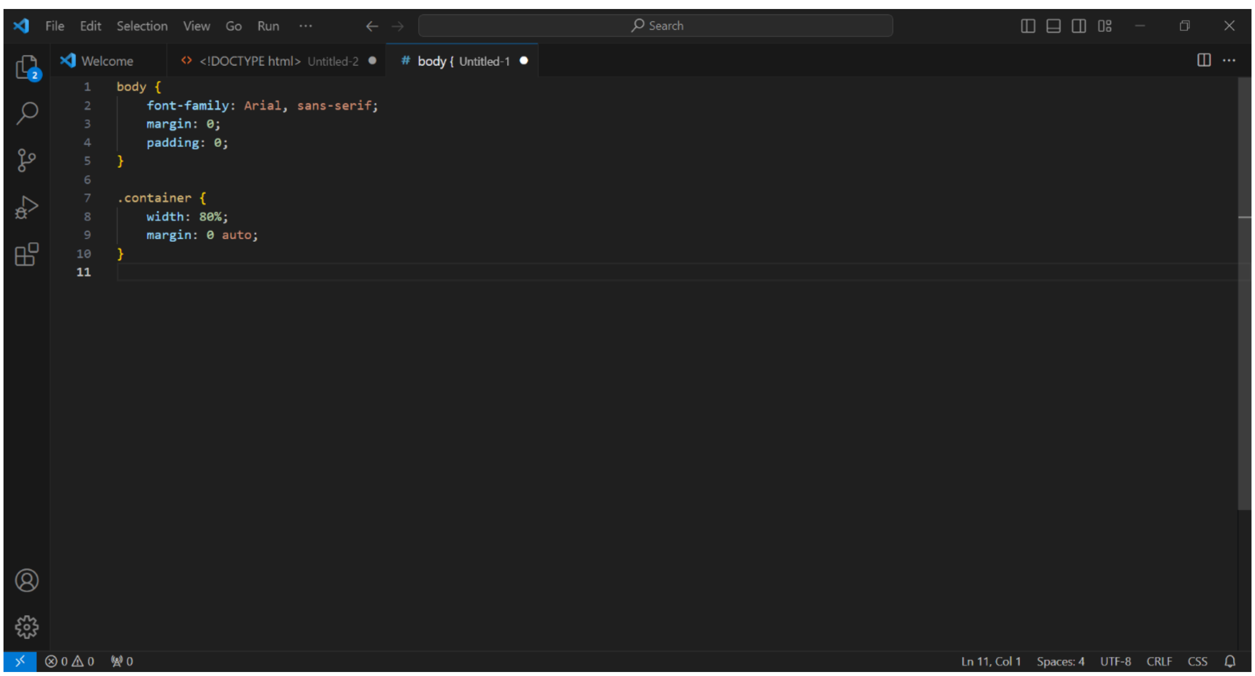

Figure 3.27: CSS For Body of Landing Page

III. Footer

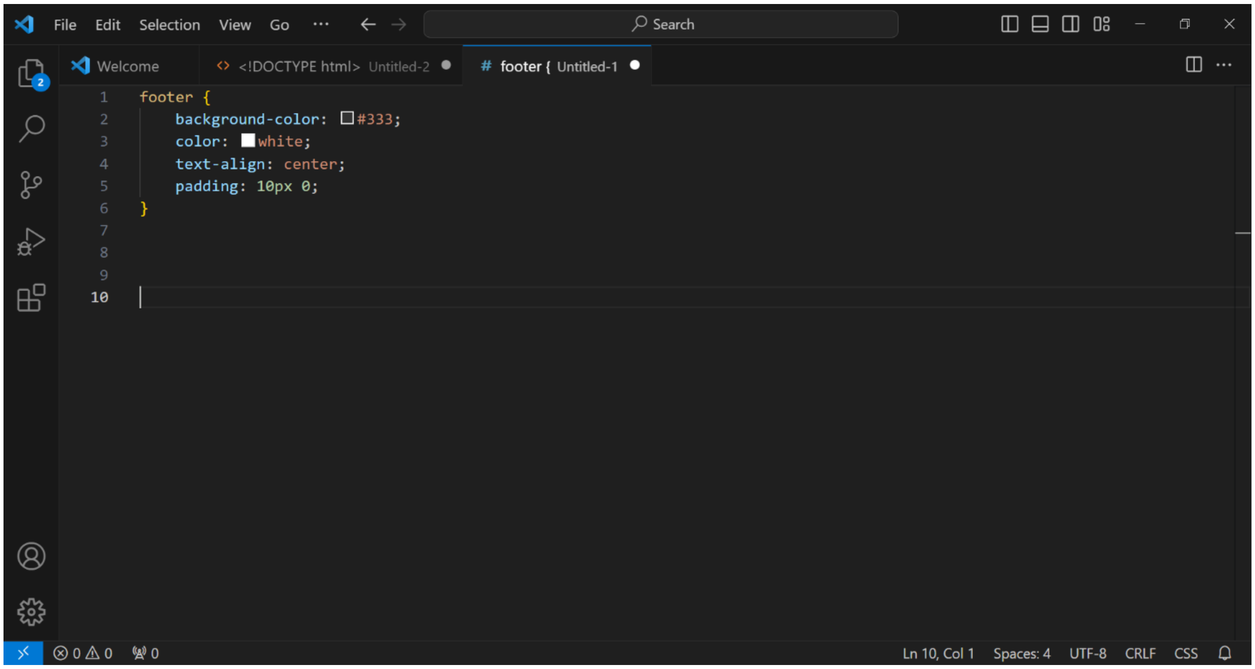

Figure 3.28: CSS For Footer of Landing Page

3.6.2 Integration

The integration process outlined provides a high-level overview of how the distinct outputs generated by (1) the Machine Learning environment and (2) PowerBI visualization will be merged. Within this integration phase, we will extract data from the input landing page and present consolidated results on the designated result page. Given the central reliance on Application Programming Interfaces (APIs) to ensure the smooth integration of these outputs into our result page, a more detailed discussion on the integration process is reserved for Section 3.8 Final Implementation.

3.6.3 Result Page

A result page in a website is a web page that displays information or outcomes based on a user's query, search, or interaction with the website. It aims to provide users with relevant information or confirm their actions on the website. The result page is where our users receive the skin care ingredient analysis results. Therefore, the visualization of the results (pie chart), ingredient details, ingredient ratings and other related information will be developed in the user interface of the result page. The interface examples are shown in section 3.4.1 (Figures 3.10 and 3.11). The reference code for the result page is as follows:

A. HTML Reference Code

I. Head

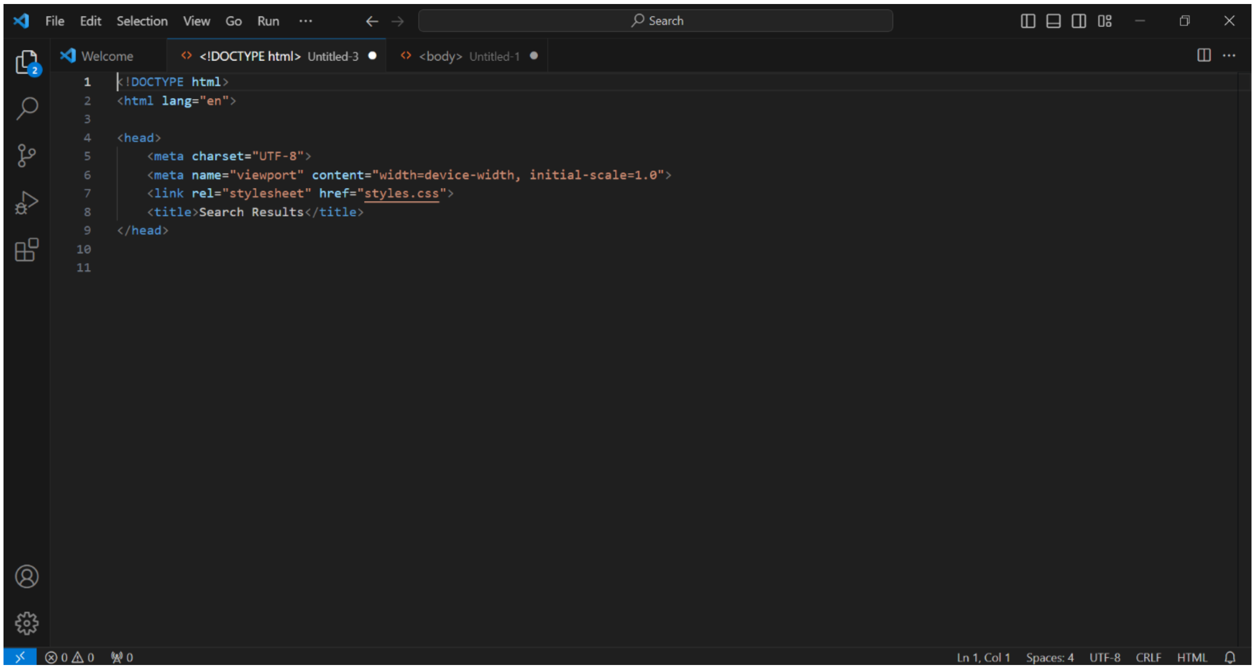

Figure 3.29: HTML For Head of Result Page

II. Body

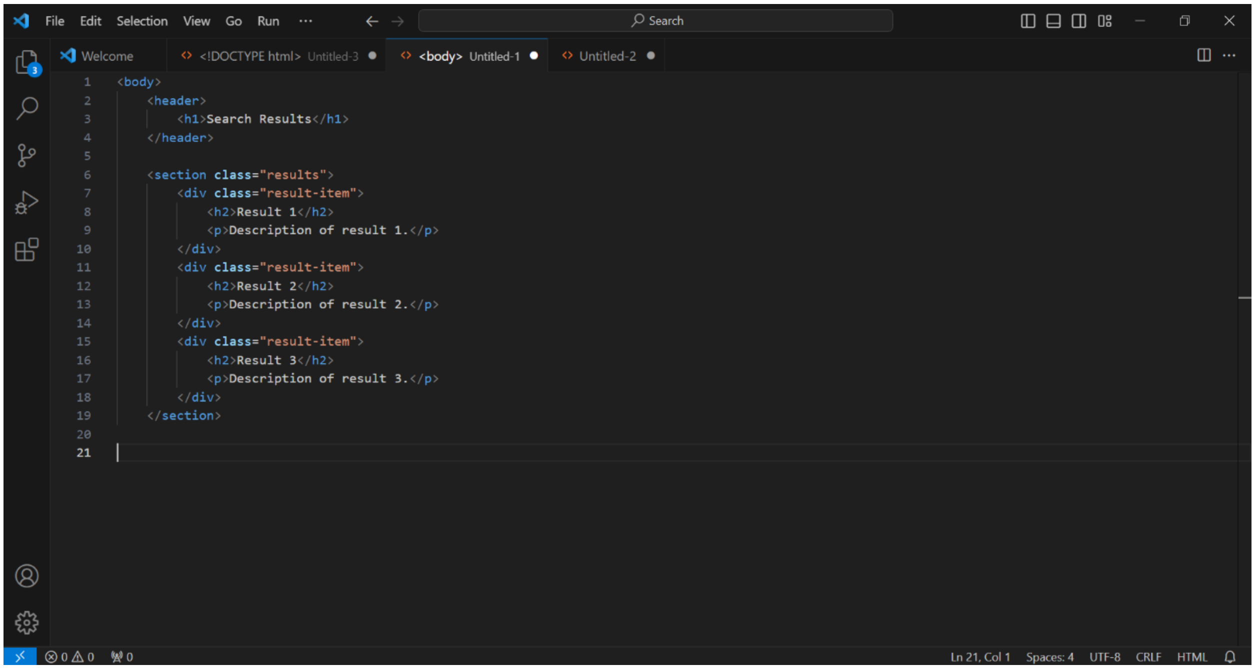

Figure 3.30: HTML For Body of Result Page

III. Footer

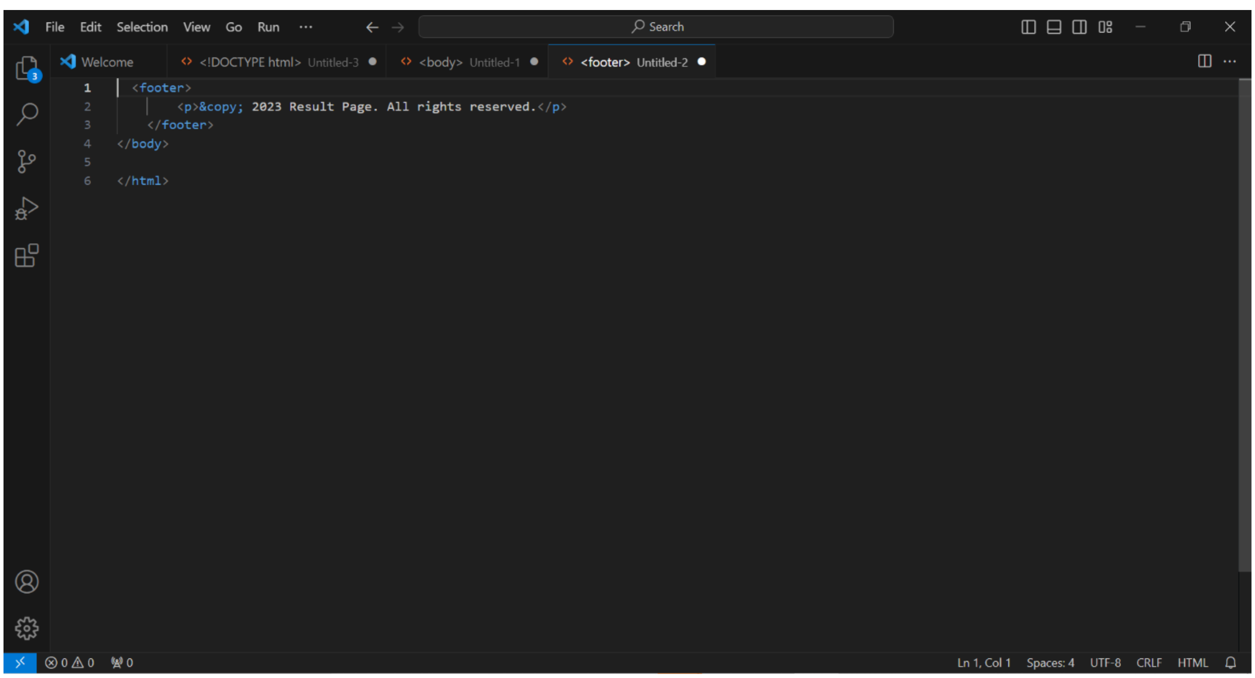

Figure 3.31: HTML For Footer of Result Page

B. CSS Reference Code

I. Head

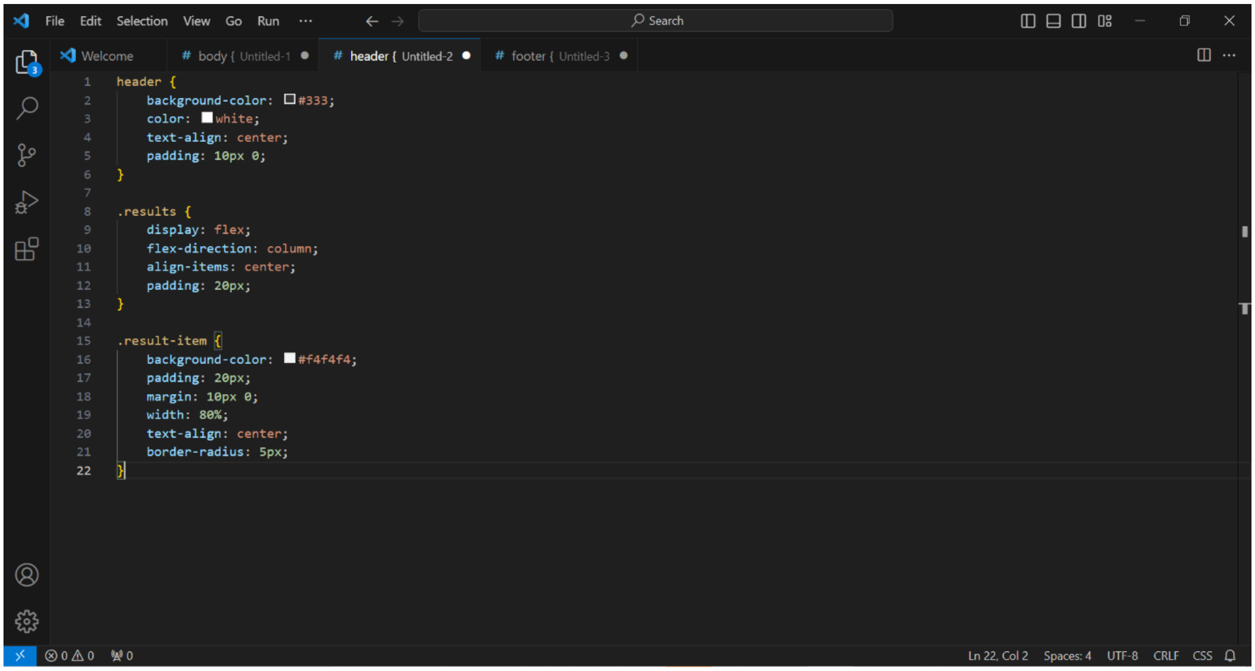

Figure 3.32: CCS For Head Result Page

II. Body

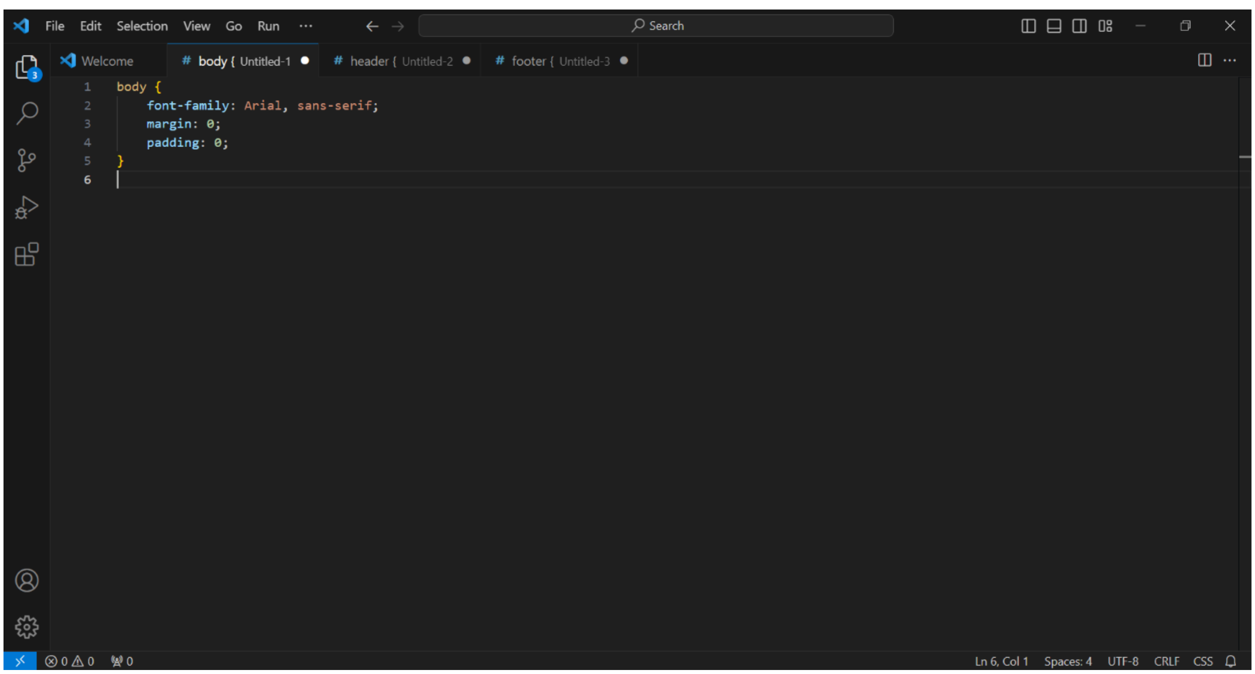

Figure 3.33: CCS For Body of Result Page

III. Footer

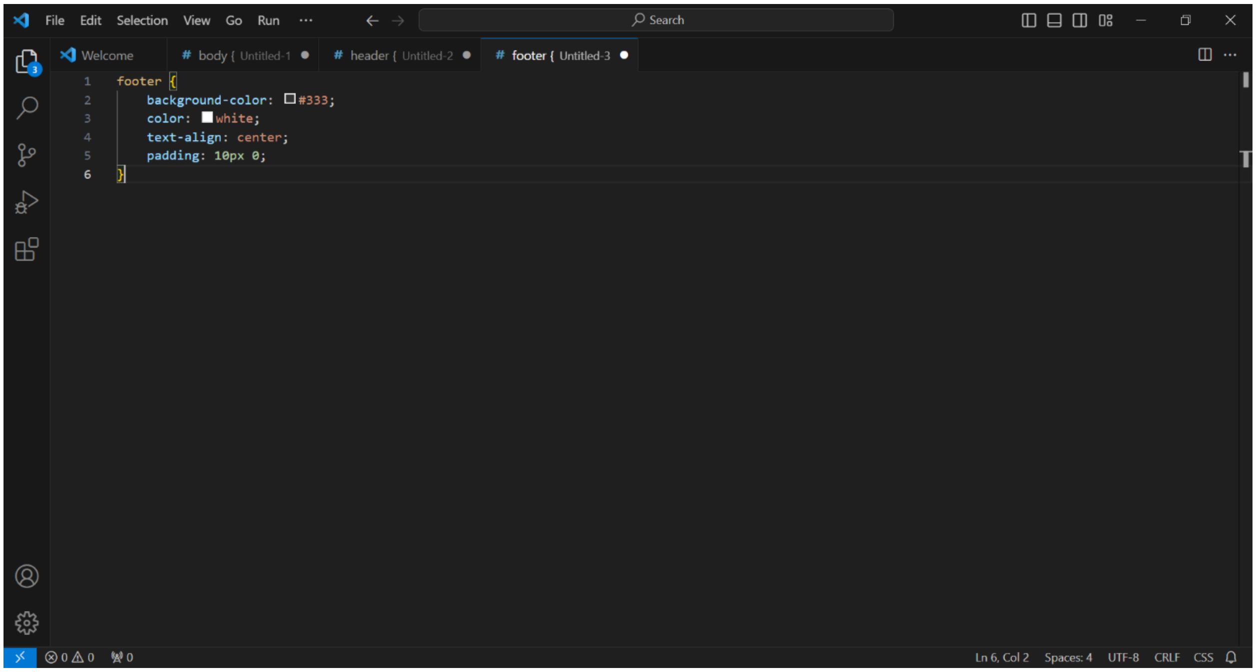

Figure 3.34: CCS For Footer of Result Page

3.7 OCR Development

This section describes the integration of Optical Character Recognition (OCR) technology into our system.

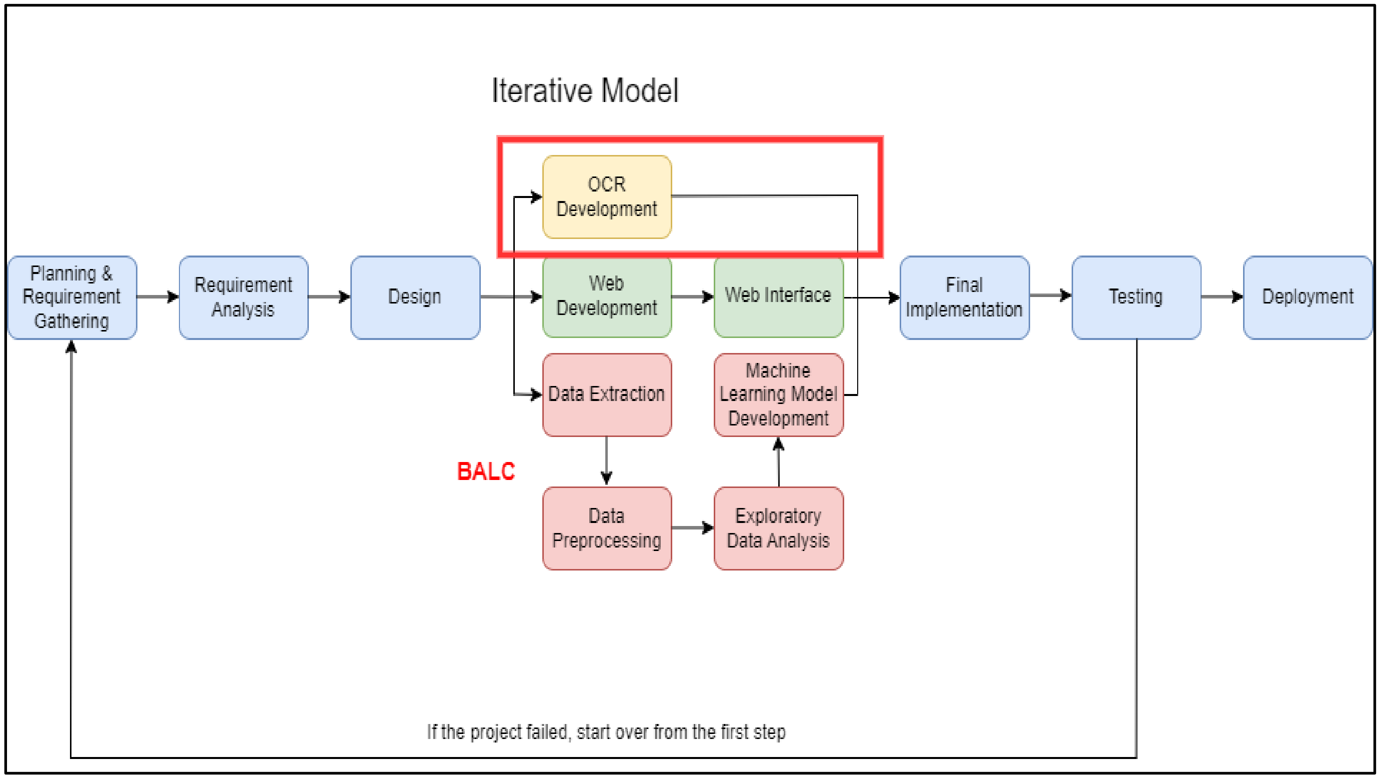

Figure 3.35: OCR Development

We have chosen to integrate the Tesseract OCR engine, an open-source OCR solution developed by Google, known for its high accuracy and multilingual support. We utilized the Python programming language and the 'pytesseract' library to ensure a seamless data flow between the OCR engine and our classification model. Additionally, preprocessing techniques and regular expressions are implemented to enhance text quality and remove irrelevant information. When users submit images of skincare product ingredient lists, the OCR is triggered to convert the image-based content into machine-readable text. Figure 3.29 shows a simplified example of OCR integration using Python's 'pytesseract' library, while Table 3.9 provides explanations of the significant pseudocode used for OCR integration.

```
[1] # Install Tesseract and Python Library
    !sudo apt install tesseract-ocr
    !pip install pytesseract

[2] # Import necessary libraries
    import pytesseract
    import shutil
    import os
    import random
    try:
     from PIL import Image
    except ImportError:
     import Image

[3] # Upload image
    from google.colab import files
    uploaded = files.upload()

    Choose Files  Screenshot... 100423.png
    • Screenshot 2023-10-11 100423.png(image/png) - 128952 bytes, last modified: 10/11/2023 - 100% done
    Saving Screenshot 2023-10-11 100423.png to Screenshot 2023-10-11 100423.png

[8] # Perform OCR on the image
    text = pytesseract.image_to_string(Image.open('Screenshot 2023-10-11 100423.png'))

[9] # Print the extracted text
    print(text)
```

Figure 3.36: Pseudocode for OCR Integration

__Table 3.8: Description of Significant Pseudocode for OCR Integration__

Significant Pseudocode	**Description**
# Install Tesseract and Python Library	● Install Tesseract OCR tool and the Python library for interfacing with Tesseract.
# Import necessary libraries	● Import essential Python libraries required for the OCR process.
# Upload image	● Use the 'files.upload()' function to upload an image for OCR.
# Perform OCR on the image	● Use the 'pytesseract.image_to_string()' function to extract text from the uploaded image.

3.8 Final Implementation

The final, yet most crucial part of the phase is to integrate all the individual processes of system development segments into the user interface or the result page. Hence, the final implementation phase will discuss how (1) text classification result generated by OCR and ML and (2) PowerBI visualization will be fed onto the result page for the display of analysis result. API will be heavily involved to ensure results under different environment can be integrated seamlessly.

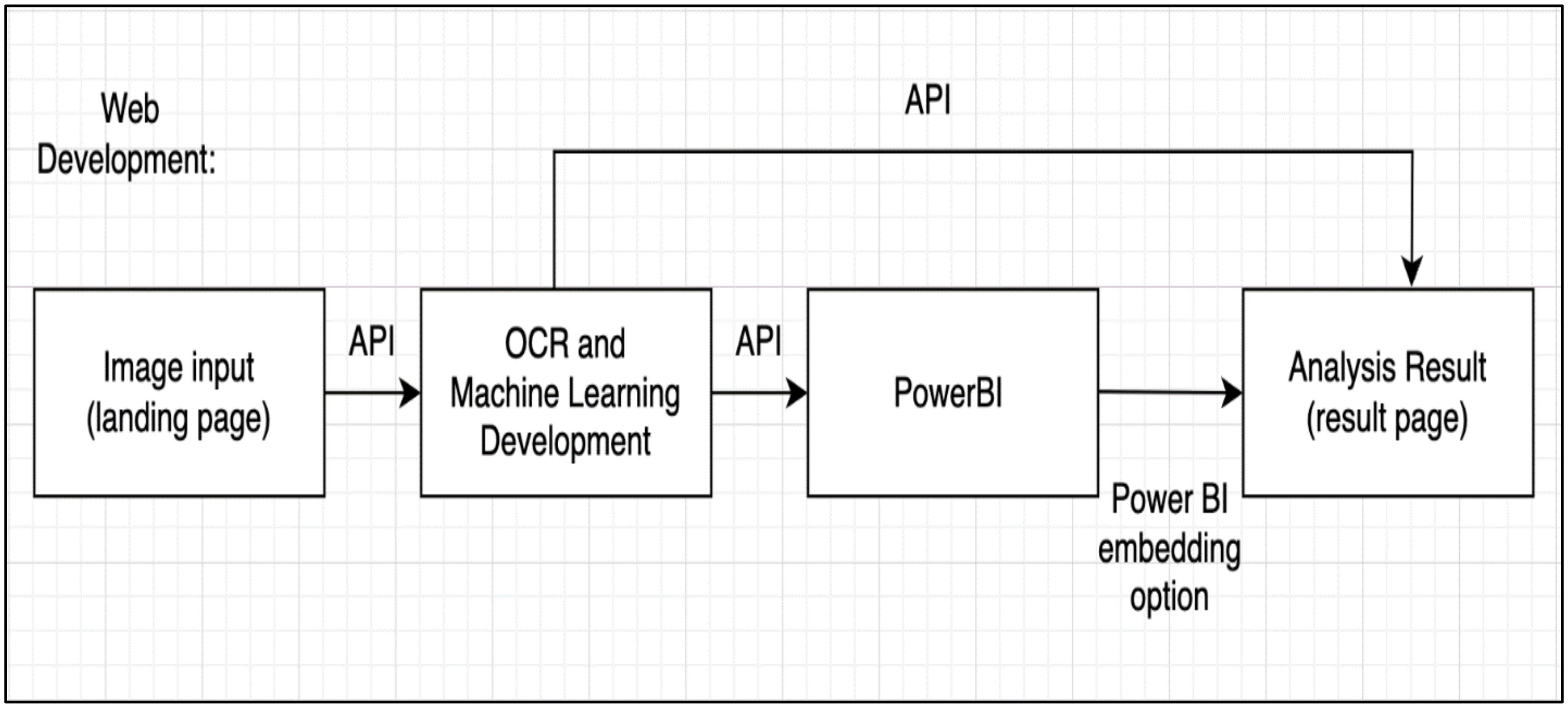

Figure 3.37: Final Implementation of Beauty Formula DermoScan

In the final implementation, we will employ Application Programming Interfaces (APIs) to seamlessly combine separate processes into the ultimate graphical user interface (GUI). The involved processes are outlined below:

1. Users will upload an image on their landing page, and the API endpoint will be set up in OCR and machine learning development environment to receive data. Which will then trigger the machine learning and OCR development environment to generate text classification results.
2. The text classification results will be sent into Power BI’s API endpoint for visualization and enhanced data representation. The classification output will also be shown on the user interface (result page) using the API endpoint.
3. For the visualization, the Power BI results will be further integrated into the skincare ingredient analysis result within the GUI using its own embedded option, ensuring a comprehensive and cohesive user experience.

Once the web interface can be executed smoothly, the machine learning model can generate accurate results, the OCR engine can effectively extract text, and the proposed project is prepared to be finally integrated into a complete system.

3.9 Testing

The testing part is separated into two parts, which is Integration Test and User Acceptance Test (UAT). The performance of the (Optical Character Recognition) OCR system and the accuracy of the Machine Learning Model will be tested in the Integration Test part, while a User Acceptance Testing flow will be included in the User Acceptance Test part.

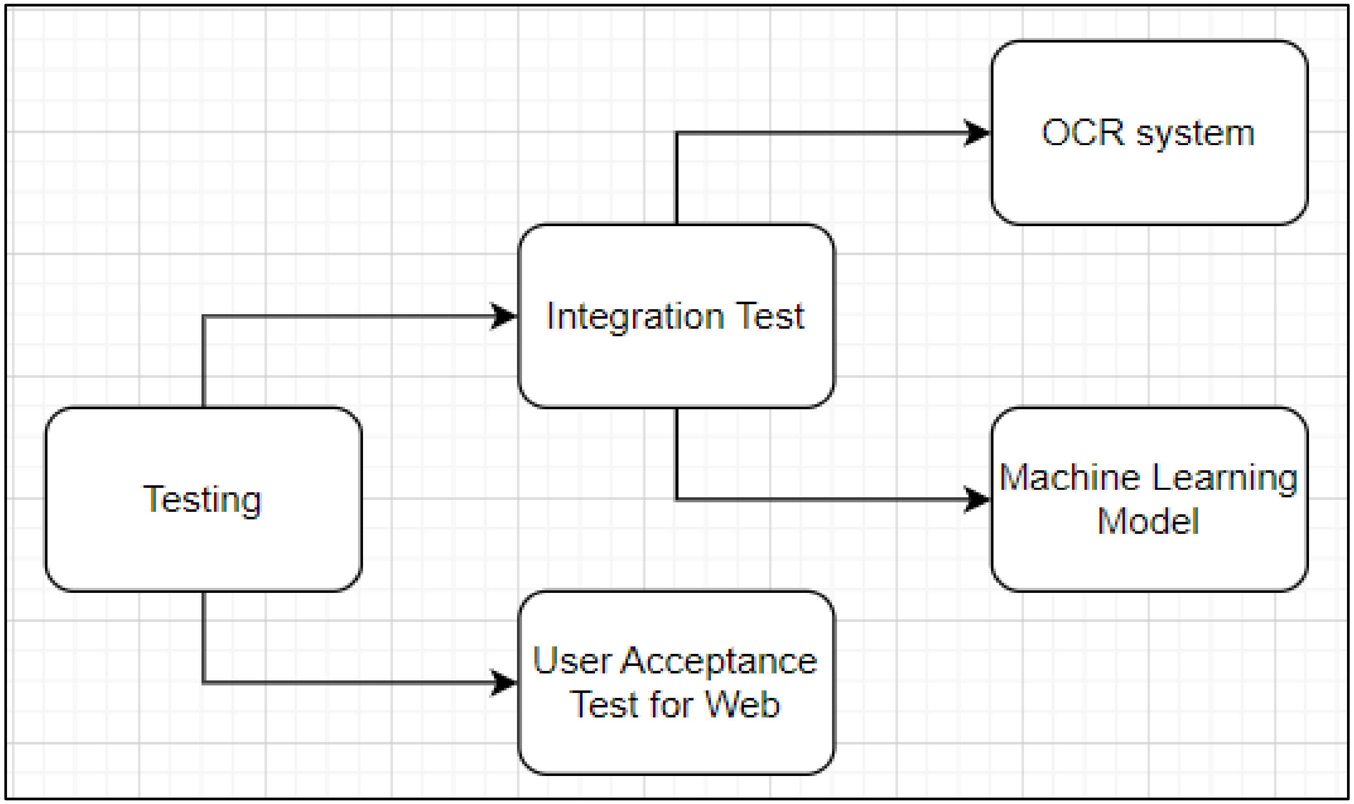

Figure 3.38: Testing Flow Overview

3.9.1 Machine Learning Model Testing

To evaluate the performance of text matching using TF-IDF and text classification using Rule-based Approach, common metrics such as accuracy, error rate, precision, recall, and F1 score are used.

Table 3.9: Description of the Evaluation Metrics for ML Model

Evaluation Metrics	Description
Accuracy	Calculate the percentage of successfully categorized ingredient texts in relation to the total number of texts examined.
Error Rate	Calculate the percentage of ingredient texts that were erroneously classified out of the total examined.
Precision	Determine the percentage of successfully classified ingredients relative to the projected classifications.
Recall	Calculate the percentage of correctly categorized ingredients in relation to the total number of actual classifications.
F1 score	Calculate the F1 score for the rule-based aspect, which reconciles precision and recall.

Data are collected and preprocessed to perform both text matching and text classification, which will be using term frequency-inverse document frequency (TF-IDF) and rule-based model, respectively. After implementing TF-IDF for text matching using Python, common metrics, including accuracy, error rate, precision, recall, and F1 score, are used to evaluate the performance of text matching, which allows comprehension of the model performance accuracy rate and completeness. After implementing rule-based text classification, the same metrics (accuracy, error rate, precision, recall, and F1 score) will be used to evaluate the results. The performance of the overall model will be reviewed by merging the results from both text matching and text classification. Components are weighted based on how important it is in a particular situation; for instance, in suitable circumstances, precision should take precedence over speed.

For the testing of the machine learning model, two machine learning models, namely the Rule-Based Approach and Support-Vector Machine (SVM), were tested and evaluated during internal examination. To conclude both models, the Rule-Based Approach is chosen in our model development due to its performance capabilities. The metrics are analyzed and interpreted to evaluate the performance of the model. Improvements will be made in order to enhance and document the model's performance in terms of accuracy. A summary of the performance for both text matching and text classification of the model will be provided for the final result. If the machine learning model satisfies the requirements, monitoring mechanisms will be deployed along with it.

3.9.2 User Acceptance Test (UAT)

The last stage of system testing is known as user acceptance testing (UAT), in which actual users test the system to determine whether it satisfies their unique requirements and performs as intended. Testing the system's functionality, usability, and overall efficacy from the perspective of the user, helps ensure that it is all set for the deployment stage. The UAT flow below is generated from the end user's perspective when they are on the website.

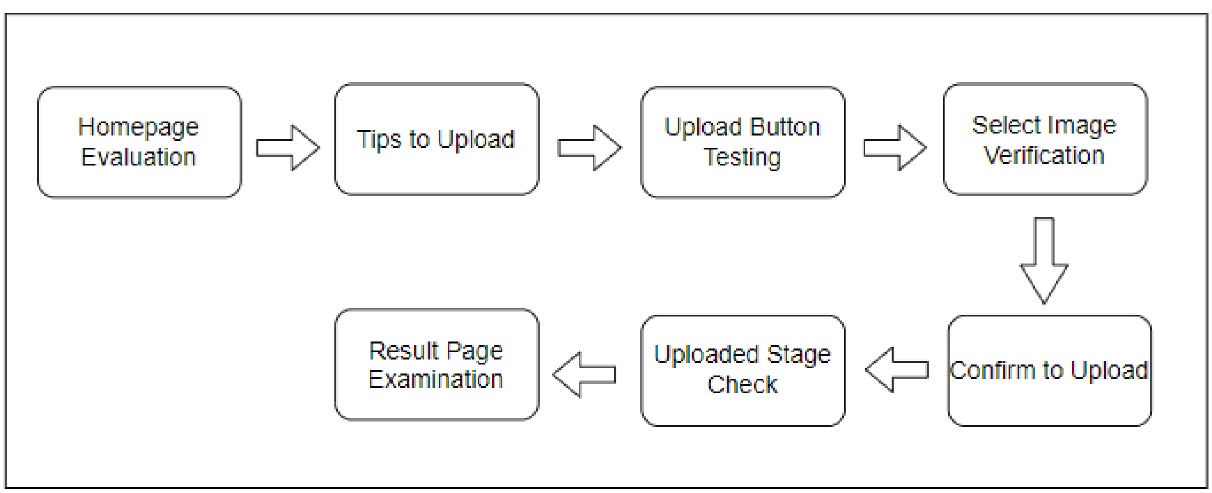

Figure 3.39: UAT Flow

1. The flow will begin with **homepage evaluation**. When the website users are on the homepage, it should be ensured that they have easy access to the relevant information and that the navigations are user-friendly.
2. After clicking on the "**tips to upload**" button, precise and useful guidance for image uploading should be provided to users. It should be ensured that the instructions for uploading are conveyed clearly.
3. The functionality of the "**upload**" button should be tested if it functions correctly and enables users to start the upload process without having any issues.
4. Verifying the "**select image**" feature ensures that users can choose the right images. In this part, image selection errors or glitches should be checked as well.
5. Next, examine if the "**confirm to upload**" button is functioning. Users should have no difficulties reviewing and confirming the image they chose.
6. Assess that the "**uploaded**" stage correctly displays the image uploaded by the users. Any mistakes or discrepancies should be addressed and resolved.

Lastly, the "**result page**" should be examined closely. A pie chart and the ingredient categories should be displayed to the users clearly based on the image they have uploaded. The name, rating, and description of the ingredients should be accurate so that the users are clear about it.

3.9.3 OCR System Testing

For the OCR system, Python will be used to implement and test the performance of it using Character Error Rate (CER) and Word Error Rate (WER). CER is the percentage of characters that contain errors when compared to the original text, while WER represents the proportion of words that the OCR system erroneously identified.

Table 3.10: Description of the Evaluation Metrics for OCR Engine

Evaluation Metrics	Description
Character Error Rate (CER)	The percentage of characters that contain errors when compared to the original text. Lower CER scores represent better performance.
Word Error Rate (WER)	The proportion of words that the OCR system erroneously identified. Lower WER scores represents better performance.
Ground Truth Text	The accurate and reference text used to make comparison with the OCR system.

The testing steps are started by collecting a dataset which contains the ingredient's names, descriptions, ratings, benefits, and risks. The dataset should contain and serve as the ground truth text, the reference text used to compare the OCR system. The text in the image uploaded by users will be extracted in the OCR system and then compared to the ground truth text to ensure accuracy.

With Python, CER and WER are calculated by the total number of characters or word errors divided by the total characters or words in the ground truth. CER and WER scores are analyzed to assess the performance of the OCR system. Higher scores represent lower performance, and vice versa. The common word and character errors made by the OCR system are examined and analyzed to make improvements and to detect the limitations the system currently has. Adjustments and improvements will be made iteratively in the OCR system in order to resolve the errors and enhance the OCR system. The testing results, including the scores of CER and WER, will be documented for future purposes, such as system performance tracking.

3.10 Deployment

The deployment stage of the software development process is the final, tested, and validated system or application that is ready to be accessed by end users. For the system's intended users to utilize all of its capabilities and features, the deployment makes sure that the software is freely accessible to them. It signifies the change from testing and development to actual use, making the system usable in real-world scenarios. The deployment stage requires meticulous implementation to preserve consistency and satisfy consumers.

After completing the testing stage, the tasks that will be carried out once the system is successfully deployed include communicating with the users regarding providing resources or training and keeping track of the system's performance, scalability, and user activity. The system needs to be maintained and improved continuously to ensure it operates effectively.

Our system, Beauty Formula DermoScan, will be deployed by a local host, a network address used to connect itself with a device. The IP address 127.0.0.1 is commonly used to represent the local host. A local host is an essential tool that is usually used in the testing and deployment stage as it ensures that the project is functioning correctly on its own devices or local system without needing external connections before making it live. It provides a convenient and safe environment for developers to work while preventing problems from occurring while sharing things online.

CHAPTER 4

EXPECTED BENEFITS

4.1 Overview

Traditionally, deciphering the chemical constituents of beauty products has presented formidable challenges to stakeholders, who historically only had access to limited resources for comprehending these complex formulations. Our research project endeavors to modernize and streamline this intricate procedure by integrating state-of-the-art Optical Character Recognition (OCR) technology into the beauty industry. This integration empowers stakeholders with a sophisticated tool that facilitates a comprehensive understanding of chemical components.

Incorporating OCR technology within our project facilitates an automated and efficient process of converting beauty product labels into coherent explanations. This transformative capacity not only enhances accessibility but also expedites decision-making for all stakeholders within the beauty industry, encompassing consumers, professionals, and businesses alike. Through a simple scan, users gain access to a wealth of information and in-depth insights concerning product ingredients, thereby enabling more informed choices and personalized product selection.

4.2 Impact to the Industry

The impact on the skincare industry is profound. By enhancing transparency, Beauty Formula DermoScan ensures that consumers have access to comprehensive data about the products they use, encouraging skincare companies to prioritize safety and transparency (Alwis et al., 2022). This increased transparency will lead to more informed purchasing decisions and raise awareness about the importance of understanding skincare ingredients. Hence, this motivates competition among manufacturers, compelling them to produce safer and more effective products with transparent ingredient labels. This, in turn, will raise the standards for skincare product quality. Overall, the impact on the skincare industry is a positive shift towards safer, more transparent, and consumer-centric practices.

The impact on the skincare industry is profound. By enhancing transparency, Beauty Formula DermoScan ensures that consumers have access to comprehensive data about the products they use, encouraging skincare companies to prioritize safety and transparency (De Alwis, 2022). This increased transparency will lead to more informed purchasing decisions and raise awareness about the importance of understanding skincare ingredients. Hence, this motivates competition among manufacturers, compelling them to produce safer and more effective products with transparent ingredient labels. This, in turn, will raise the standards for skincare product quality. Overall, the impact on the skincare industry is a positive shift towards safer, more transparent, and consumer-centric practices.

4.2.1 Organization

The Beauty Formula Dermo Scan System, enriched by the integration of OCR technology, emerges as a transformative force poised to shape the future of organizations within the beauty industry. Its multifaceted contributions offer a range of compelling benefits that are set to redefine operational paradigms and enhance competitive standing. The enhancement of customer experience through transparency and access to detailed ingredient information is paramount. This trust-building transparency positions organizations as advocates for informed decision-making in beauty product selection. Customers are more likely to return for future purchases, thus increasing customer retention and fostering brand loyalty. Organizations can leverage the wealth of ingredient data to comprehensively understand market trends, consumer preferences, and product performance. Armed with this knowledge, they can make strategic decisions that enhance product development, marketing strategies, and inventory management, ultimately driving profitability.

4.2.2 Improve Efficiency

The system could revolutionize efficiency by automating the once laborious process of manually inputting product ingredient information (Karthikeyan, 2020). With the deployment of the system, individuals can swiftly and accurately extract data, drastically reducing the time and effort involved in this task. This transformation means consumers and industry professionals can now access crucial ingredient details with remarkable speed. Consumers can make more informed choices, while industry experts can expedite their decision-making processes, resulting in heightened customer satisfaction, streamlined product development, and a significant boost in competitiveness. Overall, the Beauty Formula Dermo Scan System epitomizes the future of efficiency in the beauty industry, where precision, adaptability, and transparency combine to reshape how organizations excel in this dynamic and fiercely competitive arena.

4.3 Impact to the Model as New Knowledge

First, it helps improve the models' accuracy by incorporating information about emerging safety concerns and allergens. This means that the models can better identify potential hazards, which helps to protect consumers. As our understanding of different skin types and concerns evolves, the models can be adapted to offer tailored product suggestions that meet individual needs. This can boost user satisfaction and trust in the recommendations.

New knowledge can also benefit the TF-IDF analysis, which relies on the relevance of terms within ingredient labels. If recent research or changing consumer preferences indicate the importance of specific ingredient properties for safety or effectiveness, the TF-IDF model can be adjusted to weigh these factors accordingly. This ensures that the model's assessments align with the latest industry insights and scientific understanding.

Furthermore, new knowledge may introduce emerging skincare ingredients or innovative formulations. Both the rule-based and TF-IDF models may require updates to accurately recognize and assess these fresh ingredients, ensuring that users receive relevant information about the latest products on the market.

4.4 Summary of Expected Benefits

Our project uses OCR technology to simplify beauty product ingredient understanding and enhance transparency in the skincare industry. This fosters competition, raises industry standards, and benefits organizations by improving customer experiences and efficiency. It also adapts to new knowledge, ensuring accurate assessments and informed product suggestions, reshaping the industry with transparency and efficiency.

CHAPTER 5

PROJECT PLANNING

5.1 Overview

This chapter uses a Gantt chart to demonstrate the visual representation of our project progress, including timeframes, task delegation, and significant project highlights. On the Gantt chart, each bar's length indicates the task's projected duration, and the bars' positions on the timeframe display the task's start and end dates. The project timeframe will be displayed horizontally and is usually portrayed in a span of one week. In addition. The allocation of tasks can also be seen as assigning tasks to each member. Task distribution makes the way in accomplishing successful resource management by identifying possible constraints or deficiencies in the allocation of resources. The Gantt chart pinpoints the key project highlights, such as the dependencies of tasks, completion of the project, and allocation of resources, enabling team members to focus on crucial tasks and ensure that they stick to the project timeline and stay on track. Sequential project planning indicates that tasks are scheduled according to their dependencies, making sure that the start and end dates of each task coincide with those of tasks that come before it. Establishing a clear and planned workflow ensures that tasks are completed in the proper order and eliminates delays brought on by task dependencies.

5.2 Gantt Chart

Beauty Formula DermoScan							
No.	**Tasks**	**Assigned To**	**Start Date**	**End Date**	**Duration (Days)**	**Status**	**Phase**
1	**Brainstorming Idea**	All	11-Sep	19-Sep	8	Completed	Phase 1
2	Literature Review	All	20-Sep	29-Sep	9	Completed	
3	**Methodology**	All	30-Sep	12-Oct	12	Completed	
4	**Preperation for Report Proposal**	All	1-Oct	15-Oct	14	Completed	
5	**Development**	All	16-Oct	16-Nov	31	Not Started	Phase 2
6	**Confirmation and Deployment**	All	16-Nov	26-Nov	10	Not Started	
7	**Showcase Preperation**	All	24-Nov	11-Dec	17	Not Started	
8	**Report Finalization**	All	16-Nov	17-Dec	31	Not Started	

Figure 5.1: Overview Tasks for Gantt Chart

Figure 5.1 shows the overview tasks of the Gantt Chart, which involved 8 main tasks: brainstorming ideas, literature review, methodology, preparation for report proposal, development, confirmation, and deployment, showcase preparation, and lastly, report finalization. The first four tasks are involved in phase 1 (blue), and the remaining four tasks will be in phase 2 (orange). All of the tasks above are assigned to our group members fairly. The start date of our project is September 11, 2023, while the end date is December 17, 2023.

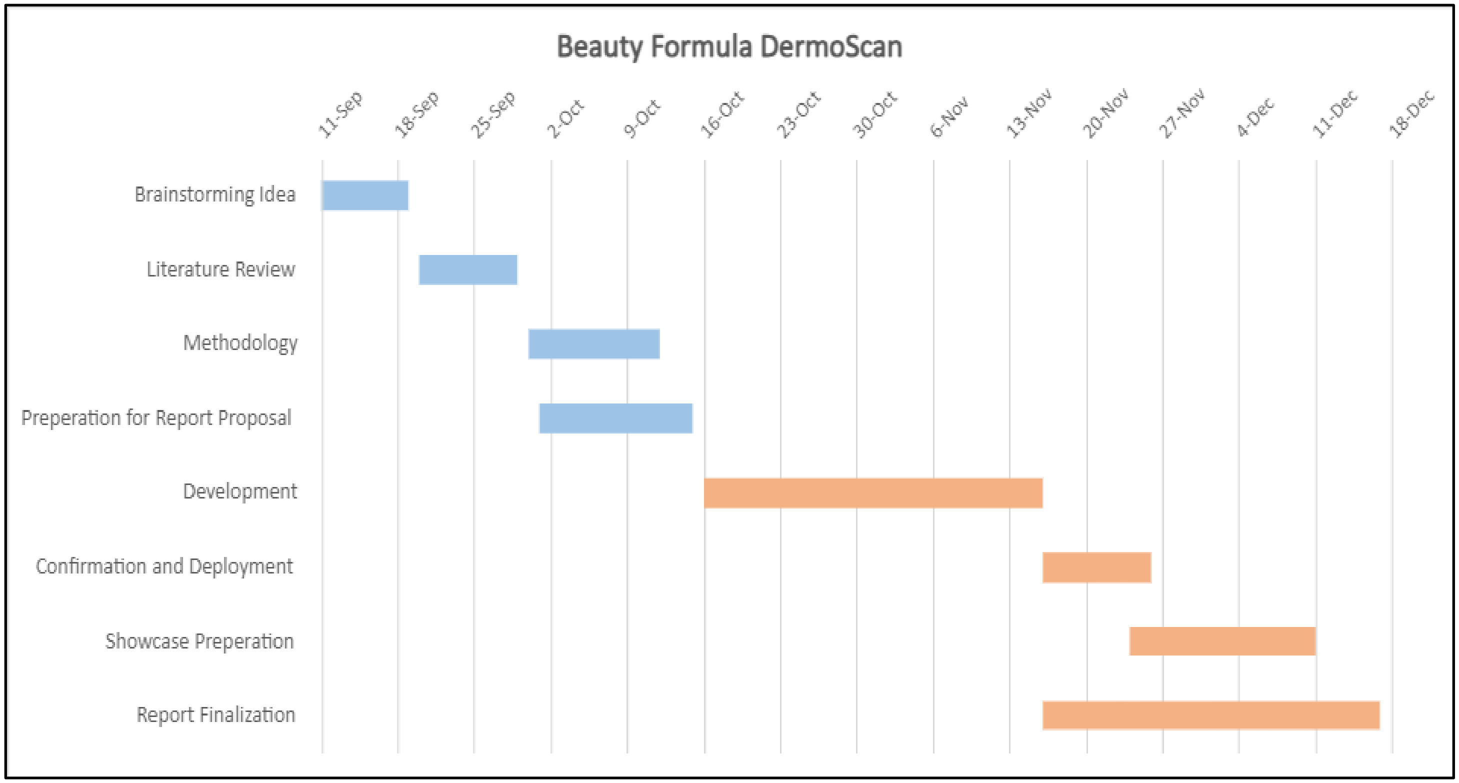

Figure 5.2: Gantt Chart for Overview Progress

Figure 5.2 above shows the overview progress of our project. Each bar length represents the projected time needed to complete each task to be done, while the position of the bars shows the start and end date for each task. It is shown that the deployment and the report finalization require the longest time, while brainstorming ideas took the least amount of time to complete.

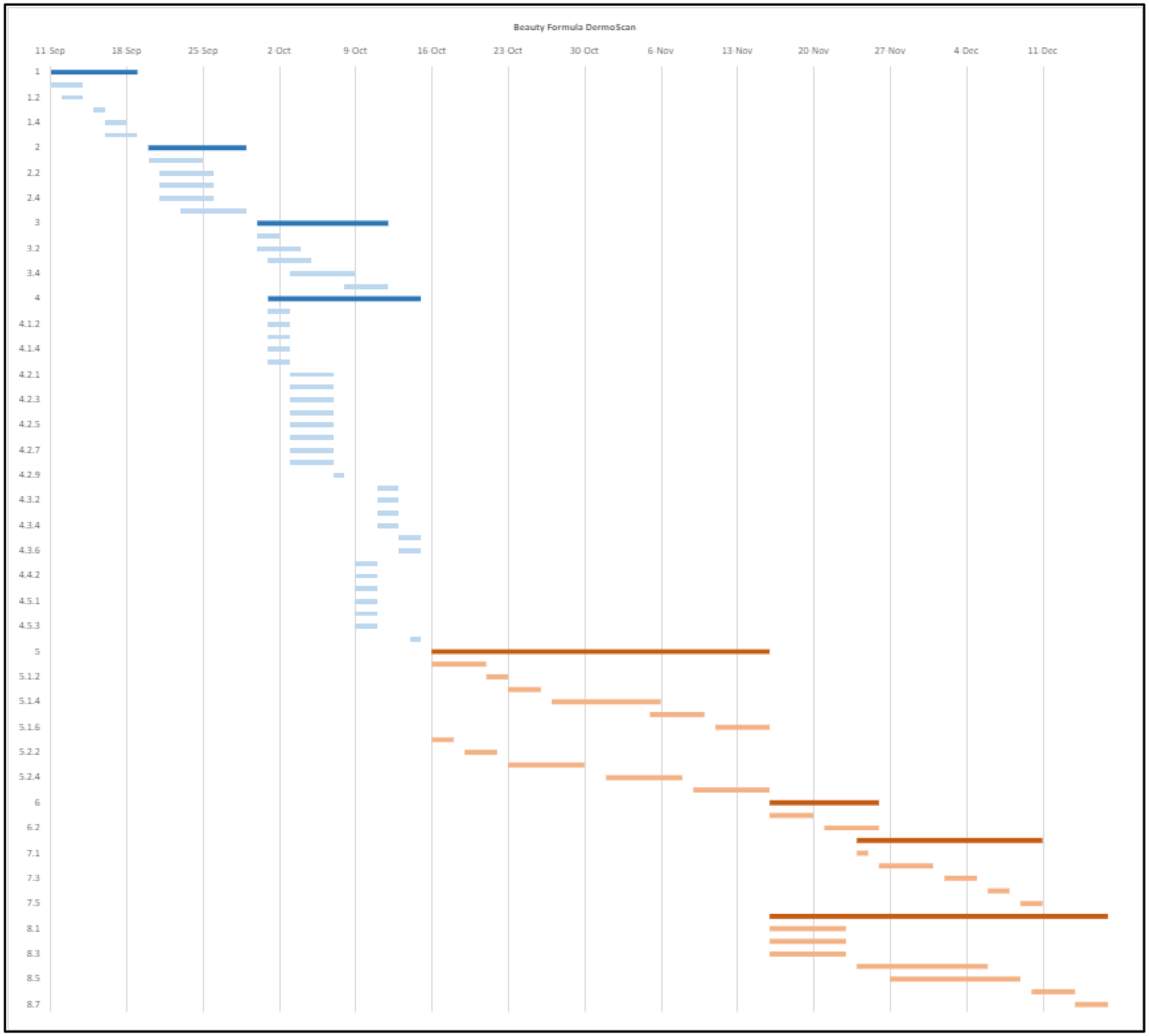

Figure 5.3: Combination of Gantt Chart Phase 1&2

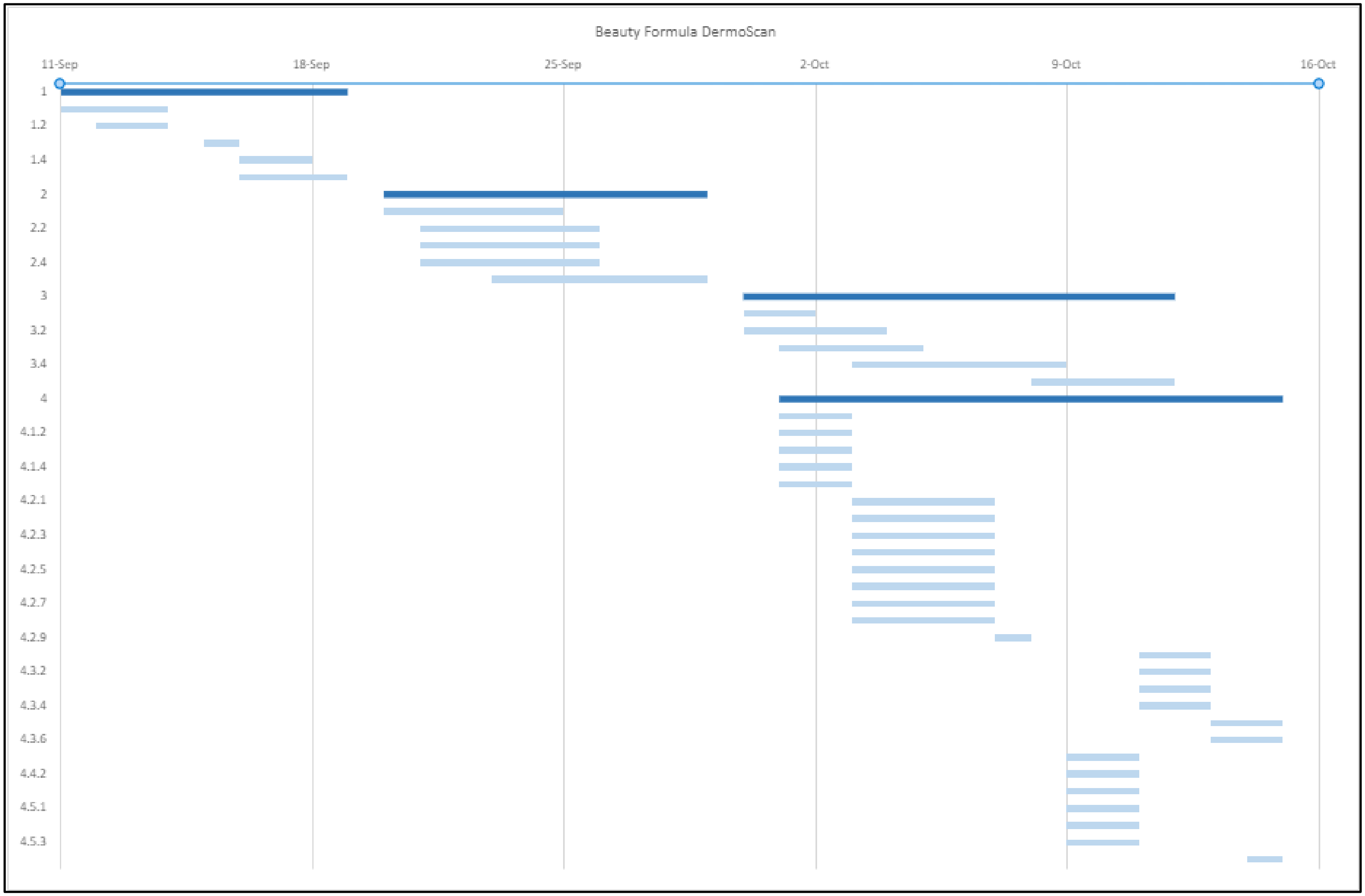

Figure 5.4: Gantt Chart Phase 1

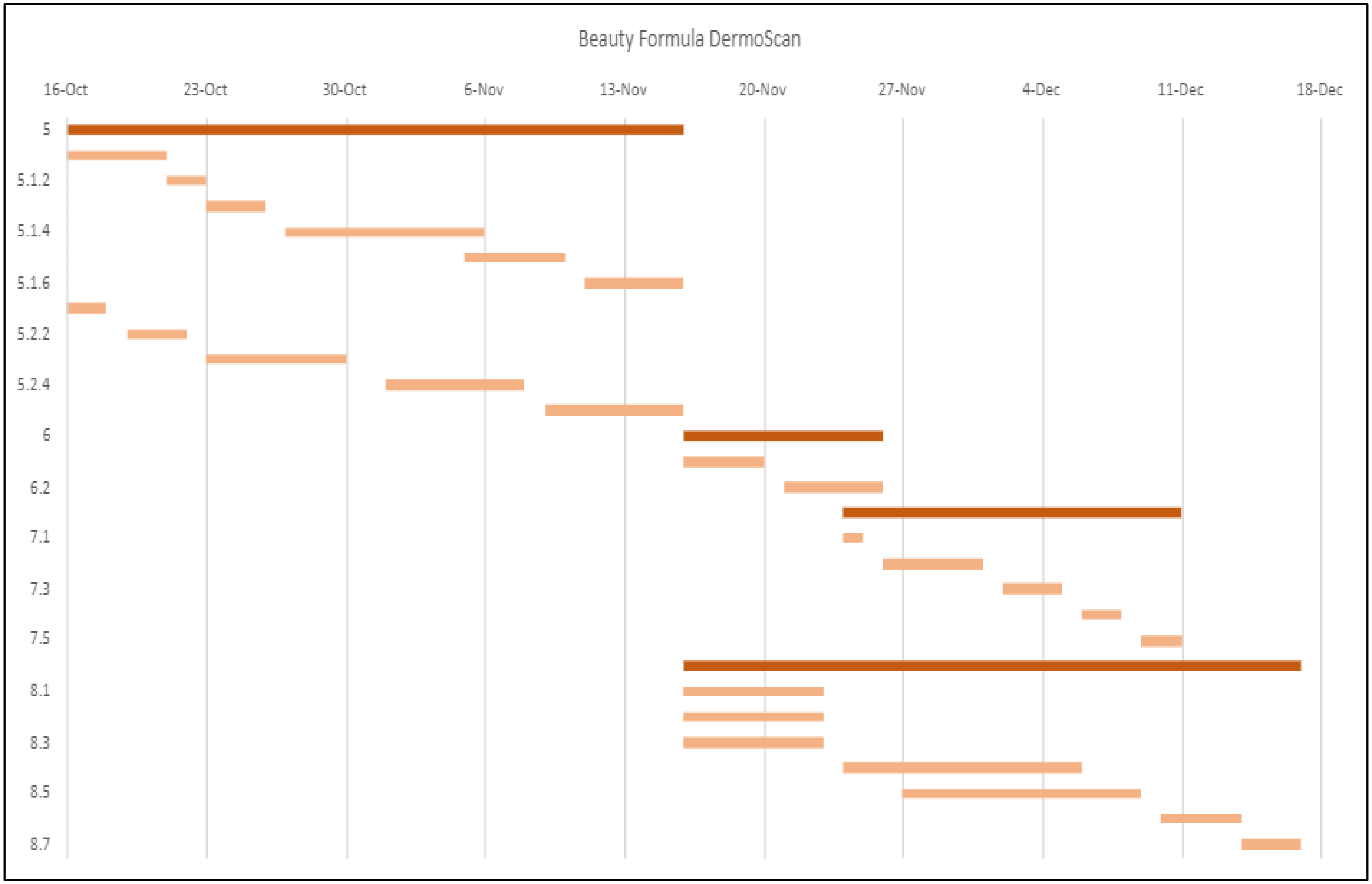

Figure 5.5: Gantt Chart Phase 2

By splitting the project into a number of tasks, the Gantt Chart helps in tracking task dependency, allocating resources, and time management. Moreover, it efficiently fosters communication and collaboration between the group members by enabling group members to share project plans with each other in an easily comprehensible format. In brief, in order for the project to be completed and executed effectively, each of the tasks listed in the Gantt chart is crucial.

5.2.1 Critical Part 1 of the Gantt Chart

Table 5.1: Critical Part 1 of the Gantt Chart

Critical Part	**Description**
Summary of project development	Model development
Focus	Using TF-IDF for text matching and rule-based approach for text classification.
Critical elements in project planning	• Data pre-processing. • Develop machine learning model. • Perform testing and evaluation on the machine learning model.
How to execute the critical elements in project planning	• Effective communication. • Adhere to the planned project timeline. • Research on best practices.

5.2.2 Critical Part 2 of the Gantt Chart

Table 5.2: Critical Part 2 of the Gantt Chart

Critical Part	Description
Summary of project development	Web Development
Focus	Creating a website for our system using HTML and CSS in Visual Studio Code.
Critical elements in project planning	• Design the webpage. • Create user-friendly interface. • Test and verify the functionality of web application.
How to execute the critical elements to drive the steps in project planning	• Effective communication. • Adhere to the planned project timeline. • Research on best practices.

5.3 Delegation of Work

Table 5.3: Delegation of work

No.	Tasks	Assigned To	Start Date	End Date	Duration (Days)	Status
1	**Brainstorming Idea**	All	11-Sep	19-Sep	8	Completed
1.1	Identify Potential Research Topics	All	11-Sep	14-Sep	3	Completed
1.2	Conduct Initial Research on Selected Topics	All	12-Sep	14-Sep	2	Completed
1.3	Evaluate and Narrow Down Research Topics	All	15-Sep	16-Sep	1	Completed
1.4	Define the Research Objectives and Questions	All	16-Sep	18-Sep	2	Completed
1.5	Propose Solution Idea	All	16-Sep	19-Sep	3	Completed
2	**Literature Review**	All	20-Sep	29-Sep	9	Completed
2.1	Discuss Domain and Scenario	Zhee Xuan	20-Sep	25-Sep	5	Completed
2.2	Tools and Techniques	Rui Xuan	21-Sep	26-Sep	5	Completed
2.3	Data Analysis	Yi Hang	21-Sep	26-Sep	5	Completed
2.4	Machine Learning Model	Yun Jie	21-Sep	26-Sep	5	Completed
2.5	Framework	James	23-Sep	29-Sep	6	Completed

3	**Methodology**	All	30-Sep	12-Oct	12	Completed
3.1	Finalize and Selection of Framework	James	30-Sep	2-Oct	2	Completed
3.2	Planning and Requirement Gathering	Yi Hang	30-Sep	4-Oct	4	Completed
3.3	Requirement Analysis and Design	Zhee Xuan	1-Oct	5-Oct	4	Completed
3.4	Development and Implementation	Rui Xuan	3-Oct	9-Oct	6	Completed
3.5	Testing and Deployment	Yun Jie	8-Oct	12-Oct	4	Completed
4	**Preparation for Report Proposal**	All	1-Oct	15-Oct	14	Completed
4.1.1	Introduction - Background of Study	Rui Xuan	1-Oct	3-Oct	2	Completed
4.1.2	Introduction - Research Motivations	Yun Jie	1-Oct	3-Oct	2	Completed
4.1.3	Introduction - Problem Statement	Yi Hang	1-Oct	3-Oct	2	Completed
4.1.4	Introduction - Research Questions and Objectives	James	1-Oct	3-Oct	2	Completed
4.1.5	Introduction - Significance of Research	Zhee Xuan	1-Oct	3-Oct	2	Completed
4.2.1	Literature Review - Overview	Rui Xuan	3-Oct	7-Oct	4	Completed
4.2.2	Literature Review - Domain and Problem Scenario	Zhee Xuan	3-Oct	7-Oct	4	Completed
4.2.3	Literature Review - Tools and Techniques	Rui Xuan	3-Oct	7-Oct	4	Completed

4.2.4	Literature Review - Data Analysis	Yi Hang	3-Oct	7-Oct	4	Completed
4.2.5	Literature Review - Machine Learning Model	Yun Jie	3-Oct	7-Oct	4	Completed
4.2.6	Literature Review - Verification and Validation	Yun Jie	3-Oct	7-Oct	4	Completed
4.2.7	Literature Review - Framework	James	3-Oct	7-Oct	4	Completed
4.2.8	Literature Review - Expected Contribution	Zhee Xuan	3-Oct	7-Oct	4	Completed
4.2.9	Literature Review - Overall Summary	Zhee Xuan	7-Oct	8-Oct	1	Completed
4.3.1	Methodology - Overview	James	11-Oct	13-Oct	2	Completed
4.3.2	Methodology - Planning and Requirement Gathering	Yi Hang	11-Oct	13-Oct	2	Completed
4.3.3	Methodology - Requirement Analysis	Zhee Xuan	11-Oct	13-Oct	2	Completed
4.3.4	Methodology - Implementation	Rui Xuan	11-Oct	13-Oct	2	Completed
4.3.5	Methodology - Testing	Yun Jie	13-Oct	15-Oct	2	Completed
4.3.6	Methodology - Deployment	James	13-Oct	15-Oct	2	Completed
4.4.1	Expected Contribution - Overview	Rui Xuan	9-Oct	11-Oct	2	Completed
4.4.2	Expected Contribution - Impact to the industry	Rui Xuan	9-Oct	11-Oct	2	Completed

4.4.3	Expected Contribution - Impact to the Model as New Knowledge	Zhee Xuan	9-Oct	11-Oct	2	Completed
4.5.1	Project Planning - Overview	Yi Hang	9-Oct	11-Oct	2	Completed
4.5.2	Project Planning - Gantt Chart	Yi Hang & James	9-Oct	11-Oct	2	Completed
4.5.3	Project Planning - Delegation of Work	Yi Hang & Yun Jie	9-Oct	11-Oct	2	Completed
4.6	Abstract	Rui Xuan	14-Oct	15-Oct	1	Completed
5	**Development**	All	16-Oct	16-Nov	31	Not Started
5.1.1	Machine Learning Model - Data Extraction	Yi Hang	16-Oct	21-Oct	5	Not Started
5.1.2	Machine Learning Model - Data Preparation	Rui Xuan	21-Oct	23-Oct	2	Not Started
5.1.3	Machine Learning Model - Data Analysis	Yi Hang	23-Oct	26-Oct	3	Not Started
5.1.4	Machine Learning Model - Model Training	Rui Xuan	27-Oct	6-Nov	10	Not Started
5.1.5	Machine Learning Model - Model Evaluation	Yun Jie	5-Nov	10-Nov	5	Not Started
5.1.6	Machine Learning Model - Model Validation	Yun Jie	11-Nov	16-Nov	5	Not Started
5.2.1	Website Development - Planning	James	16-Oct	18-Oct	2	Not Started

5.2.2	Website Development - Design and Layout	Rui Xuan	19-Oct	22-Oct	3	Not Started
5.2.3	Website Development - Front-End Development	Zhee Xuan	23-Oct	30-Oct	7	Not Started
5.2.4	Website Development - Back-End Development	James	1-Nov	8-Nov	7	Not Started
5.2.5	Website Development - Testing and Quality Assurance	James	9-Nov	16-Nov	7	Not Started
6	**Confirmation and Deployment**	All	16-Nov	26-Nov	10	Not Started
6.1	Final Implementation	All	16-Nov	20-Nov	4	Not Started
6.2	System Completion	All	21-Nov	26-Nov	5	Not Started
7	**Showcase Preparation**	All	24-Nov	11-Dec	17	Not Started
7.1	Overview plan	All	24-Nov	25-Nov	1	Not Started
7.2	Prepare Contents (Slides, Poster etc.)	All	26-Nov	1-Dec	5	Not Started
7.3	Test Run 1	All	2-Dec	5-Dec	3	Not Started
7.4	Test Run 2	All	6-Dec	8-Dec	2	Not Started
7.5	Finalize Showcase	All	9-Dec	11-Dec	2	Not Started
8	**Final Report**	All	16-Nov	17-Dec	31	Not Started

8.1	Abstract	All	16-Nov	23-Nov	7	Not Started
8.2	Introduction	All	16-Nov	23-Nov	7	Not Started
8.3	Literature Review	All	16-Nov	23-Nov	7	Not Started
8.4	Methodology	All	24-Nov	6-Dec	12	Not Started
8.5	Result and Analysis	All	27-Nov	9-Dec	12	Not Started
8.6	Conclusion and Future Work	All	10-Dec	14-Dec	4	Not Started
8.7	Finalize Report	All	14-Dec	17-Dec	3	Not Started

Workload distribution among team members is a significant part as an optimal workload distribution not only preserves the quality of performance and productivity of a project but also ensures that the quality of the tasks and the time required to complete the tasks are effective. The tasks each group member was assigned to were delegated based on our interests and personal strengths. Each group member is in charge of a different sub-section of the task they are strong on, and at the end, we will all contribute to every section of the project.

Along with individual strengths, a balanced workload is another important consideration when deciding which task to assign to each member. A balanced workload helps avoid frustration and exhaustion as team members are pretty assigned tasks by considering their availability and workload. Moreover, before delegating the tasks, the team leader will ask if the members have any preferences or choices for the tasks, so they are able to choose what they are interested in. To balance the workload, more complicated and time-consuming tasks will be assigned to two members.

5.4 Conclusion

In our effort to develop a cutting-edge skincare ingredient analyzer, diligent and thorough project planning acts as our compass. We have set ourselves up for a project that is both doable and meaningful by defining clear objectives, specifying tasks, creating realistic schedules, and efficiently allocating resources. Collaboration and communication, the two pillars of our project plan, are critical in building a cohesive and motivated team. As we move into the development phase, these concepts will remain crucial to the success of our project.

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APPENDICES

Appendix A: Gantt Chart

Capstone - Gantt
chart.xlsx